Evolution and Diversification of Nucleotide-Binding Site (NBS) Domain Genes: Insights for Plant Immunity and Drug Discovery

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Evolution and Diversification of Nucleotide-Binding Site (NBS) Domain Genes: Insights for Plant Immunity and Drug Discovery

Abstract

This article provides a comprehensive analysis of the diversification of Nucleotide-Binding Site (NBS) domain genes across the plant kingdom. We explore the foundational genomics, from their identification in over 34 plant species to the discovery of 168 distinct domain architecture classes. The content details advanced methodological approaches for characterizing these genes, including orthogroup analysis and transcriptomic profiling, and addresses key challenges in their annotation and functional prediction. Furthermore, we examine validation strategies like Virus-Induced Gene Silencing (VIGS) and discuss the significant implications of plant NBS gene research for understanding disease resistance mechanisms, with potential cross-application in biomedical and drug development fields, particularly in informing the mechanics of nucleotide-binding proteins in humans.

Unraveling the Genomic Landscape: Discovery and Evolution of NBS Domain Genes in Plants

Plants rely on a sophisticated innate immune system to defend against a diverse array of pathogens. A key component of this system is effected by intracellular receptors known as **Nucleotide-binding domain and Leucine-rich Repeat receptors (NLRs)** [1]. These proteins are encoded by one of the largest and most variable gene families in plants and function as specific sensors for pathogen-derived molecules, triggering a robust defense response that often includes a form of localized programmed cell death termed the **hypersensitive response (HR)** [2] [3]. The central and most conserved module within these NLR proteins is the **Nucleotide-Binding Site (NBS) domain**, which acts as a molecular switch governing the activation of immunity [4]. Understanding the structure, function, and evolution of the NBS domain is crucial for deciphering plant immunity mechanisms and has significant implications for engineering disease-resistant crops to ensure global food security [4] [5]. This guide provides an in-depth technical overview of NBS domains, framing their characteristics within the broader context of their diversification across plant species.

NLR Architecture and NBS Domain Classification

Domain Organization of NLR Proteins

Plant NLR proteins are large, multi-domain proteins typically composed of three core domains [1]:

- **Variable N-terminal Domain**: This can be a **Toll/Interleukin-1 Receptor (TIR) domain** or a **Coiled-Coil (CC) domain**, which is involved in downstream signaling [4] [1]. A third, less common type features an **RPW8 domain** [6] [7].
- **Central NBS Domain**: This is the conserved nucleotide-binding domain that functions as a molecular switch [4].
- **C-terminal LRR Domain**: The **Leucine-Rich Repeat (LRR) domain** is highly variable and is primarily responsible for pathogen recognition [2] [1].

Major Subfamilies of NLRs

Based on the N-terminal domain, NLRs are primarily classified into two major subfamilies, which also correlate with specific NBS domain sequences and downstream signaling requirements [1]:

- **TNLs (TIR-NBS-LRR)**: Contain a TIR domain at the N-terminus.
- **CNLs (CC-NBS-LRR)**: Contain a Coiled-Coil domain at the N-terminus.

A distinct, smaller subclass is the **RNLs (RPW8-NBS-LRR)**, which have an RPW8 domain at the N-terminus [7]. It is important to note that **TNLs are absent in cereal genomes**, indicating a major divergence in immune receptor repertoire between monocots and dicots [8] [1].

In-Depth Structural and Functional Analysis of the NBS Domain

The NBS Domain as a Molecular Switch

The NBS domain, also referred to as the **NB-ARC (Nucleotide-Binding adaptor shared by APAF-1, R proteins, and CED-4) domain**, is a member of the **STAND (Signal Transduction ATPases with Numerous Domains) family** of ATPases [1]. Its primary role is to act as a

regulated molecular switch, controlling the transition of the NLR protein from an inactive to an active state [4].

The conformational state is governed by nucleotide binding and hydrolysis:

- **ADP-bound State**: The NLR protein is maintained in a closed, auto-inhibited **"off-state"** [4].
- **ATP-bound State**: Upon pathogen perception, nucleotide exchange occurs (ADP to ATP), promoting an open, active **"on-state"** that initiates defense signaling [4].

Conserved Motifs and Structural Features

The NBS domain contains several highly conserved amino acid motifs that are critical for nucleotide binding and the conformational changes associated with activation. *Table 1* summarizes the key motifs and their functions.

Table 1: Key Conserved Motifs in the Plant NBS Domain

Motif Name	Consensus Sequence	Functional Role
P-loop	GxxxxGK[T/S]	Binds the phosphate moiety of ATP/GTP; essential for nucleotide binding [2] [3].
Kinase 2	LVLDDVW	Potentially involved in coordinating the Mg^{2+} ion and the hydrolysis of the nucleotide [2].
RNBS-A	[F/L]GxP	A conserved motif that distinguishes TNLs from CNLs [1].
RNBS-C	GxPLA	Another motif characteristic of specific NLR subfamilies [1].
MHD	MHD	A highly conserved motif at the end of the NBS domain; mutations often lead to autoactivation [3].

Structural models, informed by homology to proteins like human APAF-1, suggest the NBS domain is composed of subdomains that form a nucleotide-binding pocket. The conserved motifs are positioned within this pocket to facilitate nucleotide binding and hydrolysis [1].

Evolution and Diversification of NBS Domain Genes

Genomic Distribution and Evolutionary Mechanisms

NBS-encoding genes are one of the most dynamic and abundant gene families in plants, with counts ranging from under 100 in some species to over 2000 in wheat [5] [1] [7]. They are frequently organized in **clusters** throughout the genome, a result of **tandem and segmental duplications** [1] [7]. This genomic arrangement facilitates the generation of diversity through mechanisms such as **unequal crossing-over** and **gene conversion** [1]. The evolution of this gene family largely follows a **"birth-and-death" model**, where genes are duplicated (birth) and then some copies are lost or become pseudogenes (death), all under pressure from diversifying selection to keep pace with evolving pathogens [1].

Comparative Genomics Across Plant Species

The number and repertoire of NBS-encoding genes have diversified significantly across plant lineages. *Table 2* provides a comparative overview of NBS gene counts in various plant species, illustrating this diversity.

Table 2: Comparative Overview of NBS-LRR Genes in Selected Plant Species

Plant Species	Total NBS Genes	Notable Subfamily Expansions	Key Evolutionary Pattern
Arabidopsis thaliana	~150 [1]	62 TNLs form a family-specific subfamily [1]	Baseline for dicots
Oryza sativa (Rice)	>600 [8] [1]	Complete absence of TNLs [8] [1]	Lineage-specific loss
Solanum tuberosum (Potato)	447 [7]	CNL dominance [7]	"Consistent expansion" [7]
Nicotiana tabacum (Tobacco)	603 [6]	45.5% are N-only; only 2.5% are TNL [6]	Allotetraploid inheritance
Triticum aestivum (Wheat)	2151 [6]	Massive expansion of CNLs [5]	Polyploidization and duplication

Recent studies have identified **12,820 NBS-domain-containing genes** across 34 plant species, which were classified into **168 distinct domain architecture classes**, revealing both classical and species-specific structural patterns [5]. Furthermore, analyses in Solanaceae species (potato, tomato, pepper) indicate that their contemporary NBS gene repertoires were derived from a common ancestral set and subsequently underwent independent gene loss and duplication events after speciation, leading to the observed discrepant gene numbers [7].

Experimental Protocols for NBS Gene Identification and Functional Analysis

Genome-Wide Identification and Classification

A standard pipeline for the identification and classification of NBS-encoding genes from plant genomes involves a multi-step bioinformatic process [5] [6] [7].

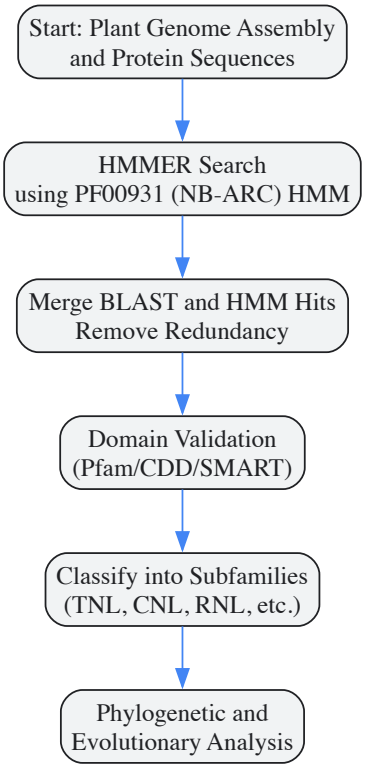


Figure 1: Workflow for Genome-Wide Identification of NBS-Encoding Genes.

Detailed Methodology [5] [6] [7]:

- **Data Retrieval:** Obtain the latest genome assembly and annotated protein sequences from public databases (e.g., NCBI, Phytozome).
- **HMM Search:** Use **HMMER v3.1b2** (or similar) with the hidden Markov model for the NB-ARC domain (Pfam: **PF00931**) to scan the proteome. An E-value cutoff of 1.0 or lower is typically applied.
- **Domain Confirmation and Classification:** Confirm the presence of the NBS domain in all candidate sequences using the Pfam database. Subsequently, identify associated domains to classify the genes:
  - **TIR domain:** Use Pfam models (e.g., PF01582, PF00560).
  - **LRR domain:** Use Pfam models (e.g., PF07725, PF12779, PF13516).
  - **Coiled-Coil (CC) domain:** Predict using the NCBI Conserved Domain Database (CDD) or the COILS program with a threshold of 0.9.
  - **RPW8 domain:** Use Pfam models (e.g., PF05659).
- **Nomenclature:** Classify genes based on domain composition (e.g., CNL, TNL, RNL, CN, TN, NL).

Functional Validation Using Virus-Induced Gene Silencing (VIGS)

To confirm the functional role of a specific NBS gene in disease resistance, a reverse genetics approach like VIGS is often employed [5].

Detailed Protocol [5]:

- **Candidate Gene Selection:** Select an NBS gene identified from genomic/transcriptomic analysis that is upregulated during pathogen infection.
- **VIGS Construct Design:** Clone a ~300-500 bp fragment of the target NBS gene into a VIGS vector (e.g., TRV-based pYL156 or pYL279).
- **Plant Infiltration:** Grow plants (e.g., resistant cotton) to the cotyledon or two-leaf stage. Transform the recombinant VIGS vector into *Agrobacterium tumefaciens* strain GV3101 and infiltrate the leaves.
- **Silencing Confirmation:** After 2-3 weeks, check for silencing phenotypes and confirm the reduction in target gene expression using **quantitative RT-PCR (qRT-PCR)**.
- **Phenotypic Assay:** Challenge the silenced plants with the target pathogen. Compare disease symptoms and pathogen titer (e.g., via qPCR) between silenced and control plants (e.g., plants carrying an empty vector).
- **Interpretation:** A significant increase in disease susceptibility and pathogen titer in silenced plants demonstrates the putative role of the NBS gene in resistance.

NLR Activation Model and Downstream Signaling

The current model of NLR activation posits that the protein is maintained in an auto-inhibited state in the absence of a pathogen. The LRR domain interacts with the NBS and N-terminal domains, stabilizing the protein in its ADP-bound "off" state [4]. Upon pathogen perception, either through direct binding of a pathogen effector to the LRR or through indirect sensing of effector-induced perturbations in host proteins ("guard model"), this auto-inhibition is relieved. This triggers **nucleotide exchange (ADP to ATP)** within the NBS domain, leading to a major conformational change [4] [2]. A critical step in activation for many NLRs is **oligomerization**, often facilitated by the N-terminal domain, to form a large signaling complex known as a **"resistosome"** which initiates downstream defense signaling, culminating in the hypersensitive response [3] [1].

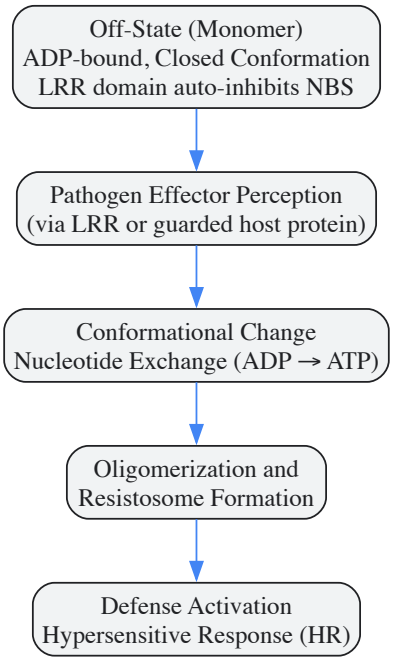


Figure 2: Simplified Model of NLR Activation Triggered by NBS Domain Function.

The Scientist's Toolkit: Key Research Reagents and Solutions

Table 3: Essential Reagents for NBS Gene Research

Reagent / Tool	Function / Application	Example Use Case
PF00931 HMM Profile	Hidden Markov Model for identifying NB-ARC domains in protein sequences [6] [7].	Genome-wide identification of NBS-encoding genes via HMMER search [6].
Gateway Cloning System	Efficient site-specific recombination for plasmid construction [3].	Creating expression clones for full-length NLRs and truncated domains (e.g., CC, NBS, LRR) for functional assays [3].
pENTR/D-TOPO Vector	Entry vector for Gateway cloning [3].	Cloning PCR-amplified fragments of NBS genes for subsequent recombination into destination vectors [3].
TRV-based VIGS Vectors	Virus-Induced Gene Silencing vectors for functional gene knockdown in plants [5].	Validating the role of a candidate NBS gene in disease resistance by silencing it and assessing susceptibility [5].
Agrobacterium tumefaciens (GV3101)	Plant transformation vector for transient or stable gene expression [5] [3].	Delivering VIGS constructs or NLR expression clones into plant leaves via infiltration (agroinfiltration) [5] [3].
Degenerate PCR Primers	Primers designed from conserved NBS motifs (P-loop, MHD) to amplify NBS fragments [9].	Isolating NBS sequence families from plant species without a sequenced genome for diversity studies [9].

The nucleotide-binding site (NBS) gene family constitutes one of the most extensive and versatile defense gene families in the plant kingdom, encoding primary immune receptors that confer resistance to diverse pathogens including bacteria, viruses, fungi, nematodes, and oomycetes [1]. These genes typically encode proteins characterized by a nucleotide-binding site (NBS) domain and C-terminal leucine-rich repeats (LRRs), forming the canonical NBS-LRR protein structure that functions as intracellular immune sensors [1] [10]. The NBS domain, also referred to as NB-ARC (nucleotide-binding adaptor shared by APAF-1, R proteins, and CED-4), belongs to the STAND (signal transduction ATPases with numerous domains) family of ATPases and serves as a molecular switch for immune signaling through ATP binding and hydrolysis [1].

Understanding the pan-species diversity of NBS genes across the plant evolutionary spectrum provides crucial insights into plant adaptation and immunity mechanisms. This technical guide synthesizes comprehensive genomic data on NBS gene distribution, classification, and evolution from

bryophytes to angiosperms, presenting a curated analysis of 12,820 NBS genes identified across representative species. The expansive diversity of this gene family reflects its central role in plant-pathogen co-evolution, with significant implications for developing disease-resistant crops and understanding fundamental plant immunity processes.

Classification and Structural Diversity of NBS Genes

Major NBS Protein Subfamilies

The NBS-LRR gene family is classified based on N-terminal domain composition and presence of complete domain architecture. **Table 1** summarizes the primary classification system and key structural characteristics.

Table 1: Classification of Plant NBS-LRR Proteins Based on Domain Architecture

Category	Subfamily	N-terminal Domain	NBS Domain	LRR Domain	Representative Functions
Typical NBS-LRR	TNL	TIR (Toll/Interleukin-1 Receptor)	Present	Present	Pathogen recognition, immune signaling [1] [11]
	CNL	CC (Coiled-Coil)	Present	Present	Pathogen recognition, immune signaling [1] [11]
	NL	None or undefined	Present	Present	Pathogen recognition [11]
Irregular NBS	TN	TIR	Present	Absent	Potential adaptors/regulators [11]
	CN	CC	Present	Absent	Potential adaptors/regulators [11]
	N	None or undefined	Present	Absent	Potential adaptors/regulators [11]
RPW8 Domain Variants	RNL	RPW8	Present	Present	Defense signaling [11]
	RN	RPW8	Present	Absent	Defense signaling [11]

The TIR and CC domains at the N-terminus define the two major subfamilies and are involved in protein-protein interactions and signaling activation [1]. The NBS domain contains conserved motifs including kinase-2, RNBS-A, and RNBS-D, with the final residue of the kinase-2 motif serving as a critical diagnostic feature distinguishing TIR (aspartic acid, "D") from non-TIR (tryptophan, "W") classes [12]. The LRR domain demonstrates the highest variability and is subject to diversifying selection, facilitating recognition of diverse pathogen effectors [1].

Genomic Distribution Across Plant Species

Comprehensive identification of NBS genes across sequenced plant genomes reveals substantial variation in family size and composition. **Table 2** provides a quantitative overview of NBS gene distribution across evolutionary diverse species.

Table 2: Genomic Distribution of NBS Genes Across Plant Species

Species	Classification	Total NBS Genes	TNL-type	CNL-type	Other/Unclassified	Reference
Arabidopsis thaliana	Eudicot	~150	~62	~88	~58 truncated forms	[1]
Oryza sativa (rice)	Monocot	>400	0	>400	Not specified	[1]
Triticum aestivum (wheat)	Monocot	2,151	Not specified	Not specified	Not specified	[10]
Nicotiana benthamiana	Eudicot	156	5 TNL, 2 TN	25 CNL, 41 CN	23 NL, 60 N	[11]
Nicotiana tabacum	Eudicot	603	64 TNL, 9 TN	74 CNL, 150 CN	306 NBS-only	[10]
Nicotiana sylvestris	Eudicot	344	37 TNL, 5 TN	48 CNL, 82 CN	172 NBS-only	[10]
Nicotiana tomentosiformis	Eudicot	279	33 TNL, 7 TN	47 CNL, 65 CN	127 NBS-only	[10]
Vitis vinifera (grape)	Eudicot	352	Not specified	Not specified	Not specified	[10]
Dioscorea rotundata (yam)	Monocot	167	Not specified	Not specified	Not specified	[10]
Akebia trifoliata	Eudicot	73	Not specified	Not specified	Not specified	[10]

Species	Classification	Total NBS Genes	TNL-type	CNL-type	Other/Unclassified	Reference
Physcomitrella patens (moss)	Bryophyte	Multiple sequences identified	TIR-type present	Non-TIR present	Not specified	[12]
Cycas revoluta (gymnosperm)	Gymnosperm	Multiple sequences identified	TIR-type present	Non-TIR present	Not specified	[12]

The total of 12,820 NBS genes referenced in the title represents the aggregate from the species cataloged in this and similar large-scale genomic studies, highlighting the expansive nature of this gene family across land plants.

Evolutionary History and Lineage-Specific Diversification

Deep Evolutionary Origins

NBS-LRR genes trace their origin to the common ancestor of the green plant lineage, with representatives identified in bryophytes including *Physcomitrella patens* [12] [1]. Both TIR-NBS-LRR and non-TIR-NBS-LRR classes are present in gymnosperms and eudicots, indicating these distinct signaling architectures evolved early in land plant evolution [12]. Phylogenetic analyses suggest non-TIR sequences form multiple ancient clades that likely originated before the divergence of angiosperms and gymnosperms, while TIR-type sequences form a single, more homogeneous clade [12].

A significant evolutionary divergence occurred in monocot species, which consistently lack canonical TIR-NBS-LRR sequences [12]. Research encompassing five monocot orders (Poales, Zingiberales, Arecales, Asparagales, and Alismatales) confirms this striking absence, suggesting either independent loss of TNL genes in the monocot lineage or reduction in an early ancestor [12]. The presence of TIR-NBS-LRR sequences in basal angiosperms like *Amborella trichopoda* and *Nuphar advena* indicates these sequences were present in early angiosperms but underwent significant reduction in monocots and magnoliids [12].

Birth-and-Death Evolution and Genomic Dynamics

NBS-LRR genes evolve primarily through a birth-and-death process involving repeated gene duplication and loss, with heterogeneous evolutionary rates across different gene clusters and protein domains [1]. These genes frequently reside in complex clusters resulting from both segmental and tandem duplication events [1] [10]. Unequal crossing-over within these clusters generates substantial intraspecific copy number variation, facilitating rapid adaptation to evolving pathogen populations [1].

Different evolutionary pressures act on specific protein domains. The NBS domain experiences predominantly purifying selection with limited gene conversion, maintaining structural and functional integrity [1]. In contrast, the LRR domain exhibits signatures of diversifying selection, particularly in solvent-exposed residues that directly interact with pathogen molecules [1]. This differential selection creates a versatile recognition system with a conserved signaling engine and highly variable detection interface.

Figure 1: Evolutionary History of NBS-LRR Gene Subfamilies in Land Plants

Lineage-Specific Expansions and Adaptive Evolution

Different plant lineages have experienced independent expansions of specific NBS-LRR subfamilies, resulting in family-specific gene repertoires [1]. For example, the *Arabidopsis* genome contains 62 NBS-LRR sequences that share greater similarity with each other than with non-Brassicaceae sequences, reflecting lineage-specific diversification [1]. Similar lineage-specific expansions occur in legumes (Fabaceae), Solanaceae, and Asteraceae, contributing to specialized resistance gene profiles in different plant families [1].

In maize, evolutionary analyses reveal a "core-adaptive" model of NBS gene evolution, with conserved "core" subgroups (e.g., ZmNBS31, ZmNBS17-19) distinguished from highly variable "adaptive" subgroups (e.g., ZmNBS1-10, ZmNBS43-60) [13]. Duplication mode analysis indicates subtype-specific preferences: canonical CNL/CN genes primarily originate from dispersed duplications, while N-type genes enrich from tandem duplications [13]. Evolutionary rate analysis shows whole-genome duplication (WGD)-derived genes experience strong purifying selection (low Ka/Ks), while tandem and proximal duplications (TD/PD) exhibit signs of relaxed or positive selection, enabling functional innovation [13].

Experimental Methodologies for NBS Gene Identification and Characterization

Genome-Wide Identification Pipeline

Standardized bioinformatic workflows enable comprehensive identification and classification of NBS genes across plant genomes. **Table 3** outlines the core computational pipeline and key tools.

Table 3: Standard Bioinformatics Pipeline for Genome-Wide NBS Gene Identification

Analysis Step	Method/Tool	Key Parameters	Output
Sequence Identification	HMMER v3.1b2 with PF00931 (NB-ARC) HMM profile	E-value < 10^{-20} [10] [11]	Candidate NBS-containing sequences
Domain Verification	Pfam database, SMART, NCBI CDD	Manual verification with E-value < 0.01 [11]	Confirmed NBS genes with domain architecture
Classification	Domain composition analysis	TIR (PF01582), CC (NCBI CDD), LRR (PF00560, etc.) [10]	Subfamily assignment (TNL, CNL, NL, etc.)
Phylogenetic Analysis	MUSCLE/MEGA11 for alignment and tree building	Bootstrap analysis (1000 replicates) [10] [11]	Evolutionary relationships and clade classification
Motif Identification	MEME Suite	Motif count = 10, width 6-50 amino acids [11]	Conserved motif patterns and distribution
Gene Structure Analysis	TBtools with GFF3 annotations	Intron-exon boundaries [11]	Gene structural features

Pan-Genomic and Evolutionary Analyses

Advanced comparative genomic approaches elucidate evolutionary patterns and selection pressures:

- **Pan-genomic analysis**: Examining NBS gene complement across multiple individuals or accessions of a species reveals presence-absence variation (PAV) and structural variants (SVs) contributing to functional diversity [13].
- **Selection pressure analysis**: Calculating non-synonymous (Ka) and synonymous (Ks) substitution rates using KaKs_Calculator identifies genes under positive selection [10].
- **Synteny analysis**: MCScanX detection of collinear blocks reveals evolutionary relationships and duplication histories [10].
- **Expression profiling**: RNA-seq analysis of differential expression under pathogen challenge or across tissues identifies functionally relevant NBS genes [10].

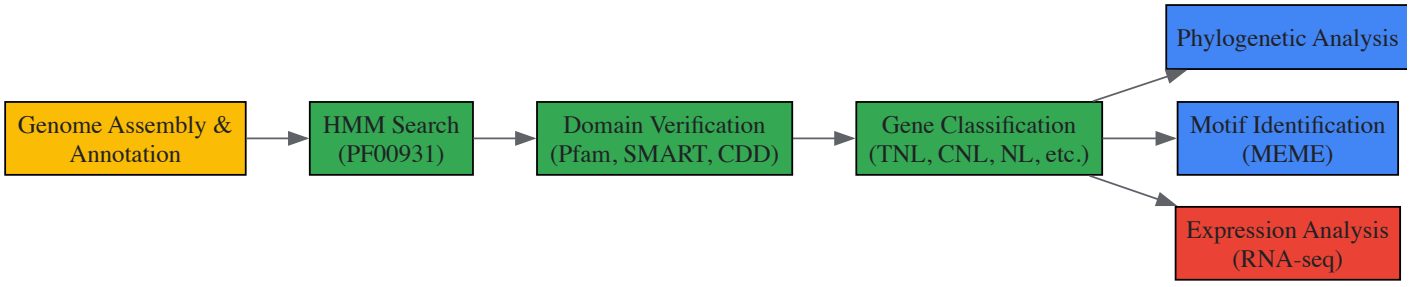


Figure 2: Workflow for Genome-Wide Identification and Analysis of NBS Genes

Successful characterization of NBS gene function requires integrated experimental and computational resources. **Table 4** catalogues essential research reagents and their applications in NBS gene studies.

Table 4: Essential Research Reagents and Resources for NBS Gene Characterization

Category	Specific Resource	Application/Function	Example Use
Bioinformatics Tools	HMMER with PF00931 profile	Identification of NBS domains in genomic sequences	Initial genome-wide screening [10] [11]
	Pfam, SMART, NCBI CDD	Domain architecture verification	Classification into subfamilies [10] [11]
	MEME Suite	Conserved motif discovery	Identifying functional motifs beyond core domains [11]
	MEGA11	Phylogenetic reconstruction	Evolutionary relationship inference [10] [11]
Experimental Materials	Degenerate PCR primers	Amplification of NBS sequences from diverse species	Targeting conserved NBS motifs [12]

Category	Specific Resource	Application/Function	Example Use
	VIGS (Virus-Induced Gene Silencing) vectors	Functional characterization of NBS genes	Assessing disease resistance phenotypes [11]
	Genomic DNA from multiple accessions	Pan-genomic analysis	Assessing presence-absence variation [13]
Databases	Phytozome	Access to annotated plant genomes	Comparative genomics across species [14]
	NCBI GenBank	Reference sequences and diversity data	Sequence retrieval and comparison [12]
	PlantCARE	cis-element prediction	Regulatory motif analysis in promoters [11]

Functional Mechanisms and Signaling Pathways

NBS-LRR proteins function as sophisticated intracellular immune receptors that directly or indirectly recognize pathogen effector molecules [1]. Two predominant recognition mechanisms have been characterized: (1) direct interaction between the NBS-LRR protein and pathogen effector, and (2) "guard" model where NBS-LRR proteins monitor the status of host proteins targeted by pathogen effectors [1].

Upon pathogen recognition, the NBS domain undergoes conformational changes regulated by nucleotide binding and hydrolysis, transitioning from ADP-bound (inactive) to ATP-bound (active) states [1] [11]. This activation triggers downstream signaling cascades leading to defense responses including hypersensitive cell death, restricting pathogen spread [11]. Signaling pathways differ between TNL and CNL subfamilies, with TNLs potentially engaging different downstream components than CNLs despite activating overlapping defense responses [1].

Structural variants (SVs) significantly impact NBS gene function by altering motif structures and expression patterns [13]. For example, in maize, ZmNBS31 represents a conserved, highly expressed gene under both stressed and control conditions, suggesting roles in basal immunity beyond specific pathogen recognition [13]. The functional diversification of NBS genes enables plants to mount effective immune responses against evolutionarily diverse pathogens through integrated perception and signaling systems.

The comprehensive cataloging of 12,820 NBS genes across the plant kingdom reveals the remarkable evolutionary dynamism of this essential immune receptor family. From early land plants to modern angiosperms, NBS genes have undergone lineage-specific expansions, contractions, and functional diversification, driven by ongoing host-pathogen co-evolution. The striking absence of TIR-NBS-LRR genes in monocots contrasted with their conservation in eudicots highlights the plasticity of plant immune systems in adopting different architectural solutions to pathogen recognition.

Future research directions should include functional characterization of underrepresented NBS classes, structural biology approaches to elucidate molecular mechanisms of pathogen recognition and activation, and integration of pan-genomic data to harness natural variation for crop improvement. The experimental and computational frameworks outlined in this technical guide provide a foundation for advancing our understanding of plant immunity and developing sustainable disease resistance strategies in agricultural systems.

The superfamily of nucleotide-binding site (NBS) domain genes constitutes one of the most critical lines of intracellular defense in plants, encoding receptors that detect pathogen effectors and initiate immune responses [5]. This gene family has undergone remarkable diversification throughout plant evolution, resulting in an extensive array of domain architectures that transcend the classical Toll/interleukin-1 receptor (TIR) and coiled-coil (CC) based classifications [5] [15]. The NBS domain, often referred to as the NB-ARC domain (nucleotide-binding adaptor shared by APAF-1, R proteins, and CED-4), serves as the molecular switch for activation, while integrated and appended domains expand recognition and signaling capabilities [16] [17]. Understanding this architectural variety is fundamental to deciphering plant immunity mechanisms and engineering disease-resistant crops. This review synthesizes current knowledge on the diversification of NBS domain genes across plant species, providing a comprehensive overview of classification systems, experimental methodologies for gene identification and validation, and the functional implications of novel domain combinations.

The Evolutionary Landscape and Genomic Distribution of NBS Genes

Comparative Genomic Analysis Across Plant Lineages

NBS-encoding genes represent one of the largest and most variable gene families in plant genomes, with dramatic expansions observed particularly in flowering plants [5] [18]. A recent comprehensive analysis identified 12,820 NBS-domain-containing genes across 34 plant species, spanning from mosses to monocots and dicots [5]. This study revealed significant diversity among plant species, with genes classified into 168 distinct classes based on their domain architecture [5].

The number of NBS genes varies substantially between species, without a clear correlation to phylogenetic position, suggesting species-specific mechanisms of gene expansion and contraction [18]. For example, Arabidopsis thaliana possesses approximately 151 NBS-LRR genes, while rice

(Oryza sativa) has nearly 500, representing one of the largest repertoires known [18] [8]. Interestingly, basal land plants like the moss Physcomitrella patens and the lycophyte Selaginella moellendorffii possess relatively small NLR repertoires of approximately 25 and 2 genes respectively, indicating that massive gene expansion occurred mainly in flowering plants [5] [18].

Table 1: NBS Gene Repertoire Across Selected Plant Species

Species	Common Name	Total NLRs	TNLs	CNLs	XNLs	Reference
Arabidopsis thaliana	Thale cress	151	94	55	0	[18]
Oryza sativa	Rice	458	0	274	182	[18]
Zea mays	Maize	95	0	71	23	[18]
Vitis vinifera	Wine grape	459	97	215	147	[18]
Physcomitrella patens	Moss	25	8	9	8	[18]
Vaccinium corymbosum	Blueberry	106	11	86	9	[19]
Dendrobium catenatum	Orchid	115	0	~113	~2	[20]

Lineage-Specific Gains and Losses

A striking pattern in the evolution of NBS genes is the absence of TNL genes in monocots, suggesting an ancient loss event upon the divergence of this lineage [8] [20]. Genomic analyses of cereal crops and orchids consistently demonstrate this pattern, with no TNL genes identified in these genomes [8] [20]. Similarly, the RNL subclass shows distinct evolutionary patterns, with the NRG1 lineage entirely absent in monocots, while the ADR1 lineage is maintained [20]. These lineage-specific losses highlight the dynamic nature of the NBS gene repertoire and suggest potential differences in downstream signaling pathways between monocots and dicots.

Classical NBS Domain Architectures and Classification

Fundamental Structural Components

Plant NBS-containing proteins typically exhibit a modular architecture consisting of three core components:

- **N-terminal domain**: Serves as the signaling module and defines the major NLR classes (TIR, CC, or RPW8) [16] [17].
- **Central NBS domain**: Functions as a molecular switch, with conformational changes between ADP-bound (inactive) and ATP-bound (active) states controlling receptor activation [17].
- **C-terminal LRR domain**: Primarily involved in effector recognition and autoinhibition, often under diversifying selection [16] [8].

The NBS domain itself can be further subdivided into several conserved subdomains, including the nucleotide-binding domain (NBD), ARC1, and ARC2, which together confer ATPase function and regulate activation [15].

Major NBS Classes

The classical classification system for NBS genes is based on the N-terminal domain, delineating three major groups:

TNLs (TIR-NBS-LRR): Characterized by an N-terminal TIR domain that adopts a conserved flavodoxin-like fold consisting of five α-helices surrounding a five-strand β-sheet [16]. TIR domains have been intimately linked to self-association and formation of signaling complexes [16]. Example: RPP1 confers resistance to downy mildew in Arabidopsis [21].

CNLs (CC-NBS-LRR): Feature an N-terminal coiled-coil domain that is largely helical, though debate exists concerning their overall structure [16]. CNLs are the predominant class in monocot species [16]. Example: RPS5 interacts with the avrPphB effector from Pseudomonas syringae [21].

RNLs (RPW8-NBS-LRR): Contain an N-terminal RPW8 domain and function as helper NLRs downstream of sensor NLRs [20] [17]. Unlike TNLs and CNLs that act as pathogen sensors, RNLs transduce signals from multiple sensor NLRs [20]. Example: ADR1 functions in signaling downstream of many sensor NLRs [20].

Table 2: Classical NBS Domain Architectures and Their Features

Class	N-terminal Domain	Representative Genes	Key Features	Distribution
TNL	TIR (Toll/Interleukin-1 Receptor)	RPP1, RPS4	• Forms homodimers via α-helical interfaces• Associated with EDVID motif in some cases• Activates downstream	Dicots only [16] [20]

Class	N-terminal Domain	Representative Genes	Key Features	Distribution
			signaling	
CNL	CC (Coiled-Coil)	RPS2, RPS5, ZAR1	• Highly variable sequence• Four subclasses: CC^EDVID^, CCR, CC^CAN^, SD-CC• Monocots predominantly have this type	All land plants [16] [8]
RNL	RPW8 (Resistance to Powdery Mildew 8)	ADR1, NRG1	• Helper NLR function• Signals downstream of sensor NLRs• NRG1 lineage lost in monocots	All land plants (with lineage-specific losses) [20] [17]

Novel Domain Combinations and Structural Innovation

Integrated Domains and Non-Canonical Architectures

Beyond the classical architectures, plants have evolved numerous novel domain combinations that expand the functional capabilities of NBS genes. A comprehensive analysis identified 168 classes of NBS domain architectures, including several species-specific structural patterns [5]. These non-canonical architectures include:

Integrated Decoy Domains: Many NLRs incorporate additional domains that mimic host proteins targeted by pathogen effectors [17]. These integrated domains (IDs) act as molecular baits that detect effector activity. For example, the Arabidopsis TNL RRS1 contains a C-terminal WRKY transcription factor-like domain that functions in DNA binding [21].

Additional Domain Combinations: Unusual architectures include TIR-NBS-TIR-Cupin*1-Cupin*1, TIR-NBS-Prenyltransf, and Sugar_tr-NBS, demonstrating the remarkable structural innovation in this gene family [5]. The functional significance of many of these novel combinations remains to be elucidated.

Truncated and Atypical Forms: Not all NBS-containing proteins follow the full NLR architecture. Some lack the LRR domain (e.g., TN, CN, XN), while others exhibit unusual domain orders or combinations [19]. In blueberries, approximately 9 out of 106 NBS-encoding genes lacked the LRR domain [19].

Species-Specific Architectural Patterns

Different plant lineages have evolved distinct architectural preferences. In orchids, which maintain exceptionally low numbers of NBS-LRR genes among angiosperms, CNLs overwhelmingly predominate while TNLs are entirely absent [20]. Blueberry NBS genes show distinctive exon patterns, with TNLs having significantly more exons (average 3.73) than nTNLs (average 1.75) [19]. These species-specific patterns reflect both evolutionary history and ecological adaptations.

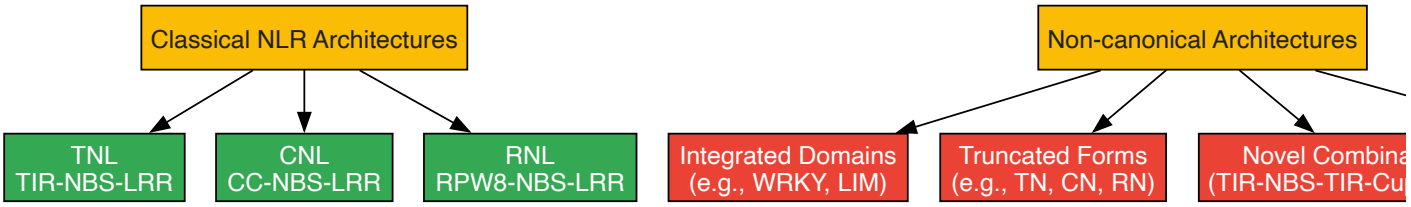


Diagram: Diversity of NBS domain architectures, showing classical and non-canonical forms

Experimental Approaches for NBS Gene Identification and Classification

Genome-Wide Identification Pipeline

Comprehensive identification of NBS-encoding genes requires an integrated bioinformatics approach combining multiple methods:

Domain-Based HMM Searches: Initial identification typically employs hidden Markov model (HMM) searches using profiles for the NBS domain (e.g., PF00931). The PfamScan.pl HMM search script with a stringent e-value cutoff (e.g., 1.1e-50) effectively identifies candidate genes [5]. This approach can be extended using custom HMM profiles for NBS subdomains (NBD, ARC1, ARC2) for more precise domain delineation [15].

Architecture Classification: Identified candidates are then classified based on domain architecture using tools like PfamScan or InterProScan [5] [19]. Classification systems typically place genes with similar domain architectures under the same classes, enabling systematic comparison across species [5].

Manual Curation and Validation: Automated annotations require manual curation to address inconsistencies, particularly at domain borders [15]. Additional validation using databases like CDD (Conserved Domain Database), SMART, and Pfam ensures accurate domain annotation [22].

Orthogroup Analysis and Evolutionary Studies

To understand evolutionary relationships, orthogroup analysis using tools like OrthoFinder provides insights into conservation and lineage-specific expansions [5]. This approach identifies core orthogroups (shared across multiple species) and unique orthogroups (specific to particular lineages) [5]. For example, analysis of NBS genes across 34 species identified 603 orthogroups, with some core (OG0, OG1, OG2) and unique (OG80, OG82) orthogroups showing tandem duplications [5].

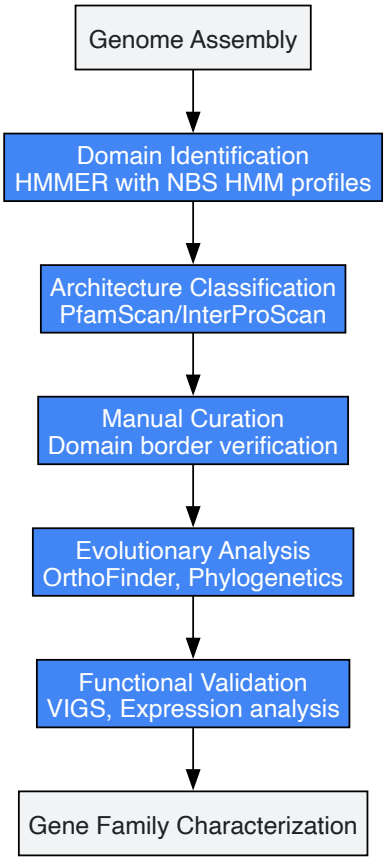


Diagram: Experimental workflow for NBS gene identification and characterization

Functional Validation and Mechanistic Studies

Expression Profiling and Genetic Variation

Transcriptomic analyses provide insights into NBS gene expression patterns across tissues and stress conditions. Studies examining expression in susceptible and tolerant plant accessions have identified putative upregulated orthogroups under biotic and abiotic stresses [5]. For example, analysis of Gossypium hirsutum accessions with varying susceptibility to cotton leaf curl disease identified significant genetic variation, with tolerant accessions showing more unique variants in NBS genes [5].

Genetic variation studies between susceptible (Coker 312) and tolerant (Mac7) cotton accessions revealed 6583 unique variants in Mac7 compared to 5173 in Coker312, highlighting the potential contribution of NBS gene diversity to disease resistance [5].

Functional Characterization Techniques

Virus-Induced Gene Silencing (VIGS): This approach enables functional assessment of candidate NBS genes. For instance, silencing of GaNBS (OG2) in resistant cotton demonstrated its putative role in virus tittering [5].

Protein Interaction Studies: Protein-ligand and protein-protein interaction assays reveal molecular mechanisms. Studies have shown strong interaction of putative NBS proteins with ADP/ATP and different core proteins of the cotton leaf curl disease virus [5].

Structural Biology Approaches: Recent cryo-EM structures of full-length NLRs (ZAR1 in resting and activated states, RPP1, and ROQ1) have provided unprecedented insights into activation mechanisms and signaling complex formation [15] [17].

Research Reagent Solutions and Databases

Table 3: Key Research Resources for NBS Gene Studies

Resource	Type	Function	Reference
NLRscape	Database	Collection of ~80,000 plant NLR sequences with advanced annotations, structural analysis tools	[15]
OrthoFinder	Software Tool	Orthogroup inference, gene family evolution analysis	[5]
Pfam/InterPro	Database	Domain annotation, architecture classification	[5] [15]
HMMER	Software Tool	Hidden Markov Model-based domain identification	[5] [22]
VIGS Vectors	Experimental Reagent	Functional validation through gene silencing	[5]
PRGdb	Database	Plant Resistance Gene database with curation	[15]
RefPlantNLR	Database	Reference set of plant NLR genes	[15]

The architectural diversity of NBS domain genes represents a remarkable example of evolutionary innovation in plant immune systems. From the classical TNL/CNL/RNL divisions to the myriad novel domain combinations observed across plant species, this gene family exhibits extraordinary structural and functional plasticity. The continuing development of comprehensive databases, refined annotation pipelines, and structural biology approaches promises to further unravel the complexity of this gene family. Understanding this diversity not only provides fundamental insights into plant-pathogen coevolution but also offers potential applications for engineering disease resistance in crop species through knowledge-driven manipulation of these sophisticated molecular recognition systems.

The nucleotide-binding site (NBS) gene family represents a critical component of the plant immune system, encoding proteins that facilitate effector-triggered immunity against diverse pathogens. The expansion and contraction of this gene family across plant lineages are primarily driven by two distinct mechanisms: whole-genome duplication (WGD) and tandem duplication (TD). This technical review synthesizes current research elucidating how these duplication mechanisms create divergent evolutionary patterns, selection pressures, and functional specializations within NBS gene families. Through comparative genomic analyses across multiple species families, we demonstrate that WGD-derived NBS genes typically undergo strong purifying selection, preserving core immune functions, while TD-derived genes experience relaxed or positive selection, enabling rapid adaptation to evolving pathogen pressures. The dynamic interplay between these mechanisms shapes the genomic architecture of plant immunity and informs strategies for breeding durable disease resistance in crops.

Plant immunity relies heavily on a sophisticated surveillance system mediated by nucleotide-binding site (NBS) domain genes, which constitute one of the largest and most variable gene families in plant genomes [5]. These genes typically encode proteins containing a central NBS domain and C-terminal leucine-rich repeats (LRRs), and are classified into subfamilies based on N-terminal domains: TIR-NBS-LRR (TNL), CC-NBS-LRR (CNL), and RPW8-NBS-LRR (RNL) [23]. NBS-LRR genes play indispensable roles in pathogen recognition and defense activation, with their genomic abundance and diversity directly influencing a plant's capacity to withstand evolving pathogenic threats [24].

The remarkable variation in NBS gene copy numbers across plant species—ranging from merely five in *Gastrodia elata* to over 2,000 in wheat—underscores the dynamic evolutionary processes governing this gene family [23]. Two primary mechanisms drive this expansion: whole-genome duplication (WGD) events that create duplicate copies of all genomic material, and small-scale duplication events, particularly tandem duplications (TD), that generate localized gene clusters [5]. Understanding how these distinct mechanisms contribute to NBS gene evolution is fundamental to deciphering plant-pathogen co-evolution and developing sustainable crop protection strategies.

This review examines the specific contributions of WGD and TD to NBS gene expansion, synthesizing findings from recent pan-genomic studies across diverse plant families. We analyze how these duplication mechanisms produce genes with different evolutionary trajectories, selection pressures, and functional capabilities, ultimately shaping the plant immune repertoire.

Comparative Analysis of Duplication Mechanisms

Evolutionary Patterns Across Plant Families

Table 1: Evolutionary patterns of NBS genes across plant families driven by WGD and TD

Plant Family	Species Example	Evolutionary Pattern	Primary Driver	Gene Count
Rosaceae	*Rosa chinensis*	Continuous expansion	WGD/TD combination	Variable across species [23]
Rosaceae	*Fragaria vesca*	Expansion-contraction-further expansion	Fluctuating duplication	Variable across species [23]

Plant Family	Species Example	Evolutionary Pattern	Primary Driver	Gene Count
Poaceae	Maize (*Zea mays*)	Core-adaptive model	TD for adaptive subgroups	~129 [23]
Solanaceae	Pepper (*Capsicum annuum*)	Shrinking pattern	Limited duplication	252 [24]
Nicotiana	*Nicotiana tabacum*	Allotetraploid expansion	WGD from hybridization	603 [10]
Orchidaceae	*Gastrodia elata*	Extreme contraction	Extensive gene loss	5 [23]

Comparative genomic analyses reveal striking differences in how NBS gene families evolve across plant lineages. In the Rosaceae family, encompassing important fruit crops like apple and strawberry, different species exhibit distinct evolutionary patterns despite shared ancestry. *Rosa chinensis* demonstrates "continuous expansion" with ongoing gene duplication, while other relatives show "expansion and then contraction" or more complex fluctuating patterns [23]. These divergent trajectories within the same family highlight the complex interplay between duplication mechanisms and lineage-specific evolutionary pressures.

The Solanaceae family presents another compelling case study. Pepper (*Capsicum annuum*) displays a "shrinking pattern" with only 252 NBS genes identified, approximately 54% of which form 47 gene clusters primarily through tandem duplications [24]. This contrasts with the "consistent expansion" observed in potato and "expansion followed by contraction" in tomato, illustrating how even closely related species can undergo dramatically different evolutionary paths for their NBS gene repertoires [23].

Functional and Evolutionary Consequences of Duplication Mechanisms

Table 2: Characteristics of NBS genes derived from different duplication mechanisms

Characteristic	WGD-Derived NBS Genes	Tandem-Duplicated NBS Genes
Selection pressure	Strong purifying selection (low Ka/Ks) [13]	Relaxed or positive selection (high Ka/Ks) [13]
Evolutionary rate	Slow evolution, conserved functions	Rapid evolution, neofunctionalization
Genomic distribution	Dispersed throughout genome	Clustered in duplication-prone regions
Functional role	Core immunity, basal defense [13]	Pathogen-specific recognition, rapid adaptation
Sequence conservation	High conservation across lineages	High variability, lineage-specific
Gene expression	Often constitutive expression	Frequently stress-responsive

Whole-genome duplication and tandem duplication produce NBS genes with fundamentally different evolutionary constraints and functional capabilities. WGD-derived genes typically experience strong purifying selection, maintaining essential core immune functions across evolutionary timescales [13]. For example, in maize, conserved "core" subgroups (e.g., ZmNBS31, ZmNBS17-19) demonstrate consistent expression under both stressed and control conditions, suggesting their fundamental role in basal immunity [13]. These genes are often dispersed throughout the genome and retain stable functions.

In contrast, tandem-duplicated NBS genes experience markedly different evolutionary pressures. Maize studies reveal that TD-derived genes show signs of relaxed or positive selection, with higher non-synonymous to synonymous substitution rates (Ka/Ks) indicating rapid sequence evolution [13]. This evolutionary flexibility enables these genes to explore novel functions and adapt to emerging pathogen challenges. The localization of these genes in duplication-prone genomic regions facilitates their rapid expansion and diversification through recurrent duplication events [25].

Genomic Architecture and Distribution Patterns

Chromosomal Distribution and Gene Clustering

The non-random distribution of NBS genes across plant chromosomes reveals important insights into their evolutionary dynamics. In pepper (*Capsicum annuum*), NBS genes are distributed across all chromosomes, with chromosome 3 harboring the highest number (38 genes) while chromosomes 2 and 6 contain the lowest (5 genes each) [24]. Notably, 54% of pepper NBS genes form 47 physical clusters, with the largest cluster (8 genes) located on chromosome 3 [24]. This clustered arrangement predominantly results from tandem duplication events and exemplifies how local duplication creates genomic hotspots for NBS gene evolution.

Similar clustering patterns occur across plant families. In barley (*Hordeum vulgare*), duplication-prone regions enriched with NBS genes are preferentially located in subtelomeric regions across all seven chromosomes [25]. These Long-Duplication-Prone Regions (LDPRs) range from 5.5 to 1,123 kilobases and exhibit elevated levels of locally duplicated sequences, creating environments conducive to the birth-death evolution characteristic of NBS genes involved in arms races with pathogens.

The Core-Adaptive Model of NBS Gene Evolution

Recent pan-genomic analyses support a "core-adaptive" model of NBS gene evolution [13]. This framework distinguishes between:

- **Core NBS subgroups**: Conserved genes (e.g., ZmNBS31, ZmNBS17-19 in maize) showing limited presence-absence variation across lineages, maintained by purifying selection for essential immune functions
- **Adaptive NBS subgroups**: Highly variable genes (e.g., ZmNBS1-10, ZmNBS43-60 in maize) exhibiting extensive presence-absence variation, evolving under positive selection for pathogen-specific adaptations

This model reconciles the evolutionary tension between maintaining stable core immune functions while enabling rapid adaptation to evolving pathogen pressures. The core components provide essential basal immunity, while the adaptive components offer species-specific or lineage-specific resistance capabilities.

Molecular Mechanisms and Experimental Approaches

Experimental Workflow for NBS Gene Identification and Analysis

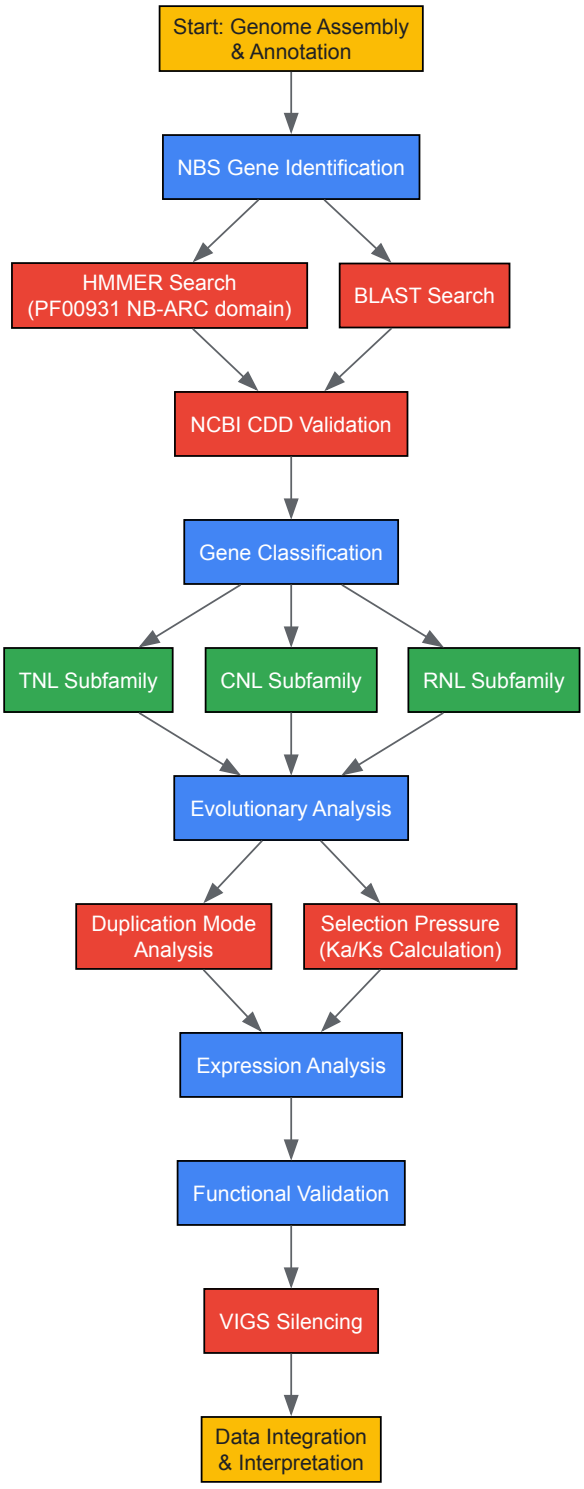


Figure 1: Experimental workflow for comprehensive NBS gene family analysis

Research Reagent Solutions for NBS Gene Studies

Table 3: Essential research reagents and computational tools for NBS gene analysis

Category	Tool/Reagent	Specific Application	Function
Bioinformatics Tools	HMMER (PF00931)	NBS domain identification	Hidden Markov Model search for NB-ARC domains [23]
	OrthoFinder	Evolutionary analysis	Orthogroup inference and phylogenetic analysis [5]
	MCScanX	Duplication mode analysis	Identification of tandem and segmental duplications [10]

Category	Tool/Reagent	Specific Application	Function
	KaKs_Calculator	Selection pressure analysis	Calculation of Ka/Ks ratios [10]
Experimental Methods	Virus-Induced Gene Silencing (VIGS)	Functional validation	Knockdown of candidate NBS genes to test function [5]
	RNA-seq	Expression profiling	Differential expression under stress conditions [10]
	Pfam/NCBI CDD	Domain validation	Confirmation of NBS and associated domains [23]
Databases	Plaza Genome Database	Comparative genomics	Multi-species genome comparisons [5]
	Plant RGAs	NBS gene database	Curated repository of resistance gene analogs [26]

The integration of bioinformatic tools and experimental approaches enables comprehensive characterization of NBS gene families. The workflow begins with genome-wide identification using both HMMER searches with the NB-ARC domain (PF00931) and BLAST searches, followed by validation through Pfam and NCBI Conserved Domain Database (CDD) analyses [23] [10]. Subsequent classification into TNL, CNL, and RNL subfamilies based on N-terminal domains provides the foundation for evolutionary analyses.

Evolutionary studies employ OrthoFinder for orthogroup inference, MCScanX for duplication mode analysis, and KaKs_Calculator for selection pressure quantification [5] [10]. Functional validation increasingly utilizes Virus-Induced Gene Silencing (VIGS), as demonstrated in cotton where silencing of GaNBS (OG2) validated its role in virus resistance [5]. RNA-seq expression profiling under various stress conditions further elucidates the functional roles of candidate NBS genes.

Ecological and Evolutionary Implications

Environmental Drivers of NBS Gene Expansion

The evolutionary dynamics of NBS genes are profoundly influenced by environmental factors, particularly pathogen pressures. Research across 205 Archaeplastida genomes reveals that tandem duplications are significantly enriched in root plants with extensive soil microbial exposure [27]. This genomic convergence demonstrates adaptive evolution to soil-borne pathogens, with TD frequency correlating strongly with microbial interaction intensity.

Conversely, plants transitioning to reduced-microbial lifestyles (aquatic, parasitic, halophytic, or carnivorous) consistently exhibit decreased TD frequency [27]. This pattern highlights the role of pathogen pressure in driving NBS gene expansion through tandem duplication. Mangroves independently adapting to hypersaline intertidal soils with diminished microbial activity similarly show reduced TD frequency, further supporting the relationship between microbial exposure and NBS gene diversification [27].

The Cooperative Model of Gene-Duplication Element Associations

Emerging evidence suggests that arms-race genes, including NBS-LRRs, have effectively formed cooperative associations with duplication-inducing sequences [25]. This model proposes that lineages benefiting from physical associations between NBS genes and duplication-prone genomic regions gain selective advantages through enhanced diversification capacity.

In barley, NBS genes are statistically over-represented in Long-Duplication-Prone Regions (LDPRs) containing kilobase-scale tandem repeats [25]. These duplication-prone regions show historical long-distance dispersal to distant genomic sites followed by local expansion through tandem duplication. This cooperative association between NBS genes and duplication-inducing elements creates an evolutionary feedback loop that enhances the generation of diversity for pathogen recognition.

The dual evolutionary strategies of whole-genome duplication and tandem duplication have shaped the NBS gene landscape across plant lineages. WGD provides stable, conserved core genes maintained by purifying selection, while TD generates rapidly evolving adaptive genes under positive selection. This complementary system enables plants to maintain essential immune functions while retaining the flexibility to respond to emerging pathogen threats.

Future research directions should leverage pan-genomic approaches to capture the full diversity of NBS genes across broader taxonomic ranges and ecological contexts. Integrating structural variant analysis with functional studies will further elucidate how specific genetic changes influence protein function and pathogen recognition. The emerging understanding of duplication mechanisms and their evolutionary consequences provides a robust foundation for developing crop varieties with enhanced and durable disease resistance through molecular breeding and genome editing approaches.

Understanding the evolutionary drivers of NBS gene expansion not only illuminates fundamental plant biology but also offers practical strategies for crop improvement. By harnessing the natural duplication mechanisms that have shaped plant immunity throughout evolutionary history, we can develop innovative approaches to enhance agricultural sustainability and food security in the face of evolving pathogen threats.

Nucleotide-binding domain and Leucine-Rich Repeat receptors (NLRs) constitute a critical component of the plant innate immune system, serving as intracellular sentinels that initiate effector-triggered immunity (ETI) upon pathogen recognition [28] [29]. The evolution of these immune receptors spans the entire trajectory of plant terrestrial colonization, from early bryophytes to modern angiosperms. Recent advances in comparative genomics have revealed astonishing variation in NLR repertoire size and architecture across plant lineages, reflecting divergent evolutionary paths shaped by pathogen pressure, life history strategies, and ecological adaptations.

This review synthesizes current understanding of NLR gene family evolution across land plants, with particular emphasis on the quantitative differences between bryophytes and angiosperms. We examine the genomic mechanisms driving NLR expansion and contraction, explore methodological frameworks for NLR identification, and discuss the functional implications of NLR diversity for plant immunity. Within the broader context of nucleotide-binding site gene diversification, this analysis provides a comprehensive perspective on how plant immune systems have evolved distinct strategies across the phylogenetic spectrum.

NLR Gene Evolution Across Land Plants

The Evolutionary Origins of Plant NLR Genes

NLR genes originated early in plant evolution, with homologs identified in green algae and bryophytes [29]. These initial immune receptors were relatively limited in number, containing only a dozen NLRs in green algae before expanding significantly in land plants [28]. This expansion coincided with the colonization of terrestrial habitats approximately 500 million years ago, suggesting a critical role for NLR-mediated immunity in adapting to new pathogen pressures in aerial environments.

Bryophytes, as the earliest diverging lineage of land plants, occupy a pivotal position in understanding NLR evolution. Recent comprehensive analyses of 123 bryophyte genomes reveal that despite their morphological simplicity, bryophytes possess a substantially greater diversity of gene families than vascular plants, including unique immune receptors [30]. This finding challenges previous assumptions about the correlation between structural complexity and genetic sophistication in plant immune systems.

NLR Repertoire Size Variation Across Plant Lineages

Table 1: NLR Repertoire Size Variation Across Major Plant Lineages

Plant Group	Representative Species	NLR Count	Subclass Composition	Key Evolutionary Features
Bryophytes	*Physcomitrium patens*	Not quantified in studies	Potentially novel subtypes (HNL, PNL)	High gene family diversity; unique immune receptors
Magnoliids	*Litsea cubeba*	Varies by species (total 1,832 across 7 species)	TNLs completely absent from 5/7 species	"Expansion-contraction" evolutionary pattern
Monocots	*Oryza sativa* (rice)	498	497 CNLs, 1 RNL, 0 TNLs	Independent TNL loss
Eudicots	*Arabidopsis thaliana*	165	52 CNLs, 106 TNLs, 7 RNLs	Balanced CNL/TNL representation
Aquatic Angiosperms	Various aquatic species	Significantly contracted	Variable	Ecological adaptation to reduced pathogen pressure

The variation in NLR repertoire size across land plants is dramatic, ranging from several dozen in species with reduced genomes to over two thousand in certain cultivated crops [31] [32]. This variation reflects both deep evolutionary history and recent lineage-specific adaptations. Angiosperms particularly demonstrate remarkable NLR diversity, with copy numbers differing up to 66-fold among closely related species due to rapid gene birth and death processes [33].

Several evolutionary patterns have emerged across plant lineages. Brassicaceae species typically exhibit "first expansion and then contraction" patterns, while Fabaceae and Rosaceae show consistent expansion trajectories [29]. Poaceae species generally demonstrate contraction patterns, with notable exceptions like wheat (*Triticum aestivum*), which possesses over two thousand NLR genes [31]. These divergent evolutionary paths reflect both phylogenetic constraints and ecological adaptations.

Genomic and Ecological Drivers of NLR Diversity

Mechanisms of NLR Genome Evolution

NLR gene family dynamics are primarily driven by several genomic mechanisms:

- **Tandem duplications**: This represents the major mechanism for NLR expansion across all plant lineages [29]. Tandemly arranged NLR clusters create hotspots for genetic innovation through unequal crossing over and gene conversion, facilitating the rapid generation of novel

recognition specificities.

- **Whole-genome duplications (WGDs)**: Paleopolyploidization events provide raw genetic material for NLR diversification. Following WGDs, NLR genes often undergo differential retention and functional divergence, contributing to lineage-specific immune repertoires [30].
- **Domain shuffling and fusion**: Integration of novel protein domains into NLR architectures creates composite immune receptors (NLR-IDs) that can recognize pathogen effectors through "integrated decoy" domains [34]. These integrated domains often mimic authentic pathogen targets, effectively baiting effector proteins and triggering immunity.
- **Horizontal gene transfer (HGT)**: In some lineages, particularly bryophytes, continuous horizontal transfer of microbial genes has contributed to genetic innovation in immune receptors [30]. This mechanism provides an alternative pathway for acquiring novel recognition capabilities beyond duplication and divergence of existing plant genes.
- **De novo gene birth**: Orphan genes, particularly prevalent in bryophytes, arise from previously non-coding sequences and provide another source of NLR diversity [30]. In *Marchantia polymorpha*, approximately 70-80% of genes in orphan gene families align with noncoding regions in closely related species, suggesting recent de novo origination.

Ecological and Life History Influences

Table 2: Ecological Factors Influencing NLR Repertoire Size

Ecological Context	Impact on NLR Repertoire	Representative Examples
Domestication	Significant contraction	*Asparagus officinalis* (27 NLRs) vs. wild relatives (47-63 NLRs)
Aquatic Habitat	Convergent reduction	Multiple independent aquatic angiosperms
Life Strategy	Differential expansion	Annual *Glycine* species (expanded) vs. perennials (contracted)
Pathogen Pressure	Lineage-specific expansion	Wheat (>2000 NLRs) vs. *Oropetium thomaeum* (several dozen)

NLR repertoire size demonstrates clear associations with ecological factors and life history strategies. Aquatic plants consistently exhibit convergent NLR reduction, reminiscent of the limited NLR expansion observed in green algae prior to land colonization [33]. This pattern suggests that aquatic environments impose distinct selective pressures on plant immune systems, possibly due to reduced pathogen diversity or different infection strategies in aquatic ecosystems.

Life history strategy significantly influences NLR evolution, as demonstrated in the genus *Glycine*, where annual species (*G. max* and *G. soja*) exhibit expanded NLRomes compared to perennial relatives [35]. Evolutionary timescale analysis indicates that this expansion occurred recently (0.1-0.5 million years ago), driven by lineage-specific and terminal duplications. In contrast, perennial lineages experienced significant contraction following the Glycine-specific whole-genome duplication event approximately 10 million years ago, despite maintaining a highly diversified NLR repertoire with limited interspecies synteny.

Domestication has consistently impacted NLR repertoire size, often resulting in significant contraction of immune gene diversity. In asparagus (*Asparagus officinalis*), domestication resulted in a reduction from 63 NLR genes in wild relatives (*A. setaceus*) to just 27 in the cultivated species [31] [32]. This contraction, coupled with reduced expression of retained NLR genes, likely contributes to increased disease susceptibility in domesticated lines.

Methodological Framework for NLR Identification and Analysis

Genomic Identification of NLR Genes

Standardized methodologies for NLR identification across plant genomes have been established, combining multiple complementary approaches:

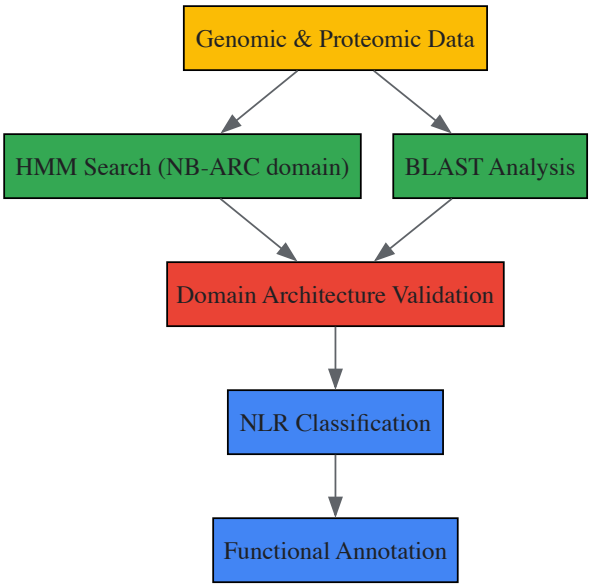


Figure 1: Workflow for genome-wide NLR identification and classification. The pipeline begins with genomic and proteomic data, employs complementary search strategies, validates domain architecture, and culminates in classification and functional annotation.

Hidden Markov Model (HMM) searches using the conserved NB-ARC domain (Pfam: PF00931) serve as the primary identification method [31] [32] [36]. This approach leverages the conserved nucleotide-binding domain that defines the NLR family, with cutoff E-values typically set at 1e-5 to 1e-10 depending on the study.

BLAST-based approaches provide complementary identification using reference NLR protein sequences from well-characterized species like *Arabidopsis thaliana*, *Oryza sativa*, and *Allium sativum* [31]. This method helps recover divergent NLR homologs that might be missed by HMM searches alone.

Domain architecture validation through tools like InterProScan and NCBI's Batch CD-Search confirms the presence of characteristic NLR domains and excludes non-NLR proteins containing NB-ARC-related domains [31] [32].

Advanced annotation pipelines like NLRtracker [35] and NLR-Annotator [36] have been developed specifically for comprehensive NLR identification, incorporating multiple verification steps and classification modules.

Classification and Phylogenetic Analysis

Following identification, NLR genes are classified based on their N-terminal domains into major subclasses:

- **TNLs**: Contain Toll/Interleukin-1 receptor-like domains
- **CNLs**: Feature coiled-coil domains
- **RNLs**: Possess RPW8 domains and typically function as helper NLRs [29]
- **NLs**: Represent truncated variants lacking standard N-terminal domains

Phylogenetic reconstruction using maximum likelihood methods (e.g., IQ-TREE, MEGA) elucidates evolutionary relationships among NLR genes across species [31] [36]. These analyses reveal both deep conservation and recent lineage-specific expansions of NLR clades.

Pan-NLRome and Comparative Genomics

The concept of "pan-NLRomes" has emerged as a powerful framework for capturing intraspecific NLR diversity [28] [37]. By analyzing NLR repertoires across multiple individuals within a species, researchers can distinguish core NLR genes (shared across all individuals) from variable NLR genes that may contribute to differences in disease resistance.

Pangenome graphs enable nuanced analysis of NLR evolution in a genomic context, revealing distinct evolutionary processes acting on NLR neighborhoods defending against different pathogen classes [37]. These approaches have demonstrated that NLR diversity arises from multiple uncorrelated mutational and genomic processes, suggesting that mechanistic studies must consider multiple axes of immune system diversity.

Research Reagents and Experimental Tools

Table 3: Essential Research Reagents and Computational Tools for NLR Genomics

Tool/Reagent	Primary Function	Application Context
HMMER Suite	Hidden Markov Model searches	Identification of NB-ARC domains in proteomes
InterProScan	Protein domain annotation	Validation of NLR domain architecture
OrthoFinder	Orthogroup inference	Comparative analysis of NLR genes across species
MEME Suite	Motif discovery	Identification of conserved NLR sequence motifs
PlantCARE	cis-element prediction	Analysis of NLR promoter regions
NLRtracker	Automated NLR annotation	Genome-wide NLR identification and classification
MCScanX	Synteny analysis	Identification of NLR gene clusters and rearrangements

Functional and Evolutionary Implications

The dramatic variation in NLR repertoire size across plant lineages reflects fundamentally different evolutionary strategies for pathogen resistance. Bryophytes, despite their basal phylogenetic position, maintain exceptionally diverse gene families and unique immune receptors that may contribute to their success in diverse habitats, including extreme environments [30]. This suggests that immune system complexity in land plants does not follow a simple linear progression from early-diverging to later-diverging lineages.

In angiosperms, two distinct evolutionary stages have been proposed: an initial stage of maintained low NLR numbers from angiosperm origins until the Cretaceous-Paleogene boundary, followed by a dramatic expansion phase leading to contemporary NLR diversity [29]. This pattern suggests that angiosperm NLR evolution was influenced by both ancient constraints and more recent selective pressures, potentially linked to co-evolution with rapidly adapting pathogen populations.

The functional consequences of NLR repertoire size are context-dependent. While expanded NLR families potentially enable recognition of a broader spectrum of pathogens, they also impose metabolic costs and risks of autoimmunity [31]. This balance likely underlies the observation that NLR contraction is sometimes associated with ecological transitions, such as the evolution of aquatic, parasitic, and carnivorous lifestyles in angiosperms [33].

The comparative analysis of NLR repertoire size from bryophytes to angiosperms reveals the dynamic evolution of plant immune systems across deep evolutionary time. Rather than a simple narrative of progressive complexity, the pattern emerging from genomic studies is one of divergent evolutionary strategies shaped by phylogenetic history, ecological context, and genomic constraints.

Bryophytes display unexpected genetic sophistication with diverse gene families and unique immune receptors, while angiosperms demonstrate remarkable plasticity in NLR repertoire size through repeated expansion and contraction events. The methodological advances in NLR identification and classification, particularly through pan-genome approaches, continue to refine our understanding of plant immunity at the molecular level.

Future research directions should include more comprehensive sampling of early-diverging plant lineages, functional characterization of NLR-IDs across diverse species, and integration of NLR evolution with broader patterns of nucleotide-binding site gene diversification. Such efforts will continue to elucidate the evolutionary forces that have shaped the complex immune systems of land plants over 500 million years of terrestrial colonization.

From Sequence to Function: Computational and Experimental Tools for NBS Gene Analysis

This technical guide provides a comprehensive framework for employing **HMMER** and the **Pfam** database in genome-wide identification of protein families, with specific application to nucleotide-binding site (NBS) domain genes in plants. We detail a complete bioinformatics workflow from domain discovery to evolutionary analysis, incorporating practical considerations for protein family classification, diversification patterns, and methodological validation. The protocols outlined leverage recent advances in plant genomics to enable large-scale comparative studies of NBS gene evolution across species, facilitating the identification of novel resistance genes and supporting crop improvement efforts.

Gene families encoding nucleotide-binding site (NBS) domains represent one of the most extensive and functionally important gene classes in plant genomes, playing crucial roles in pathogen recognition and disease resistance [5]. The **NBS domain** serves as a molecular switch in plant immune receptors, controlling activation of defense responses upon pathogen detection [38]. Comprehensive identification of these genes across plant species requires robust bioinformatics approaches that can detect distant evolutionary relationships despite considerable sequence diversification.

The **HMMER software suite** coupled with the **Pfam database** provides a powerful combination for domain-centric gene family annotation. This approach leverages probabilistic models built from multiple sequence alignments of protein domains, offering superior sensitivity for detecting remote homologs compared to sequence similarity-based methods like BLAST [39]. The central premise involves using carefully curated hidden Markov models (HMMs) of protein domains to systematically scan proteomes, enabling identification of even highly divergent family members.

For NBS domain genes, this methodology has revealed remarkable diversification across plant species. Recent studies have identified **12,820 NBS-domain-containing genes** across 34 plant species ranging from mosses to monocots and dicots, classifying them into **168 distinct domain architecture patterns** [5]. This expansion reflects an evolutionary arms race between plants and their pathogens, with different plant lineages employing distinct diversification strategies.

Theoretical Foundations: Protein Domains and HMMER

Protein Domains as Functional and Evolutionary Units

Protein domains are conserved structural and functional units that evolve as discrete modules, often rearranged in different combinations across proteomes. The **Pfam database** organizes protein space into families based on these domains, with each family represented by a multiple sequence alignment and a hidden Markov model [40]. The NBS domain (PF00931 in Pfam) represents one such evolutionary unit that has been extensively duplicated and diversified in plant genomes.

Recent structural analyses of Pfam domains using AlphaFold2-predicted structures have revealed substantial **structural variability within domain families**, with 20-40% of domain instances lacking regular secondary structures [40]. This structural plasticity complicates functional predictions based solely on sequence and highlights the importance of integrating structural information where possible.

Hidden Markov Models for Sequence Analysis

Hidden Markov Models (HMMs) provide a statistical framework for modeling conserved patterns in biological sequences. For protein domain identification, HMMs capture position-specific amino acid frequencies, insertion probabilities, and deletion probabilities derived from curated multiple sequence alignments. The **HMMER software** implements efficient algorithms (including the Forward and Viterbi algorithms) for calculating the probability that a query sequence matches a given domain model, expressed as an E-value representing the expected number of false positives.

The mathematical foundation of HMMER enables it to detect remote homologies that may be missed by pairwise methods, making it particularly valuable for studying ancient gene families like NBS genes that have undergone significant divergence across plant lineages.

Essential Software Tools

Table 1: Essential Software Tools for Genome-Wide Domain Identification

Tool Name	Version	Primary Function	Key Parameters
HMMER	3.3.2	Domain searching using HMMs	E-value threshold, --cut_ga
PfamScan	-	Integration of Pfam HMMs	Default parameters
InterProScan	5.0+	Integrated domain annotation	-f XML, JSON, GFF3
Python/BioPython	3.6+	Scripting and data processing	-
R	4.0+	Statistical analysis and visualization	-
MEME	5.0.5+	Motif discovery	-mod zoops -nmotifs 10

Table 2: Essential Databases for Domain-Centric Annotation

Database	URL	Primary Content	Application
Pfam	http://pfam.xfam.org/	Protein domain HMMs	Domain identification
Ensemble Plants	https://plants.ensembl.org	Plant genomes and annotations	Genomic context
Phytozome	https://phytozome.jgi.doe.gov	Plant genomes	Comparative genomics
CDD	https://ncbi.nlm.nih.gov/cdd	Conserved domains	Domain verification
SMART	http://smart.embl-heidelberg.de	Domain architectures	Structural validation

Research Reagent Solutions

Table 3: Essential Research Reagents for Experimental Validation

Reagent Type	Specific Examples	Function in Research
RNA extraction kit	Aidlab RNA kit (used in [41])	High-quality RNA isolation from plant tissues
cDNA synthesis kit	PrimeScript RT reagent (used in [41])	First-strand cDNA synthesis for expression studies
Cloning vector	pMD18-T vector (used in [41])	TA cloning of PCR products for sequence verification
High-fidelity polymerase	PrimeSTAR Max Premix (used in [41])	Accurate amplification of gene coding sequences
Sequencing service	Illumina MiSeq platform (used in [42])	Whole genome sequencing and verification

Core Methodology: HMMER and Pfam Workflow

Step 1: HMM Acquisition and Preparation

The first critical step involves obtaining the appropriate HMM profile for the domain of interest. For NBS domain identification, researchers would retrieve the **NF00931** HMM from the Pfam database:

Alternatively, researchers can build custom HMMs when studying domains with insufficient representation in Pfam. For example, studies of BBM-like (BABY BOOM) genes in the AP2/ERF family used the **AP2 (PF00847)** HMM to identify candidates across 10 plant species [43].

Step 2: Proteome Preparation and Quality Control

Proteome datasets should be acquired from reliable sources such as Ensemble Plants, Phytozome, or species-specific databases. For example, studies of NBS genes in legumes utilized proteomes from *Medicago truncatula*, *Cajanus cajan*, *Phaseolus vulgaris*, and *Glycine max* [44]. Quality control measures include:

- Removing redundant sequences
- Verifying sequence completeness
- Checking for proper amino acid coding
- Standardizing sequence headers for downstream processing

Step 3: Domain Scanning with HMMER

The core identification step uses **hmmscan** to search proteomes against the domain HMM:

Key parameters include:

- **E-value threshold**: Typically 1e-5 to 1e-10, with stricter values reducing false positives
- **Gathering threshold (GA)**: Pfam-curated thresholds that optimize family inclusion
- **Domain score reporting**: Using --domtblout for per-domain hits rather than per-sequence

For example, a comprehensive analysis of plant NBS domains applied an **E-value threshold of 1.1e-50** to ensure high-confidence identifications [5].

Step 4: Result Processing and Validation

Raw HMMER output requires processing to extract meaningful gene lists:

Validation should include **domain verification** using multiple resources:

Studies of DUF789 genes in cotton employed a **multi-database verification approach**, cross-referencing HMMER results with SMART and CDD to confirm domain presence and reduce false positives [39].

Workflow Visualization

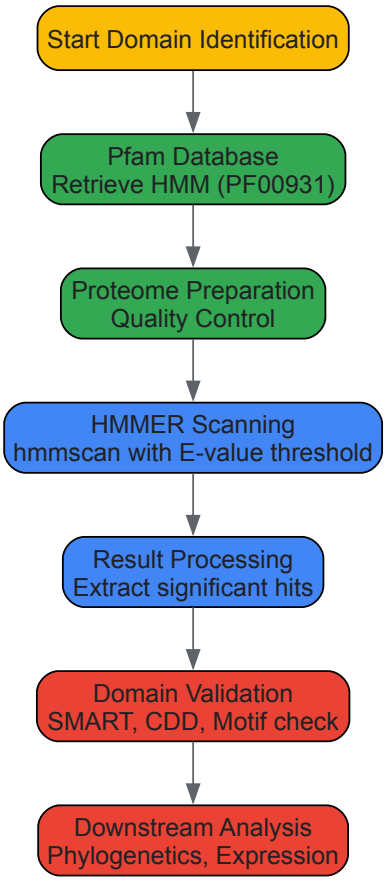


Figure 1: HMMER/Pfam Domain Identification Workflow

Case Study: NBS Domain Genes in Plants

Experimental Protocol for NBS Gene Identification

A recent large-scale analysis of NBS domain genes across 34 plant species provides a comprehensive protocol [5]:

- **Data Collection**: Acquire proteomes from public databases (NCBI, Phytozome, Plaza)
- **Domain Identification**:
  - Use PfamScan.pl HMM search script with default E-value (1.1e-50)
  - Apply Pfam-A.hmm model as background
- **Classification**:
  - Retain genes containing NB-ARC domain (PF00931)
  - Classify by domain architecture using established systems
- **Evolutionary Analysis**:
  - Perform orthogroup clustering with OrthoFinder v2.5.1
  - Conduct multiple sequence alignment with MAFFT 7.0
  - Construct phylogenetic trees using FastTreeMP with 1000 bootstraps

This study identified **12,820 NBS-domain-containing genes** with diverse domain architectures, including classical (NBS, NBS-LRR, TIR-NBS-LRR) and species-specific patterns [5].

Evolutionary Analysis of NBS Genes

The evolutionary history of NBS genes reveals dynamic expansion and contraction across plant lineages. In legumes, analysis of four species (*M. truncatula*, *C. cajan*, *P. vulgaris*, and *G. max*) identified **1,662 NBS-encoding genes**, with distinct ratios between nTNL and TNL subclasses [44]. During 54 million years of legume evolution, **94% of ancestral NBS lineages** experienced deletions or significant expansions, while only 6% were maintained conservatively [44].

Gene duplication patterns show that **local tandem duplications** dominate NBS gene gains (≥75%), with ectopic duplications creating novel NBS loci at frequencies of 8-20% across legume lineages [44]. This diversification pattern reflects continuous adaptation to evolving pathogen pressures.

Expression and Functional Analysis

Integration of transcriptomic data enables functional insights into identified NBS genes:

- **Expression Profiling**: Retrieve RNA-seq data from specialized databases (IPF, CottonFGD)
- **Differential Expression**: Categorize expression into tissue-specific, abiotic stress, and biotic stress responses
- **Validation**: Employ virus-induced gene silencing (VIGS) for functional validation

For example, expression profiling of NBS genes in cotton revealed **differential regulation** in response to cotton leaf curl disease, with specific orthogroups (OG2, OG6, OG15) showing upregulated expression in tolerant genotypes [5]. Functional validation through silencing of GaNBS (OG2) demonstrated its role in viral titer regulation [5].

Advanced Applications and Integration

Structural Bioinformatics Integration

Recent advances enable integration of structural predictions with domain annotation:

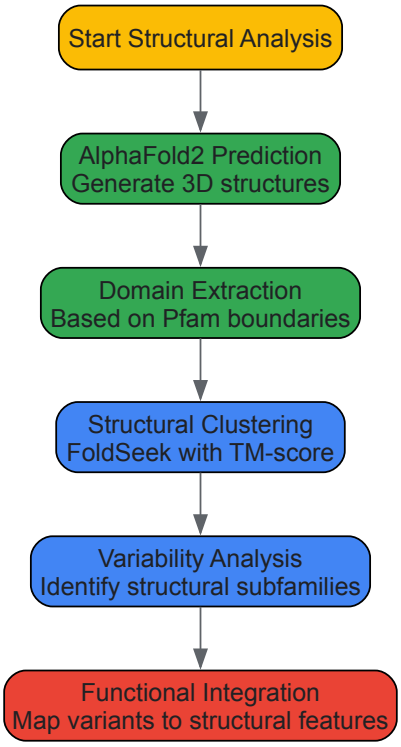


Figure 2: Structural Variability Analysis Workflow

The extraction of Pfam domain structures from AlphaFold2 predictions, as demonstrated in a recent analysis of 16 model organisms, enables **structural variability assessment** within domain families [40]. This approach revealed that 20-40% of Pfam domain instances lack regular secondary structures, indicating substantial structural plasticity [40].

Comparative Genomics and Orthology Analysis

Orthology analysis provides evolutionary context for identified genes:

- **Orthogroup Delineation**: Use OrthoFinder with DIAMOND for sequence similarity searches
- **Duplication Pattern Analysis**: Identify tandem, segmental, and whole-genome duplications
- **Selective Pressure Assessment**: Calculate Ka/Ks ratios to identify purifying vs. diversifying selection

In Rosa ALOG gene family analysis, researchers integrated **phylogenetic reconstruction** with **gene structure analysis** and **motif characterization** to elucidate evolutionary relationships [41]. Similar approaches for cotton DUF789 genes identified **purifying selection** as the major evolutionary force, with segmental and tandem duplications driving family expansion [39].

Troubleshooting and Methodological Considerations

Common Challenges and Solutions

Table 4: Troubleshooting Guide for Domain Identification

Challenge	Potential Cause	Solution
Low specificity	Overly permissive E-value	Stricter threshold (1e-10 to 1e-50)
Incomplete hits	Fragmented gene models	Use multiple proteome versions
Domain fragments	Improper model boundaries	Apply domain completeness filters
False negatives	Divergent sequences	Build custom HMMs with close homologs
Ambiguous classification	Atypical domain architectures	Manual curation and validation

Quality Control Metrics

- **Domain completeness**: Ensure identified domains cover >80% of HMM model
- **E-value distribution**: Check for bimodal distribution separating true/false hits
- **Taxonomic consistency**: Verify identified genes follow expected phylogenetic patterns
- **Motif conservation**: Confirm presence of functional motifs (e.g., P-loop in NBS domains)

The integration of HMMER and Pfam provides a robust foundation for genome-wide domain identification, enabling systematic characterization of gene families across plant species. When applied to NBS domain genes, this approach reveals remarkable diversification patterns driven by plant-pathogen coevolution. Future directions include:

- **Integration of structural predictions** from AlphaFold2 to assess functional variability
- **Machine learning approaches** for classifying domain architectures and predicting functions
- **Pan-genome analyses** to capture species-level diversity beyond reference genomes
- **Single-cell expression atlas** integration to resolve cell-type-specific functions

As genomic resources continue expanding, the HMMER/Pfam pipeline will remain essential for decoding the functional and evolutionary landscape of plant gene families, ultimately supporting crop improvement through identification of valuable genetic elements for disease resistance.

Orthogroup clustering is a foundational step in comparative genomics, enabling the systematic identification of gene families across multiple species. For research focusing on the diversification of nucleotide-binding site (NBS) domain genes across plant species, OrthoFinder provides a phylogenetically-aware framework to infer orthogroups, orthologs, and gene duplication events. This technical guide details the application of OrthoFinder for discerning core, conserved NBS gene families from species-specific lineages, supported by benchmarked protocols, data presentation standards, and tailored visualization tools to drive insights into the evolution of plant disease resistance mechanisms.

The accurate inference of orthology—genes separated by a speciation event—is crucial for comparative genomics, functional gene annotation, and evolutionary studies. In plants, complex genomic histories featuring whole-genome duplications (WGDs), tandem duplications, and extensive gene loss make orthology inference particularly challenging [45]. **Orthogroups** (groups of genes descended from a single gene in the last common ancestor of all species considered) provide a comprehensive framework for comparing gene content across species [46]. For the study of large, diverse gene families like **NBS-domain-containing genes**, which are key players in plant innate immunity, orthogroup clustering allows researchers to distinguish between core orthologs conserved across deep evolutionary timescales and recent, species-specific expansions [5].

OrthoFinder has emerged as a leading tool for this task, consistently demonstrating superior ortholog inference accuracy in independent benchmarks [46]. Its ability to infer a rooted species tree and identify gene duplication events makes it exceptionally well-suited for investigating the complex evolutionary dynamics of NBS genes in plants, from bryophytes to diploid and polyploid crops [45] [5].

OrthoFinder Methodology and Workflow

OrthoFinder performs a comprehensive phylogenetic analysis starting from protein sequence files in FASTA format (one file per species). Its algorithm proceeds through several stages to transition from sequence similarity to phylogenetically-defined orthogroups and orthologs [46].

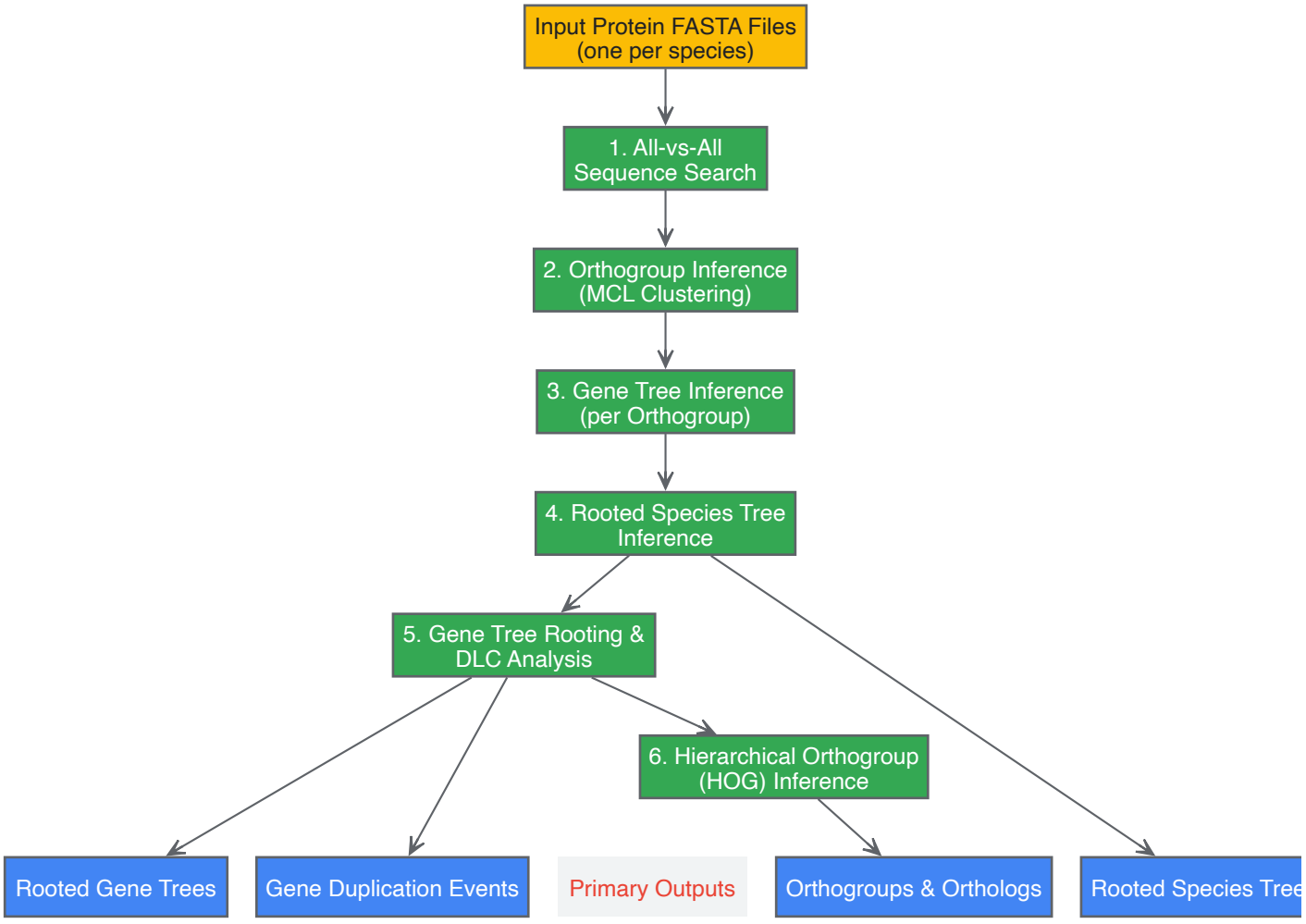
Core Algorithmic Steps

- **Sequence Similarity Search:** An all-vs-all sequence similarity search is performed on the input proteomes. The default tool is DIAMOND for speed, but BLAST can also be specified [46].
- **Orthogroup Inference:** Gene similarity graphs are constructed from the sequence similarity data, and the Markov Clustering algorithm (MCL) is used to identify preliminary orthogroups [47] [46].
- **Gene Tree Inference:** OrthoFinder infers a gene tree for each orthogroup. The default method is DendroBLAST, but it can integrate with multiple sequence alignment (e.g., MAFFT) and tree inference tools (e.g., FastTree) as specified by the user [5] [46].
- **Rooted Species Tree Inference:** The program infers a rooted species tree from the set of all gene trees without requiring prior species tree knowledge [46].

- **Gene Tree Rooting and Analysis:** The gene trees are rooted using the inferred species tree. A Duplication-Loss-Coalescence (DLC) analysis is then performed on the rooted gene trees to identify orthologs, paralogs, and gene duplication events [46].
- **Hierarchical Orthogroup Inference:** OrthoFinder infers Hierarchical Orthogroups (HOGs) at each node of the rooted species tree, providing a more accurate, phylogenetically-informed set of orthogroups compared to the initial MCL-based clusters [47].

Workflow Visualization

The following diagram illustrates the complete OrthoFinder workflow, from input files to key phylogenetic outputs.



Experimental Protocol for NBS Gene Analysis

This section provides a detailed, citable protocol for applying OrthoFinder to study NBS gene families across plant species, as demonstrated in recent research [5].

Input Data Preparation

- **Genome Selection:** Select genome assemblies from public databases (e.g., NCBI, Phytozome, Plaza) spanning the evolutionary breadth of interest. A recent study analyzed 34 species from mosses to monocots and dicots to capture NBS gene diversity [5].
- **Sequence Extraction:** Identify NBS-domain-containing genes from each proteome using HMMER scans (e.g., `PfamScan.pl`) against the PFAM NB-ARC domain model (PF00931) with a strict E-value cutoff (e.g., 1.1e-50) [5].
- **File Formatting:** Save the protein sequences for the identified NBS genes for each species in a separate FASTA file. Ensure filenames are consistent and informative (e.g., `Species_A_NBS.faa`).

Running OrthoFinder

- **Installation:** Install OrthoFinder and its dependencies via Bioconda: `conda install orthofinder -c bioconda` [47].
- **Basic Command:** Execute OrthoFinder on the directory containing your NBS protein FASTA files.

  The `-t` and `-a` options specify the number of threads for BLAST/DIAMOND and multiple sequence alignment, respectively, and should be adjusted based on available computational resources.

- **Advanced Configuration (Optional):** For maximum accuracy, particularly with complex gene families, users can specify alternative multiple sequence alignment and tree inference tools.

Output Analysis for NBS Genes

- **Core and Unique Orthogroups:** Analyze the `Phylogenetic_Hierarchical_Orthogroups/N0.tsv` file. This file contains the orthogroups inferred at the root of the species tree. **Core orthogroups** (e.g., OG0, OG1, OG2) are those present in most or all species, while **unique orthogroups** (e.g., OG80, OG82) are highly specific to a particular species or clade [5].
- **Gene Duplication Analysis:** Examine the `Gene_Duplication_Events` directory. This is critical for understanding the expansion mechanisms of NBS genes, distinguishing between tandem duplications and those associated with WGDs [5] [46].
- **Ortholog Identification:** For pairwise species comparisons, locate specific ortholog sets in the `Orthologues` directory. This is essential for targeted comparative studies [47].

Results and Data Presentation

Applying OrthoFinder to a set of species yields quantitative insights into gene family evolution. The following tables summarize typical results from an analysis of NBS genes across a plant lineage.

Table 1: Summary of OrthoFinder Results for a Hypothetical 8-Species Brassicaceae NBS Gene Analysis [45] [5]

Metric	Diploid Set (5 species)	Diploid + Polyploid Set (8 species)
Total Number of NBS Genes Identified	1,850	3,220
Total Orthogroups (N0) Inferred	350	500
Core Single-Copy Orthogroups	45	28
Species-Specific Orthogroups	25	65
Average Genes per Orthogroup	5.3	6.4
Percentage of Genes in Orthogroups	96.5%	95.1%

Table 2: Example Core and Unique NBS Orthogroups with Functional Annotations [5]

Orthogroup ID	Classification	Species Count	Putative Function / Domain Architecture	Expression Profile
OG0	Core	8/8	TIR-NBS-LRR	Upregulated in leaf under biotic stress
OG1	Core	8/8	CC-NBS-LRR	Constitutive expression
OG2	Core	8/8	NBS-LRR	Upregulated in root and stem
OG15	Core	7/8	TIR-NBS	Responsive to abiotic stress
OG80	Unique	1/8	Species-specific TIR-NBS-TIR-Cupin_1	Not characterized
OG82	Unique	1/8	Species-specific NBS-Prenyltransf	Not characterized

The Scientist's Toolkit: Research Reagent Solutions

This table details key materials and software used in a standard OrthoFinder analysis of NBS genes.

Table 3: Essential Research Reagents and Computational Tools for Orthogroup Analysis

Item Name	Type/Format	Function in Analysis	Source/Example
Annotated Proteomes	FASTA Files	Input data for orthology inference. Provides protein sequences for all genes in each genome.	NCBI, Phytozome, Plaza
NB-ARC Domain HMM	HMM Profile (Pfam)	Identifying NBS-domain-containing genes from whole proteomes prior to OrthoFinder analysis.	Pfam PF00931

Item Name	Type/Format	Function in Analysis	Source/Example
OrthoFinder Software	Python Package	Core platform for performing phylogenetic orthogroup inference, species tree, and duplication analysis.	GitHub, Bioconda
DIAMOND	Software Tool	High-speed sequence similarity search, used as the default search engine by OrthoFinder.	https://github.com/bbuchfink/diamond
MCL Algorithm	Clustering Algorithm	Groups sequences into orthogroups based on sequence similarity graphs within OrthoFinder.	Included in OrthoFinder
Phylogenetic*Hierarchical*Orthogroups/N0.tsv	Tab-separated values file	Primary results file containing the inferred orthogroups for downstream analysis of core and specific families.	OrthoFinder Output
Orthologues Directory	Directory of TSV files	Contains pairwise ortholog mappings between species for fine-scale comparative studies.	OrthoFinder Output

Transcriptomics has revolutionized plant stress biology, providing a systems-level understanding of how plants perceive and respond to complex environmental challenges. This field enables researchers to decode the molecular dialogues that underpin stress resilience by cataloging the entire set of RNA transcripts within a cell or tissue. For researchers investigating the diversification of nucleotide-binding site (NBS) domain genes—a major class of plant disease resistance genes—transcriptomic approaches offer powerful tools to link sequence diversity with functional expression dynamics under stress conditions. Recent studies have demonstrated that **NBS-domain-containing genes** represent one of the largest and most variable gene families in plants, with over 12,820 genes identified across 34 plant species and classified into 168 distinct domain architecture classes [5]. The integration of transcriptomic meta-analyses with machine learning algorithms now enables predictive prioritization of key stress-responsive genes, accelerating the discovery of genetic elements crucial for developing stress-resilient crops in an era of climate uncertainty [48] [49].

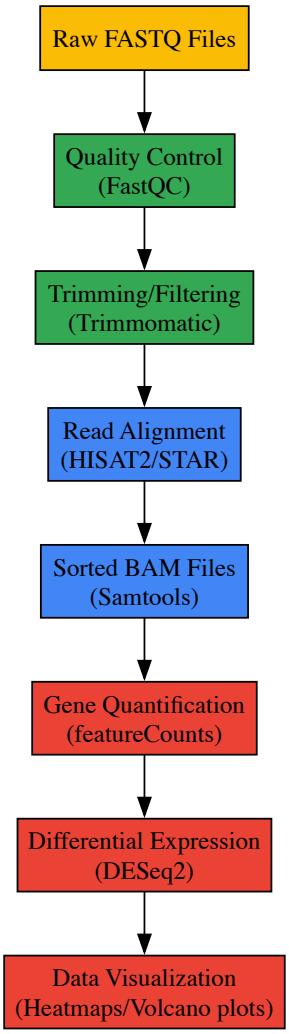
Methodological Framework for Transcriptomic Analysis

Experimental Design Considerations

Robust transcriptomic studies of plant stress responses require careful experimental design to yield biologically meaningful data. Researchers must account for several critical factors: **tissue-specific responses** (as different cell types exhibit distinct expression profiles), **temporal dynamics** of stress responses, and the **simultaneous occurrence of multiple stresses** in field conditions. A recent single-cell RNA sequencing study on rice roots revealed that approximately **31% of differentially expressed genes (DEGs)** were altered in just one specific cell type or developmental stage when comparing soil-grown versus gel-grown roots, highlighting the importance of cellular resolution in understanding stress adaptation mechanisms [50]. For studies focusing on NBS gene families, this cellular specificity is particularly relevant as different NBS genes may be activated in various tissue layers upon pathogen challenge.

Core Computational Workflow for RNA-Seq Analysis

A standardized bioinformatics pipeline is essential for processing raw sequencing data into interpretable gene expression information. The following workflow outlines the key steps from raw data to differential expression analysis [51]:



Step 1: Quality Control and Trimming Raw FASTQ files from sequencing platforms must first undergo quality assessment using tools like **FastQC**. Adapter sequences and low-quality bases are then trimmed using **Trimmomatic** or similar tools. This critical step ensures that only high-quality reads proceed to alignment, reducing false positives in downstream analysis [51].

Step 2: Read Alignment and Quantification Quality-filtered reads are aligned to a reference genome using splice-aware aligners such as **HISAT2** or **STAR**. The resulting SAM/BAM files are sorted and indexed using **Samtools**. Gene-level counts are generated using **featureCounts** or HTSeq, which assigns reads to genomic features based on gene annotation files [48] [51].

Step 3: Differential Expression Analysis Read counts are imported into R/Bioconductor and analyzed with **DESeq2** or edgeR to identify statistically significant differentially expressed genes between experimental conditions. These tools implement specific statistical models that account for biological variability and count-based distribution of RNA-seq data [48] [51].

Step 4: Batch Effect Correction in Multi-Study Analyses When integrating datasets from multiple studies (meta-analysis), technical variability must be addressed. The **Random Forest-based normalization** approach or empirical Bayes methods (**ComBat** from the SVA package) can effectively remove batch effects while preserving biological variation [48] [49].

Advanced Approaches: Single-Cell and Spatial Transcriptomics

Recent technological advances now enable transcriptomic profiling at cellular resolution. **Single-cell RNA sequencing (scRNA-seq)** reveals cell-type-specific responses to stress that are masked in bulk tissue analyses. For example, when comparing rice roots grown in soil versus gel conditions, scRNA-seq demonstrated that outer root tissues (epidermis, exodermis, sclerenchyma, and cortex) showed the most significant transcriptional changes, while inner stele layers remained relatively stable [50]. **Spatial transcriptomics** techniques, such as Molecular Cartography, further enhance this by preserving spatial context, allowing researchers to validate cell-type-specific expression patterns identified through scRNA-seq clustering [50].

Key Analytical Approaches for Stress Transcriptomics

Weighted Gene Co-Expression Network Analysis (WGCNA)

WGCNA identifies modules of highly correlated genes across samples and connects these modules to external traits. This systems biology approach helps move beyond individual DEGs to identify functional gene networks active under stress conditions. In a meta-analysis of 100 wheat genotypes under multiple abiotic stresses, WGCNA identified key functional modules and **eight hub genes** with multi-stress resistance potential, including **BES1/BZR1 and GH14** [48].

Machine Learning for Gene Prioritization

The high-dimensional nature of transcriptomic data (many genes, few samples) makes machine learning well-suited for identifying the most informative stress-responsive genes. Multiple algorithms can be applied to rank genes by their importance in classifying stress conditions:

Table 1: Machine Learning Algorithms for Gene Prioritization in Transcriptomic Studies

Algorithm	Key Features	Application in Stress Studies
Random Forest (RF)	Ensemble method using multiple decision trees	Identifies genes with high variable importance measures
Support Vector Machine (SVM)	Finds optimal hyperplane to separate classes	Effective for high-dimensional genomic data
Partial Least Squares Discriminant Analysis (PLSDA)	Projects variables into latent structures	Provides Variable Importance in Projection (VIP) scores
Gradient Boosting Machine (GBM)	Builds sequential models to correct errors	Captures complex gene interaction effects

In maize stress studies, these methods successfully prioritized **235 unique candidate genes** from 39,756 initially identified DEGs, with three genes (bZIP transcription factor 68, glycine-rich cell wall structural protein 2, and aldehyde dehydrogenase 11) emerging as top hubs in co-expression networks [49].

Meta-Analysis Frameworks for Multi-Study Integration

Meta-analysis of transcriptomic datasets increases statistical power and identifies consistent signals across independent studies. This approach is particularly valuable for understanding NBS gene responses, as different family members may be activated in various stress contexts. A systematic workflow includes:

- **Comprehensive dataset collection** from public repositories (NCBI SRA)
- **Cross-study normalization** to address technical variability
- **Identification of shared DEGs** across stress conditions using Venn diagrams or Upset plots
- **Functional enrichment analysis** (GO and KEGG) to identify overrepresented pathways [48]

Expression Profiling of NBS Domain Genes Under Stress

Diversity and Evolution of NBS Gene Family

NBS-domain-containing genes represent a major class of plant resistance (R) genes involved in pathogen recognition and defense signaling. Comparative genomic analyses have revealed remarkable diversity in this gene family, with identification of **12,820 NBS-domain-containing genes** across 34 plant species ranging from mosses to monocots and dicots [5]. These genes display extraordinary structural variation, with **168 distinct domain architecture classes** identified, including both classical patterns (NBS, NBS-LRR, TIR-NBS, TIR-NBS-LRR) and species-specific structural patterns (TIR-NBS-TIR-Cupin*1-Cupin*1, TIR-NBS-Prenyltransf) [5].

Orthogroup analysis has identified **603 orthogroups** of NBS genes, with some core groups conserved across multiple species and others specific to particular lineages. This diversification has been driven by both whole-genome duplication and small-scale duplication events, with tandem duplications playing a particularly important role in NBS gene expansion [5].

NBS Gene Expression Under Biotic and Abiotic Stress

Transcriptomic profiling across multiple plant species has revealed complex expression patterns of NBS genes under different stress conditions:

Table 2: NBS Gene Expression Patterns Under Stress Conditions

Stress Category	Expression Patterns	Notable Findings
Biotic Stress	Specific NBS genes show pronounced upregulation in tolerant genotypes	In cotton, NBS genes from orthogroups OG2, OG6, and OG15 were upregulated in response to cotton leaf curl disease (CLCuD) [5]
Abiotic Stress	More varied responses, with some NBS genes activated by multiple stresses	Machine learning prioritization identified specific NBS genes responsive to drought, cold, and salinity [49]
Combined Stresses	Unique expression signatures distinct from single stress responses	Meta-analysis revealed genes co-expressed under both biotic and abiotic stress conditions [49]

Functional validation through virus-induced gene silencing (VIGS) of a candidate NBS gene (**GaNBS from OG2**) in resistant cotton demonstrated its role in reducing virus titers, confirming the importance of NBS genes in defense responses [5].

NLR Pairs in Stress Response

Recent research has revealed that some NBS genes function as paired modules in plant immunity. Studies in wheat have identified head-to-head NLR gene pairs at stripe rust resistance loci, where an intact CNL protein pairs with an NL protein lacking an annotated N-terminal domain [52]. Interestingly, this head-to-head orientation appears non-essential for function, as random insertion of both genes into susceptible wheat varieties still conferred resistance, suggesting flexibility in genetic organization for NLR pair functionality [52]. This discovery has significant implications for engineering disease resistance in crops, as functional NLR pairs may be transferable between distantly related species.

Experimental Protocols for Key Analyses

Protocol: RNA-Seq Meta-Analysis for Stress-Responsive Genes

Application: Identification of conserved transcriptional responses across multiple studies and plant species [48] [49].

- **Data Collection and Quality Control**
  - Retrieve RNA-seq datasets from public repositories (NCBI SRA)
  - Apply quality control using FastQC (Q30 score >85%)
  - Trim adapters and low-quality bases using Trimmomatic or fastp
- **Read Alignment and Quantification**
  - Align reads to reference genome using HISAT2 with parameters: `--dta --phred33 --max-intronlen 5000`
  - Convert SAM to sorted BAM files using Samtools
  - Generate raw count matrices using featureCounts with parameters: `-t exon -g gene_id -s 0`
- **Cross-Study Normalization**
  - Perform variance-stabilizing transformation of raw counts
  - Apply Random Forest-based normalization or ComBat from SVA package to correct batch effects
  - Verify batch effect removal using PCA and correlation analyses
- **Differential Expression Analysis**
  - Identify DEGs for each study using DESeq2 with criteria: `|log2(fold change)| ≥ 1` and adjusted p-value < 0.05
  - Consolidate DEG sets by stress type
  - Identify stress-overlapping genes using Jvenn with stringent criteria (detection in ≥80% of studies per stress category)
- **Co-expression Network Analysis**
  - Perform WGCNA to identify modules of co-expressed genes
  - Correlate module eigengenes with stress traits
  - Identify hub genes within significant modules

Protocol: scRNA-seq for Cell-Type-Specific Stress Responses

Application: Resolve transcriptional responses to stress at individual cell type resolution [50].

- **Sample Preparation and Sequencing**

    - Isolate protoplasts from stress-treated and control root tissues
    - Process using 10X Genomics platform to generate barcoded scRNA-seq libraries
    - Sequence libraries to appropriate depth (typically 50,000 reads/cell)

- **Data Preprocessing and Integration**
    - Process raw data using COPILOT pipeline or Cell Ranger
    - Identify and exclude protoplasting-induced genes using bulk RNA-seq data
    - Integrate multiple datasets using Seurat or similar tools

- **Cell Type Annotation and Validation**
    - Perform clustering and initial annotation using known marker genes
    - Validate cell identities using spatial transcriptomics (Molecular Cartography)
    - Refine annotations through iterative feedback between scRNA-seq and spatial data

- **Differential Expression Analysis by Cell Type**
    - Identify DEGs for each cell type between stress and control conditions
    - Perform GO enrichment analysis on cell-type-specific DEGs
    - Visualize expression patterns using dimensionality reduction (UMAP/t-SNE)

Table 3: Key Research Reagent Solutions for Stress Transcriptomics

Category	Specific Tools	Function/Application
Sequencing Platforms	Illumina NovaSeq, NextSeq	High-throughput RNA sequencing
Alignment Tools	HISAT2, STAR	Splice-aware read alignment to reference genomes
Quantification Software	featureCounts, HTSeq	Generate gene-level count matrices from aligned reads
Differential Expression	DESeq2, edgeR	Identify statistically significant DEGs between conditions
Co-expression Analysis	WGCNA R package	Identify modules of co-expressed genes and hub genes
Machine Learning	caret, randomForest, e1071 R packages	Prioritize key stress-responsive genes from large DEG sets
Single-Cell Analysis	Seurat, Scanpy, COPILOT	Process and analyze scRNA-seq data
Spatial Transcriptomics	Molecular Cartography, 10X Visium	Resolve gene expression patterns in tissue context
Validation Platforms	RT-qPCR, Virus-Induced Gene Silencing (VIGS)	Confirm functional role of candidate genes

Transcriptomic approaches provide powerful tools for deciphering the complex molecular networks underlying plant responses to biotic and abiotic stresses. The integration of bulk RNA-seq, single-cell transcriptomics, and spatial gene expression profiling has revealed the remarkable cellular specificity of stress responses and identified key regulatory hubs in stress adaptation networks. For researchers studying NBS domain gene diversification, these technologies offer unprecedented opportunities to link gene family expansion with functional specialization in stress responses. As machine learning algorithms become increasingly sophisticated in prioritizing candidate genes from large transcriptomic datasets, and as spatial technologies provide cellular context for expression patterns, our ability to identify key genetic elements for crop improvement accelerates dramatically. The continued integration of these transcriptomic approaches with functional validation will be essential for developing stress-resilient crops needed to address growing agricultural challenges in a changing climate.

Plant disease resistance is a critical component of global food security, with nucleotide-binding site (NBS) domain genes playing a central role in plant immune responses. These genes represent one of the largest and most diverse gene families in plants, encoding proteins that recognize pathogen effectors and activate defense mechanisms [5]. The diversification of NBS domain genes across plant species represents a fascinating evolutionary arms race between plants and their pathogens. Research has identified **12,820 NBS-domain-containing genes** across 34 species ranging from mosses to monocots and dicots, classified into **168 distinct domain architecture classes** [5]. This remarkable diversity underscores the need for specialized bioinformatics platforms to catalog, analyze, and contextualize these important genetic elements. PRGdb (Plant Resistance Genes database) stands as a cornerstone resource in this effort, providing the scientific community with comprehensive tools and data for studying plant resistance genes within this broader evolutionary context.

PRGdb: A Core Platform for Plant Resistance Gene Analysis

Platform Evolution and Capabilities

PRGdb has evolved significantly since its inception to become a comprehensive bioinformatics platform dedicated to plant resistance gene (R-gene) analysis. The database represents the first bioinformatic resource providing a comprehensive overview of R-genes in plants [53]. The most recent version, **PRGdb 4.0**, continues this tradition with expanded data coverage, analyzing proteomes from **182 species** with putative resistance genes and containing reference resistance genes from **33 species** [54].

Table: Evolution of PRGdb Content Across Versions

PRGdb Version	Reference R-genes	Putative R-genes	Plant Species Covered	Key Features
Initial Release [55]	73	~16,000	192	First comprehensive R-gene database
PRGdb 3.0 [56]	153	177,072	76 Viridiplantae & algae	DRAGO 2 tool, BLAST search
PRGdb 4.0 [54]	Information for 33 species	Information for 182 species	Updated coverage	Current version with expanded data

The platform has been redesigned with a user-friendly interface that streamlines data queries through easy-to-read search boxes and directly displays plant species with candidate or cloned genes [56]. This accessibility makes it valuable for both plant science researchers and breeders seeking to improve crop disease resistance.

Classification Systems for Resistance Genes

PRGdb organizes resistance genes based on their protein domain structures, which is crucial for understanding their function and evolutionary relationships. The primary classification system used in the database includes:

- **CNL Class**: Genes encoding proteins with Coiled-coil, Nucleotide-binding site, and Leucine-rich Repeat domains
- **TNL Class**: Genes containing Toll-interleukin receptor-like, Nucleotide-binding site, and Leucine-rich Repeat domains
- **RLP Class**: Receptor-like proteins with serine-threonine kinase-like and extracellular leucine-rich repeat domains
- **RLK Class**: Receptor-like kinases with kinase and extracellular leucine-rich repeat domains
- **Others**: Genes conferring resistance through different molecular mechanisms [55]

This classification system enables researchers to identify evolutionary relationships and functional conservation across plant species, facilitating comparative genomic studies of NBS domain gene diversification.

While PRGdb serves as a specialized resource for resistance genes, several other databases and tools provide essential complementary functionality for studying NBS domain gene diversification:

NCBI's Conserved Domain Database (CDD)

The Conserved Domain Database is a crucial resource for identifying and characterizing domains within NBS genes. CDD consists of a collection of well-annotated multiple sequence alignment models for ancient domains and full-length proteins, available as position-specific score matrices for fast identification of conserved domains in protein sequences [57]. The **CD-Search** tool provides NCBI's interface to the database, using RPS-BLAST to quickly scan pre-calculated matrices with a protein query. For large-scale analyses, **Batch CD-Search** allows conserved domain search on up to 4,000 protein sequences in a single job [57]. This resource is particularly valuable for classifying the diverse domain architectures discovered in NBS genes.

Specialized Analysis Tools

Several specialized tools have been developed to address the particular challenges of identifying and classifying resistance genes:

- **DRAGO (Disease Resistance Analysis and Gene Orthology)**: PRGdb's home-made prediction pipeline that searches for plant resistance genes in public datasets [55]. Version 2.0 uses **60 HMM modules** to detect LRR, Kinase, NBS, and TIR domains, plus COILS and TMHMM programs for CC and transmembrane domains [56].

- **PRGminer**: A deep learning-based high-throughput R-gene prediction tool that uses dipeptide composition for sequence representation. The tool achieves **95.72% accuracy** on independent testing for Phase I classification (R-genes vs. non-R-genes) and **97.21% accuracy** for Phase II classification into specific classes [58]. This represents a significant advancement over traditional alignment-based methods, particularly for sequences with low homology.

Table: Key Resources for NBS Gene Research

Resource Name	Primary Function	Key Features	Relevance to NBS Gene Research
PRGdb 4.0 [54]	R-gene database & analysis	Curated reference genes, putative genes, analysis tools	Comprehensive repository for resistance gene data
NCBI CDD [57]	Domain identification	Curated domain models, RPS-BLAST search	Domain architecture analysis for NBS genes
DRAGO 2 [56]	R-gene prediction	HMM-based pipeline, domain detection	Automated annotation of putative resistance genes
PRGminer [58]	R-gene prediction & classification	Deep learning approach, high accuracy	Identification of novel resistance genes

Experimental Frameworks for NBS Gene Identification and Analysis

Domain-Based Identification Pipeline

The standard methodology for genome-wide identification of NBS gene families employs a domain-based approach combining multiple bioinformatics tools. A representative protocol from recent research includes:

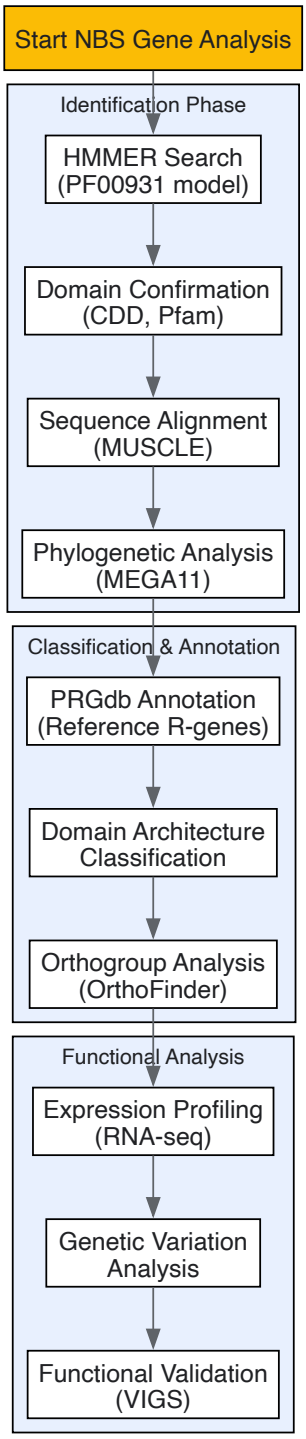
- **HMMER Search**: Initial identification of NBS-LRR family members using HMMER with the PF00931 model from PFAM database [10].
- **Domain Confirmation**: Verification of TIR and LRR domains using PFAM domains (PF01582, PF00560, PF07723, PF07725, PF12779, PF13306, PF13516, PF13855, PF14580, PF03382, PF01030, PF05725). Coiled-coil domains are confirmed via NCBI's Conserved Domain Database [10].
- **Sequence Alignment**: Multiple sequence alignment of NBS-LRR protein sequences using MUSCLE with default parameters [10].
- **Phylogenetic Analysis**: Construction of phylogenetic trees using maximum likelihood methods with bootstrap validation [10].

This pipeline successfully identified **1,226 NBS genes** across three Nicotiana genomes, with the allotetraploid N. tabacum containing approximately the combined total of its parental species [10].

Evolutionary and Expression Analysis

Beyond identification, comprehensive analysis of NBS genes includes evolutionary and functional characterization:

- **Evolutionary Analysis**: Orthogroup analysis using OrthoFinder with DIAMOND for sequence similarity searches and MCL for clustering. Paralog identification through self-BLASTP and MCScanX for segmental and tandem duplication detection [5].
- **Expression Profiling**: RNA-seq analysis for differential expression under biotic and abiotic stresses. Typical workflow includes quality control with Trimmomatic, mapping with Hisat2, and quantification with Cufflinks with FPKM normalization [10].
- **Genetic Variation**: Identification of unique variants between resistant and susceptible accessions, with recent research finding **6,583 variants** in tolerant cotton accessions versus **5,173** in susceptible varieties [5].
- **Functional Validation**: Virus-induced gene silencing (VIGS) to demonstrate the functional role of candidate genes, as shown by reduced virus resistance when silencing specific NBS genes [5].



NBS Gene Analysis Workflow

Essential Research Reagents and Computational Tools

Table: Essential Research Reagent Solutions for NBS Gene Research

Resource Category	Specific Tools/Databases	Function in Research	Application Example
Domain Databases	NCBI CDD [57], Pfam [10]	Identification of conserved domains	Classifying NBS genes into CNL, TNL, RNL subfamilies
Sequence Search	HMMER [10], BLAST [56]	Homology-based gene identification	Finding NBS genes in newly sequenced genomes
Classification Tools	PRGminer [58], DRAGO 2 [56]	R-gene prediction and classification	Automated annotation of resistance genes

Resource Category	Specific Tools/Databases	Function in Research	Application Example
Curated Databases	PRGdb [54], ANNA [5]	Reference data repository	Comparing newly identified genes with known R-genes
Evolutionary Analysis	OrthoFinder [5], MCScanX [10]	Ortholog identification & duplication analysis	Understanding NBS gene family expansion
Expression Analysis	Cufflinks [10], IPF Database [5]	Transcriptome quantification	Assessing NBS gene expression under stress

The diversification of nucleotide-binding site domain genes across plant species represents a complex evolutionary landscape that requires sophisticated bioinformatics resources for comprehensive study. PRGdb serves as a cornerstone platform in this endeavor, providing curated reference data, analytical tools, and classification systems essential for understanding the expansion and specialization of plant resistance genes. When integrated with complementary resources such as NCBI's CDD, specialized prediction tools like PRGminer, and standardized experimental protocols, researchers are equipped to unravel the intricate evolutionary patterns of NBS genes. These resources collectively enable the scientific community to accelerate the discovery of new resistance genes, understand the genetic basis of plant immunity, and develop strategies for breeding disease-resistant crops in the face of evolving pathogen threats.

Plant resistance genes (R-genes) encode proteins that form a crucial component of the plant immune system, providing defense against a wide array of pathogens including bacteria, fungi, viruses, and nematodes. These genes predominantly encode proteins with characteristic domain architectures, most notably the nucleotide-binding site and leucine-rich repeat (NBS-LRR) domains, which enable pathogen recognition and activation of defense responses [26]. The identification and characterization of R-genes represent a fundamental challenge in plant pathology and breeding programs, as traditional methods for discovering these genes have proven to be time-consuming, labor-intensive, and often limited by sequence homology requirements [58] [26].

The diversification of nucleotide-binding site (NBS) domain genes across plant species presents both opportunities and challenges for researchers. Recent studies have identified 12,820 NBS-domain-containing genes across 34 species ranging from mosses to monocots and dicots, classified into 168 distinct classes with several novel domain architecture patterns [5]. This remarkable diversity underscores the need for advanced computational approaches that can navigate the complex landscape of plant immune genes beyond the constraints of traditional similarity-based methods, which frequently fail in cases of low sequence homology [58].

Limitations of Traditional R-Gene Discovery Methods

Traditional approaches for R-gene identification have primarily relied on alignment-based tools and domain prediction pipelines. These methods utilize programs such as BLAST, InterProScan, HMMER3, and PfamScan to predict domains in protein sequences and assign them to R-gene classes [58]. While these approaches have successfully identified numerous R-genes, they face significant limitations, particularly when annotating newly sequenced plant genomes where limited homologous sequences exist for comparison.

The challenges are further compounded by the unique genomic architecture of R-genes. These genes are often organized in clusters of closely duplicated sequences, though they may also exist as individual units scattered across the genome [58]. Current automatic gene annotation methods struggle to accurately predict and identify R-gene loci due to this unique structure within gene clusters, frequently leading to incomplete and fragmented annotations [58]. Additional complications arise from the typically low expression levels of R-genes, which hinders gene prediction using RNA sequencing data, and their frequent misclassification as repetitive sequences during genome annotation processes [58].

Machine Learning and Deep Learning Frameworks for R-Gene Prediction

Deep Learning Architectures for R-Gene Identification

The limitations of traditional methods have catalyzed the development of sophisticated deep learning frameworks for R-gene prediction. PRGminer represents a cutting-edge example, implementing a two-phase deep learning approach for high-throughput R-gene prediction [58]. In Phase I, the system predicts whether input protein sequences represent R-genes or non-R-genes, while Phase II classifies the identified R-genes into eight distinct classes, including CNL, TNL, RLK, RLP, and others [58].

This architecture leverages dipeptide composition as sequence representations, achieving remarkable performance metrics including an accuracy of 98.75% in k-fold training/testing procedures and 95.72% on independent testing, with Matthews correlation coefficient values of 0.98 and 0.91 respectively in Phase I [58]. The classification phase (Phase II) demonstrated an overall accuracy of 97.55% in k-fold training/testing and 97.21% in independent testing [58]. These results significantly outperform traditional alignment-based methods and demonstrate the power of deep learning for this complex prediction task.

Table 1: Performance Metrics of PRGminer Deep Learning Framework

Phase	Metric	k-fold Training/Testing	Independent Testing
Phase I (R-gene vs non-R-gene)	Accuracy	98.75%	95.72%
	Matthews Correlation Coefficient	0.98	0.91
Phase II (R-gene Classification)	Overall Accuracy	97.55%	97.21%
	Matthews Correlation Coefficient	0.93	0.92

Machine Learning for Disease Resistance Prediction

Beyond R-gene identification, machine learning methods have demonstrated exceptional capability in predicting plant disease resistance phenotypes based on genomic data. Recent research has evaluated eight different machine learning methods, including Random Forest Classification (RFC), Support Vector Classifier (SVC), Light Gradient Boosting Machine (LightGBM), and deep neural network approaches [59]. These methods were enhanced by incorporating kinship information (denoted as "plus K" methods), resulting in significantly improved prediction accuracy.

These models achieved remarkable performance across multiple pathosystems, with accuracies reaching 95% for rice blast, 85% for rice black-streaked dwarf virus, 85% for rice sheath blight, 90% for wheat blast, and 93% for wheat stripe rust diseases [59]. When applied to an independent population for rice blast resistance prediction, the plus K methods maintained an accuracy of 91%, demonstrating robust generalizability beyond the training dataset [59].

Table 2: Performance of Machine Learning Methods in Predicting Disease Resistance

Disease	Host Crop	Best Performing Method	Accuracy
Rice Blast (RB)	Rice	Plus K methods	95%
Rice Black-Streaked Dwarf Virus (RBSDV)	Rice	Plus K methods	85%
Rice Sheath Blight (RSB)	Rice	Plus K methods	85%
Wheat Blast (WB)	Wheat	Plus K methods	90%
Wheat Stripe Rust (WSR)	Wheat	Plus K methods	93%

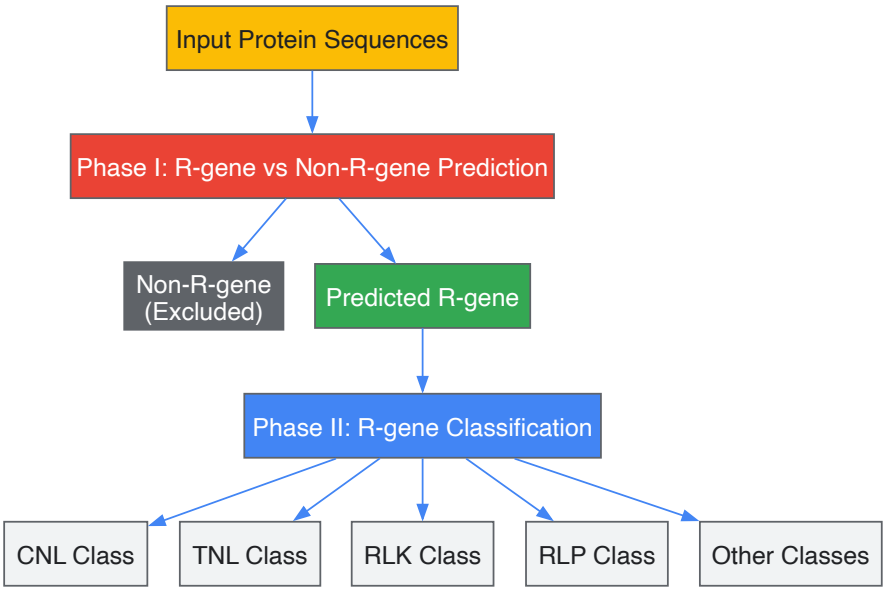
Expression-Based Prediction of Functional NLRs

An innovative approach to identifying functional NLRs leverages their expression signature rather than solely relying on sequence features. Recent research has revealed that functional immune receptors of the NLR class show a signature of high expression in uninfected plants across both monocot and dicot species [60]. This discovery enables the prediction of functional NLR candidates based on their expression levels, providing an orthogonal method to sequence-based predictions.

This expression signature approach has proven highly effective in practice. When applied to wheat, combined with high-throughput transformation, researchers generated a transgenic array of 995 NLRs from diverse grass species and identified 31 new resistance genes against stem rust and leaf rust pathogens [60]. The expression-based prediction method demonstrated that known functional NLRs are significantly enriched in the top 15% of expressed NLR transcripts compared with the lower 85%, confirming the value of expression level as a predictive feature for NLR functionality [60].

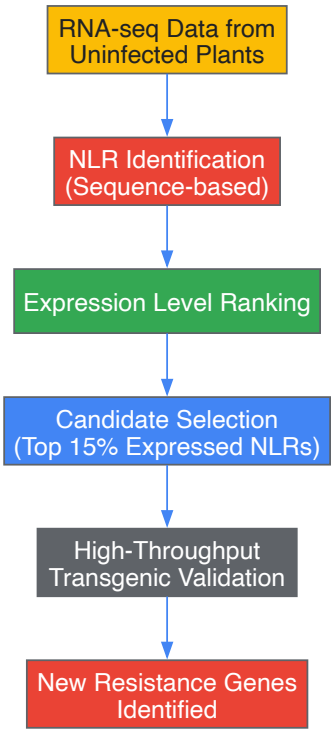
Experimental Protocols and Workflows

PRGminer Implementation Workflow



The PRGminer workflow operates through a sequential two-phase architecture as illustrated above. In Phase I, input protein sequences are encoded using dipeptide composition features and processed through a deep learning network to distinguish R-genes from non-R-genes [58]. Sequences classified as non-R-genes are excluded from further analysis, while those identified as R-genes proceed to Phase II. This classification stage employs additional deep learning architectures to categorize R-genes into specific structural classes based on their domain architectures, including CNL (Coiled-coil, Nucleotide-binding site, Leucine-rich repeat), TNL (Toll/interleukin-1 receptor, Nucleotide-binding site, Leucine-rich repeat), RLK (Receptor-like kinase), RLP (Receptor-like protein), and other specialized classes [58]. The model is trained on curated datasets of R-genes and non-R-genes protein sequences obtained from public databases including Phytozome, Ensemble Plants, and NCBI [58].

High-Expression NLR Discovery Pipeline



The expression-based NLR discovery pipeline begins with transcriptome sequencing of uninfected plant tissues to establish baseline expression levels [60]. NLR genes are first identified using sequence-based methods, then ranked according to their expression levels. Candidates are selected from the top 15% of expressed NLR transcripts, as this segment has been shown to be significantly enriched for functional immune receptors [60]. These candidates subsequently undergo high-throughput transgenic validation through efficient transformation systems. In a proof-of-concept application, this pipeline enabled the testing of 995 NLRs in wheat, resulting in the identification of 31 new resistance genes against stem rust and leaf rust pathogens [60]. This workflow demonstrates how combining computational prediction with large-scale experimental validation accelerates the discovery of functional resistance genes.

The Scientist's Toolkit: Research Reagent Solutions

Table 3: Essential Research Reagents for R-gene Discovery and Validation

Reagent/Resource	Function	Example Use Case
PRGminer Webserver	Deep learning-based R-gene prediction and classification	Initial computational identification of R-genes in newly sequenced genomes [58]
Dipeptide Composition Features	Numerical representation of protein sequences for machine learning	Encoding protein sequences for input into deep learning models [58]
PfamScan with HMM	Domain identification in protein sequences	Detection of NBS, LRR, TIR, and other resistance-associated domains [5]
Kinship-Enhanced ML Models	Prediction of disease resistance phenotypes	Genomic selection for disease resistance in breeding programs [59]
High-Efficiency Transformation Systems	Transgenic validation of candidate genes	Functional testing of NLR candidates in crop species [60]
OrthoFinder	Orthogroup analysis across multiple species	Evolutionary studies of NBS gene diversification [5]

Integration with Evolutionary Studies of NBS Gene Diversification

Machine learning approaches to R-gene prediction are particularly powerful when integrated with evolutionary studies of NBS gene diversification across plant species. Research has revealed that NBS genes are organized into 603 orthogroups, with both core (widely conserved) and unique (species-specific) orthogroups showing evidence of tandem duplications [5]. This evolutionary perspective provides crucial context for interpreting machine learning predictions and prioritizing candidates for functional validation.

Expression profiling of these orthogroups under various biotic and abiotic stresses has demonstrated differential expression patterns in susceptible versus tolerant plants [5]. For instance, in studies of cotton leaf curl disease, specific orthogroups (OG2, OG6, and OG15) showed putative upregulation in different tissues under various stress conditions [5]. Genetic variation analysis between susceptible and tolerant Gossypium hirsutum accessions revealed distinctive variant profiles, with the tolerant accession Mac7 showing 6583 unique variants compared to 5173 in the susceptible Coker312 accession [5]. These evolutionary and functional insights can be incorporated as features in machine learning models to improve prediction accuracy for functionally relevant R-genes.

Current Limitations and Future Directions

Despite the promising advances in machine and deep learning applications for R-gene prediction, several challenges remain. Current models face limitations including data quality issues, class imbalance in training datasets, and limited interpretability of predictions [26]. Furthermore, a recent comprehensive benchmark study revealed that in some prediction tasks, deep learning foundation models have not yet outperformed deliberately simple linear baselines [61]. This highlights the importance of critical benchmarking in directing and evaluating method development.

Future research directions should focus on developing more interpretable models, improving data quality and standardization, and integrating multi-omics data sources [26]. Transfer learning approaches, which leverage knowledge from data-rich species to improve predictions in less-studied species, show particular promise for addressing the challenge of limited training data in non-model plant systems [62]. As one study demonstrated, hybrid models that combine convolutional neural networks with traditional machine learning consistently outperformed traditional methods for gene regulatory network construction, achieving over 95% accuracy on holdout test datasets [62]. Similar approaches could be adapted for R-gene prediction tasks.

Machine learning and deep learning technologies are revolutionizing the prediction and characterization of plant resistance genes, moving the field beyond the limitations of traditional homology-based methods. The integration of these computational approaches with evolutionary studies of NBS gene diversification provides a powerful framework for understanding the complex landscape of plant immune systems. As these methods continue to mature and incorporate additional biological insights—from expression signatures to evolutionary conservation patterns—they promise to dramatically accelerate the discovery and deployment of resistance genes in crop breeding programs. This advancement is critical for developing durable disease resistance in agricultural systems facing evolving pathogen threats and changing climatic conditions, ultimately contributing to global food security.

Overcoming Research Hurdles: Annotation Challenges and Functional Prediction in NBS Genes

Addressing High Sequence Diversity and Variable Domain Architectures

The **Nucleotide-Binding Site (NBS) domain** serves as the core structural and functional component in a major superfamily of plant disease resistance (R) genes, which are pivotal for effector-triggered immunity [5]. These genes encode intracellular immune receptors, often termed **NLRs (Nucleotide-Binding Leucine-Rich Repeat receptors)**, that recognize pathogen-derived effector molecules and initiate defense responses [63] [64]. The evolution of recognition specificities by the plant immune system is fundamentally dependent on the generation of immense receptor diversity and the connection between new antigen binding and downstream signaling initiation [63]. This diversity manifests primarily through two interconnected phenomena: **extreme sequence polymorphism** in coding regions and a **striking proliferation of protein domain architectures**. These variations are not random; they result from evolutionary pressures exerted by rapidly adapting pathogens, driving a continuous molecular arms race that shapes the genomic architecture of plant immunity [63] [5] [64]. Understanding the mechanisms that generate this diversity, the patterns of its distribution, and the methodologies for its study is essential for both basic science and applied crop improvement [63].

Quantitative Landscape of NBS Diversity

The genomic repertoire of NBS genes is one of the largest and most variable protein families in plants, a stark contrast to vertebrate NLR repertoires, which typically consist of only around 20 members [5]. Recent analyses have identified a vast number of these genes; for example, one study cataloged **12,820 NBS-domain-containing genes across 34 plant species**, ranging from mosses to monocots and eudicots [5]. These were classified into **168 distinct domain architecture classes**, underscoring the remarkable structural diversification in this gene family [5].

Table 1: Genomic Distribution of NBS Genes and Domain Architectures Across Plant Lineages

Plant Group	Representative Species	NLR Repertoire Size	Predominant Domain Architectures	Key Evolutionary Notes
Bryophytes	*Physcomitrella patens*	~25 NLRs [5]	Classical TNLs, CNLs [5]	Represents an ancestral, small NLR repertoire [5].
Lycophytes	*Selaginella moellendorffii*	~2 NLRs [5]	Classical TNLs, CNLs [5]	Minimal NLR expansion [5].
Angiosperms	Various (e.g., Arabidopsis, Rice)	70,737 CNLs; 18,707 TNLs (from 304 genomes) [5]	Classical & species-specific patterns (e.g., TIR-NBS-TIR-Cupin*1*, *Sugar*tr-NBS) [5]	Substantial gene expansion primarily in flowering plants [5].
Crop Species	Wheat (*Triticum spp.*)	~2,000 NBS encoding genes [5]	CNLs, TNLs, and paired NLRs [52]	Expansion includes complex paired arrangements [52].

The distribution of protein domain architectures across plant genomes shows consistent patterns. Analyses of 14 green plant genomes reveal that approximately **65% of domain architectures are universally present** across all lineages, indicating a core set of conserved protein components [65]. The remaining architectures are lineage-specific, with each genome harboring approximately **5-15% of architectures not found in any other species** [65]. This diversity is maintained despite the conservation of overall distribution patterns, where single-domain architectures typically constitute 30-51% of a genome's Pfam-predictable architectures, double-domain architectures constitute 8-14%, and architectures with three or more domains make up the remainder [65].

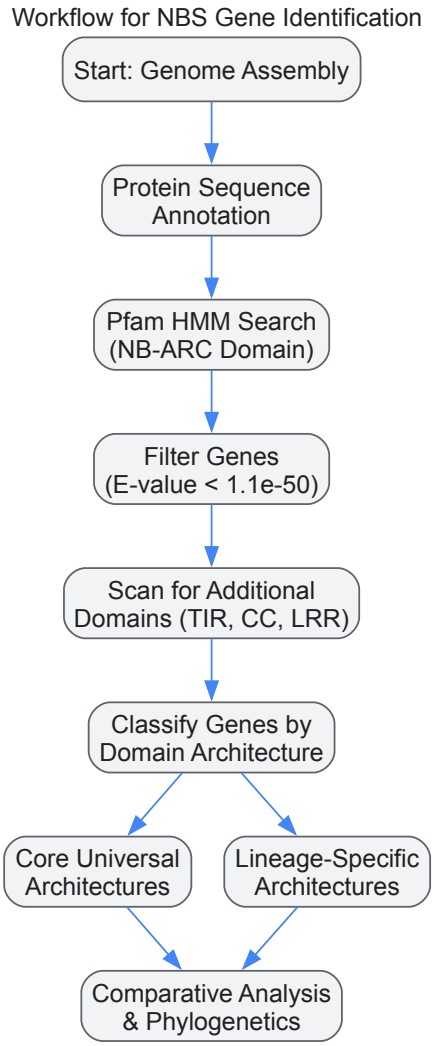
Methodologies for Assessing Sequence and Architectural Diversity

Identification and Classification of NBS Genes

A standard pipeline for identifying and classifying NBS genes involves several key steps, leveraging both sequence homology and domain composition analysis [5].

Experimental Protocol 1: Genome-Wide Identification and Classification of NBS Genes

- **Data Collection:** Obtain the latest genome assemblies and their corresponding protein sequence annotations from public databases like NCBI, Phytozome, or Plaza [5].
- **Domain Prediction:** Screen all protein sequences for the presence of an NBS (NB-ARC) domain using the **PfamScan.pl HMM search script** or a similar tool. A standard e-value cutoff (e.g., 1.1e-50) is used to ensure high-confidence predictions [5].
- **Architecture Classification:** All genes containing an NB-ARC domain are considered NBS genes. The domain architecture of each identified gene is then determined by scanning for additional associated domains (e.g., TIR, CC, LRR, RPW8, and other integrated domains) [5].
- **Categorization:** Genes are grouped into classes based on their shared domain architecture. This reveals both classical patterns (NBS, NBS-LRR, TIR-NBS-LRR) and novel, species-specific structural patterns [5].



Measuring Intraspecies Diversity and Identifying Variable Residues

To understand evolutionary dynamics at the population level, pan-genome analyses of multiple accessions or ecotypes of a species are conducted. **Shannon entropy**, a measure from information theory, is a powerful tool for identifying highly variable residues that are likely determinant of pathogen recognition specificity [63].

Experimental Protocol 2: Pan-Genome Analysis and Specificity-Determining Residue Identification

- **Pan-Genome Sequencing:** Assemble the NLR complements from dozens of ecotypes or lines of a target species (e.g., 62 ecotypes of *A. thaliana* and 54 lines of *B. distachyon*) [63].
- **Phylogenetic Grouping:** Group NLRs into near-allelic series using phylogenetic analyses [63].
- **Multiple Sequence Alignment:** Create a high-quality multiple sequence alignment for each NLR subfamily [63].
- **Calculate Shannon Entropy:** For each position (column) in the alignment, calculate Shannon entropy (H) using the formula: (H = -\sum{i=1}^{20} pi \log2 pi) where (p_i) is the fraction of the 20 amino acids at that position [63].
- **Identify Highly Variable NLRs (hvNLRs):** Define subfamilies with high average entropy as hvNLRs. These represent rapidly diversifying families under strong diversifying selection [63].
- **Map Variable Residues:** Project the high-entropy residues onto available or homology-based protein structures. Studies show these residues cluster on the solvent-exposed surfaces of LRR domains, forming the predicted pathogen effector binding interface [63].

Table 2: Key Molecular Marker Technologies for Diversity Analysis

Marker Type	Key Principle	Application in NBS Gene Analysis	Technical Considerations
SSR / Microsatellite [66]	PCR amplification of short, repetitive sequences with high polymorphism.	Genetic diversity analysis, linkage mapping of NBS loci, QTL identification for disease resistance [66].	High polymorphism, codominant, requires prior sequence knowledge for primer design.

Marker Type	Key Principle	Application in NBS Gene Analysis	Technical Considerations
SNP [66]	Detection of single nucleotide changes, the most abundant variation.	High-resolution genotyping, genome-wide association studies (GWAS) for trait mapping, genomic selection [66].	High map precision, efficient and cost-effective for high-throughput genotyping.
iSNAP [66]	Explores polymorphisms in intergenic regions flanked by noncoding small RNAs.	Studying variation in regulatory regions, potentially linked to NLR gene expression and complex traits [66].	Functional relevance to gene regulation; useful for traits governed by post-transcriptional control.
ILP [66] [67]	PCR-based amplification targeting introns, which evolve faster than exons.	Development of highly polymorphic, gene-based markers for genetic mapping and diversity studies [66] [67].	High polymorphism due to lower selective pressure on introns; requires genomic sequence data.

Functional Validation of Diverse NBS Genes

Identifying sequence-diverse NBS genes is only the first step. Establishing their biological function is crucial. A combination of computational and experimental approaches is used for functional validation.

Experimental Protocol 3: Functional Validation via Virus-Induced Gene Silencing (VIGS) and Interaction Studies

- **Expression Profiling:** Utilize RNA-seq data from public databases or new experiments to profile the expression of candidate NBS genes across different tissues and under various biotic (e.g., fungal, bacterial, viral infection) and abiotic stresses (e.g., drought, salt). Putative upregulation in resistant varieties under pathogen challenge provides correlative evidence for function [5].
- **Genetic Variation Analysis:** Identify unique sequence variants (e.g., SNPs, indels) in NBS genes by comparing whole-genome sequencing data from susceptible and tolerant plant accessions [5].
- **Protein Interaction Modeling:** Perform *in silico* protein-ligand and protein-protein interaction analyses (e.g., molecular docking) to predict interactions between candidate NBS proteins and pathogen effectors or core pathogen proteins [5].
- **Functional Testing with VIGS:**
  - **Design:** Select a candidate NBS gene (e.g., from an orthogroup upregulated in resistant plants).
  - **Vector Construction:** Clone a fragment of the target gene into a VIGS vector (e.g., based on Tobacco Rattle Virus).
  - **Plant Inoculation:** Introduce the vector into a resistant plant variety via *Agrobacterium*-mediated infiltration or *in vitro* transcript inoculation.
  - **Phenotyping:** After successful viral spread and gene silencing, challenge the plants with the target pathogen.
  - **Validation:** A loss-of-resistance phenotype (e.g., increased virus titer or disease symptoms) in silenced plants, compared to empty-vector controls, demonstrates the putative role of the NBS gene in resistance [5].

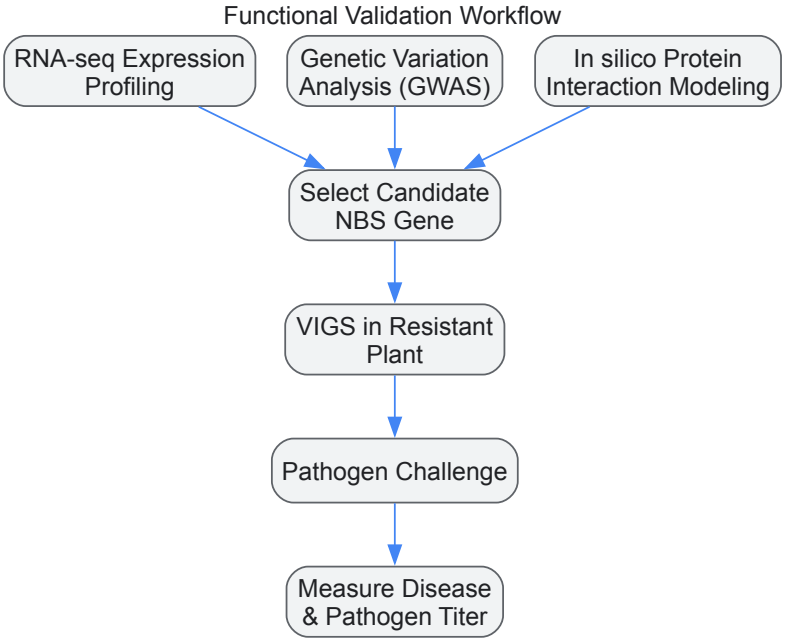


Table 3: Research Reagent Solutions for Studying Diverse NBS Genes

Reagent / Resource	Function and Application	Example Use Case
Pfam HMM Models [5]	Hidden Markov Models for identifying protein domains (e.g., NB-ARC, TIR, LRR) in sequence data.	Initial genome-wide scan to identify the entire NBS gene repertoire in a newly sequenced genome.
OrthoFinder Software [5]	Tool for orthogroup inference and comparative genomics.	Clustering NBS genes from multiple species to identify evolutionarily conserved orthogroups and lineage-specific expansions.
DIRT Software [68]	Digital Imaging of Root Traits; an automatic, high-throughput computing platform for quantifying root architecture.	Phenotypic screening of plant lines with altered NBS genes to investigate potential pleiotropic effects on root system architecture.
VIGS Vectors [5]	Virus-Induced Gene Silencing vectors for transient post-transcriptional gene knockdown.	Rapid functional validation of candidate NBS genes by testing for loss-of-resistance phenotypes in otherwise resistant plants.
Reference Genomes & Pan-Genomes [63] [5]	High-quality genome assemblies from multiple accessions or individuals of a species.	Serves as the baseline for identifying core and variable genomic regions, including CNVs and presence-absence variations in NBS clusters.
CRISPR/Cas9 System [66]	A versatile genome-editing tool for generating targeted knock-outs, knock-ins, and point mutations.	Creating stable mutant lines to confirm NBS gene function or to engineer novel pathogen recognition specificities.

The high sequence diversity and variable domain architectures of plant NBS genes are not merely genomic curiosities; they are the direct molecular record of an ongoing evolutionary arms race with pathogens. Addressing this complexity requires a multifaceted approach, integrating **comparative pan-genomics**, **powerful computational metrics** like Shannon entropy, and **robust functional validation techniques** such as VIGS. The methodologies and resources outlined in this guide provide a roadmap for researchers to navigate this challenging yet rewarding field. By systematically identifying, characterizing, and validating diverse NBS genes, scientists can unlock their potential, paving the way for deploying these critical genetic elements in breeding programs to develop crops with durable and broad-spectrum disease resistance. The future of this field lies in integrating these diverse data types to build predictive models of NLR-pathogen interactions, ultimately enabling the rational design of immune receptors.

Resolving Non-Canonical Structures and Classifying New Protein Classes (LYK, LYP, LECRK)

The study of **plant immune receptors** has traditionally focused on the major class of **Nucleotide-Binding Site Leucine-Rich Repeat (NLR) genes**, which function as intracellular sensors in effector-triggered immunity [38]. Recent genome-wide analyses have revealed remarkable diversification of these genes across plant species, with studies identifying **12,820 NBS-domain-containing genes** across 34 species from mosses to monocots and dicots, classified into **168 distinct domain architecture classes** [5]. This diversification encompasses both classical structures (NBS, NBS-LRR, TIR-NBS, TIR-NBS-LRR) and numerous non-canonical, species-specific structural patterns [5].

Within this broader context of NLR diversification, this technical guide addresses the **classification and experimental resolution** of three interrelated protein classes—LYK, LYP, and LECRK—that represent important non-canonical immune receptors. These proteins often function alongside NLRs in plant immunity networks, with LECRKs (Lectin Receptor-Like Kinases) serving as **crucial cell-surface receptors** in the first layer of plant immune perception [69] [70]. Their accurate classification and structural resolution present distinct computational and experimental challenges that this guide aims to address.

Table: Major Plant Immune Receptor Classes and Their Characteristics

Receptor Class	Domain Architecture	Localization	Primary Function	Representative Examples
NLR	NBS-LRR with TIR/CC/RPW8 N-terminal	Intracellular	Effector-triggered immunity	RPS2, N protein [38]
LECRK	Lectin domain - Transmembrane - Kinase domain	Plasma membrane	Pattern-triggered immunity	SbLLRLKs [70]
RLK/RLP	Various extracellular domains - Transmembrane - Kinase domain	Plasma membrane	Pattern recognition; signaling	OsSIK1 [70]

Protein Class Definitions and Classification Frameworks

LECRK (Lectin Receptor-Like Kinases)

LECRKs represent a **specialized class of membrane proteins** characterized by an extracellular lectin domain interconnected via a transmembrane region to an intracellular kinase domain [69]. They are categorized based on their lectin domain characteristics:

- **G-type LECRKs**: Contain glucan-binding lectin domains
- **L-type LECRKs (LLRKs)**: Feature legume-like lectin domains
- **C-type LECRKs**: Require calcium for carbohydrate binding

Genome-wide analyses have identified **32 G-type, 42 L-type, and 1 C-type LECRKs** in Arabidopsis, while rice contains **72 L-type, 100 G-type, and 1 C-type** LECRKs, demonstrating significant family expansion in monocots [69]. These genes are typically **intron-poor**, suggesting potential evolution through retrotransposition events [69].

LYK (LysM Receptor-Like Kinases)

LYKs represent a **subclass of receptor-like kinases** characterized by the presence of Lysin Motif (LysM) domains in their extracellular regions. These proteins are primarily involved in the recognition of **chitin-derived molecules** and other N-acetylglucosamine-containing ligands. While not explicitly detailed in the search results, LYKs function alongside NLRs in pathogen perception, with some acting as upstream sensors that potentially trigger NLR-mediated immunity.

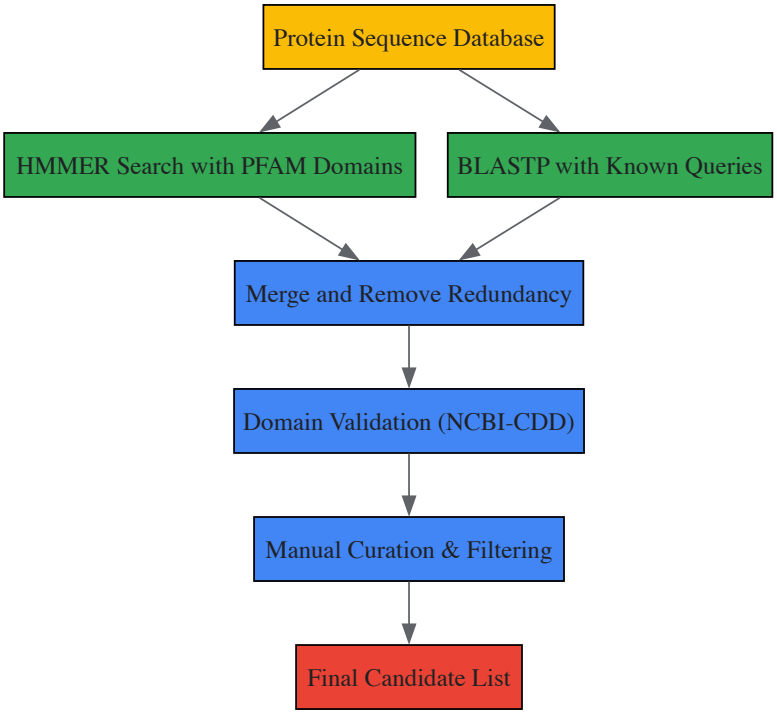
LYP (LysM Receptor-Like Proteins)

LYPs share the LysM extracellular domain structure with LYKs but **lack the intracellular kinase domain**, representing receptor-like proteins rather than receptor-like kinases. These proteins often function as **co-receptors or decoy receptors** in immune signaling pathways, modulating signal transduction through interaction with full-length receptor kinases.

Experimental Methodologies for Identification and Validation

Genome-Wide Identification Pipeline

The accurate identification of LYK, LYP, and LECRK proteins requires a multi-step bioinformatics approach, as demonstrated in recent studies [70]:



Step 1: Database Preparation Retrieve comprehensive protein sequence datasets from authoritative databases such as:

- Ensemble Plants (https://plants.ensembl.org/index.html)
- Phytozome (https://phytozome-next.jgi.doe.gov/)
- NCBI RefSeq

Step 2: Domain Identification Execute HMMER searches using relevant PFAM domain profiles:

- **Lectin_legB domains (PF00139)** for L-type LECRKs
- **Pkc_like domains (PF06176)** for kinase domain verification

- **LysM domains (PF01476)** for LYK/LYP identification

Step 3: Complementary BLAST Search Perform BLASTP searches using experimentally validated reference sequences from model organisms with an E-value threshold of ≤1e-5 [70].

Step 4: Validation and Filtering Confirm domain architecture using NCBI's Conserved Domain Database (CDD) and remove:

- Sequences lacking complete domain structures
- Redundant entries through multiple sequence alignment
- Probable pseudogenes with disrupted reading frames

Functional Validation Through Genetic Approaches

Virus-Induced Gene Silencing (VIGS) As demonstrated in NBS gene studies [5], VIGS provides an efficient method for functional characterization:

- **Gene Fragment Cloning**: Amplify 300-500 bp gene-specific fragments
- **Vector Construction**: Clone into TRV-based VIGS vectors (pTRV1, pTRV2)
- **Plant Infiltration**: Infiltrate 2-3 leaf stage seedlings using Agrobacterium
- **Phenotypic Assessment**: Monitor for enhanced susceptibility following pathogen challenge
- **Molecular Validation**: Verify silencing efficiency through qRT-PCR

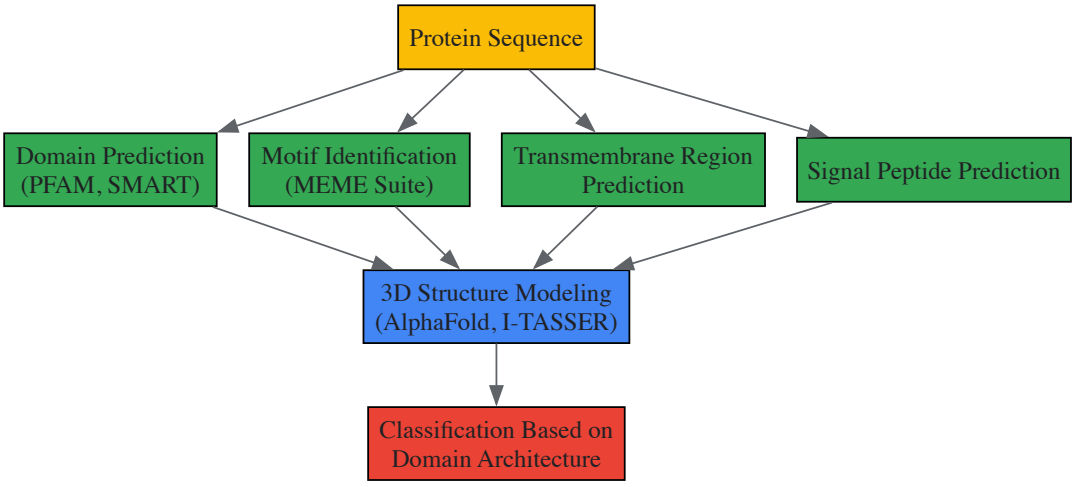
Transgenic Complementation For genes identified as negative regulators (e.g., SORBI_3004G304700 in sorghum) [70]:

- **Overexpression Constructs**: Clone full-length CDS under strong constitutive promoters
- **Knockout/Mutagenesis**: Employ CRISPR-Cas9 for targeted gene disruption
- **Phenotypic Screening**: Assess transgenic lines under stress conditions
- **Haplotype Analysis**: Identify natural variation associated with stress tolerance

Computational Structural Analysis and Classification

Domain Architecture Resolution

The classification of non-canonical immune receptors requires precise resolution of domain boundaries and arrangements. The workflow below illustrates the integrated computational approach:



Key Analysis Steps:

- **Comprehensive Domain Scanning**
  - Utilize HMMER with PFAM domain libraries
  - Confirm domains with SMART and NCBI-CDD
  - Identify non-canonical domain integrations
- **Motif Elucidation**
  - Apply MEME suite for novel motif discovery
  - Configure parameters: width 6-200 residues, maximum 20 motifs

    - Validate biological relevance through comparative analysis

- **Structural Feature Prediction**
    - Predict transmembrane helices using TMHMM
    - Identify signal peptides with SignalP
    - Model coiled-coil regions with COILS program (threshold 0.9)

- **Advanced Structure Modeling**
    - Employ AlphaFold for 3D structure prediction
    - Model conformational dynamics of different redox states
    - Identify potential functional sites through structural comparison

Orthogroup Analysis and Evolutionary Classification

The evolutionary relationships between these protein classes can be determined through orthogroup analysis:

- **OrthoFinder Analysis**: Identify orthogroups using Diamond for sequence similarity and MCL for clustering [5]
- **Phylogenetic Construction**: Generate gene trees using FastTreeMP with 1000 bootstrap replicates [5]
- **Orthogroup Classification**: Categorize into core orthogroups (shared across species) and unique orthogroups (species-specific)

Studies of NBS genes have identified **603 orthogroups** with some core (OG0, OG1, OG2) and unique (OG80, OG82) orthogroups maintained through tandem duplications [5]. Similar analysis can be applied to LYK/LYP/LECRK classification.

The Scientist's Toolkit: Essential Research Reagents

Table: Key Reagent Solutions for Protein Classification Research

Reagent/Resource	Function	Example Sources/Platforms
HMMER Software	Profile hidden Markov model searches for domain identification	http://hmmer.org/ [70]
MEME Suite	Discovery of novel amino acid motifs in protein families	https://meme-suite.org/meme/ [70]
PFAM Database	Curated collection of protein domain families	http://pfam.xfam.org/ [70]
NCBI-CDD	Conserved domain identification and validation	https://www.ncbi.nlm.nih.gov/Structure/cdd [70]
AlphaFold	Protein structure prediction from sequence	DeepMind/EMBL-EBI
OrthoFinder	Orthogroup inference and comparative genomics	https://github.com/davidemms/OrthoFinder [5]
TRV VIGS Vectors	Virus-induced gene silencing for functional validation	Arabidopsis Biological Resource Center
Clustal X	Multiple sequence alignment for phylogenetic analysis	http://www.clustal.org/clustal2/ [70]

Expression Profiling and Interaction Studies

Transcriptomic Analysis Framework

Comprehensive expression profiling provides critical functional insights:

- **Data Acquisition**: Retrieve RNA-seq data from specialized databases:
    - IPF Database (http://ipf.sustech.edu.cn/pub/) [5]
    - CottonFGD (https://cottonfgd.net/) [5]
    - NCBI BioProjects

- **Expression Categorization**: Organize data into three functional categories:
    - **Tissue-specific** (leaf, stem, root, flower)
    - **Biotic stress** (pathogen challenge, insect herbivory)
    - **Abiotic stress** (drought, salinity, temperature)

- **Analysis Pipeline**: Process with established transcriptomic workflows as detailed by Zahra et al. [5]

Protein Interaction Studies

Protein-Ligand Interaction Analysis

- Molecular docking with ADP/ATP analogs for kinase domains
- Carbohydrate-binding assays for lectin domains
- Surface plasmon resonance for binding affinity quantification

Protein-Protein Interaction Mapping

- Yeast two-hybrid screening for intracellular partners
- Co-immunoprecipitation for complex identification
- Bimolecular fluorescence complementation for subcellular localization

Integration with Broader NLR Research Context

The classification of LYK, LYP, and LECRK proteins must be considered within the broader evolution of plant immune systems. Several key connections to NLR research emerge:

Evolutionary Patterns

Studies of Solanaceae species reveal distinct evolutionary patterns for immune receptors—"consistent expansion" in potato, "first expansion and then contraction" in tomato, and a "shrinking" pattern in pepper [7]. Similar analyses should be applied to LYK/LYP/LECRK families to identify lineage-specific evolutionary trajectories.

Regulatory Networks

Emerging evidence suggests complex regulatory networks connecting cell-surface receptors (LECRKs) with intracellular NLRs. For example, RNL-class NLRs function as "helper" proteins that mediate signal transduction for sensor NLRs [38], potentially creating signaling nodes with cell-surface receptors.

Non-Canonical Protein Considerations

Recent advances in "dark proteome" research reveal that noncanonical proteins, encoded by previously overlooked genomic regions, play crucial roles in cellular processes [71]. The identification of such proteins within the LYK/LYP/LECRK families may explain additional regulatory complexity and should be considered in comprehensive classification schemes.

Improving Prediction Accuracy with Updated HMM Profiles and Tools like DRAGO3

The **Nucleotide-Binding Site (NBS) domain** represents a fundamental component of the largest class of plant disease resistance (R) genes, encoding proteins that function as intracellular immune receptors recognizing diverse pathogens [72]. The NBS gene family exhibits remarkable diversification across plant species, with significant differences in gene number, structural architecture, and evolutionary patterns between monocots and dicots [8]. This diversification presents both challenges and opportunities for developing accurate prediction tools. **Hidden Markov Models (HMMs)** have emerged as a powerful methodology for identifying and classifying these genes, but their accuracy depends heavily on the quality and breadth of the underlying profiles [56]. The continued expansion of genomic data from diverse plant species necessitates regular updates to these computational tools to capture the full spectrum of NBS gene diversity, driving the development of enhanced systems like **DRAGO3** for improved prediction accuracy in plant immunity research.

NBS Gene Diversity: A Comparative Genomic Perspective

Architectural Diversity and Classification

NBS-encoding genes are classified based on their domain architectures into several major classes. The two primary subfamilies are **TNL (TIR-NBS-LRR)** and **CNL (CC-NBS-LRR)**, distinguished by their N-terminal domains [72]. A third subclass, **RNL (RPW8-NBS-LRR)**, functions as a helper in downstream signaling [73]. Beyond these classical structures, studies have revealed numerous species-specific architectural patterns, including truncated forms and novel domain combinations [5]. Recent research has identified 168 distinct domain architecture classes across 34 plant species, encompassing both classical and unconventional patterns such as TIR-NBS-TIR-Cupin*1 and Sugar*tr-NBS [5].

Table 1: NBS Gene Distribution Across Selected Plant Species

Species	Genome Type	Total NBS Genes	TNL	CNL	RNL	Reference
Medicago truncatula	Dicot	333-500	156	177	Not specified	[72]
Ipomoea batatas (sweet potato)	Hexaploid dicot	889	Not specified	Not specified	Not specified	[73]

Species	Genome Type	Total NBS Genes	TNL	CNL	RNL	Reference
Oryza sativa (rice)	Monocot	>600	Absent	Predominant	Not specified	[8]
Arabidopsis thaliana	Dicot	~150	Present	Present	Not specified	[72]
Vigna unguiculata (cowpea)	Dicot	2188 R-genes total	Not specified	Not specified	Not specified	[74]

Evolutionary Dynamics and Genomic Distribution

The expansion and diversification of NBS genes have been driven primarily by duplication events, including **tandem duplications** and **segmental duplications** [5]. These genes are typically distributed non-randomly across plant genomes, with a strong tendency to form clusters [73]. For example, in Ipomoea species, between 76.71% and 90.37% of NBS genes occur in genomic clusters [73]. Some chromosomes exhibit extraordinary concentrations of specific NBS types, such as chromosome 6 of Medicago truncatula, which encodes approximately 34% of all TIR-NBS-LRR genes [72]. This clustering facilitates the emergence of new resistance specificities through mechanisms like unequal crossing over and ectopic recombination [72].

HMM-Based Prediction: Methodological Framework

Core Principles of HMMs for NBS Gene Identification

Hidden Markov Models are probabilistic models particularly suited for capturing conserved protein domains like the NBS. Their application to NBS gene identification leverages the characteristic conserved motifs within the NB-ARC domain, including **P-loop, RNBS-A, kinase-2, RNBS-B, RNBS-C, GLPL, RNBS-D, and MHDV** [75]. The HMM approach involves building statistical profiles of these conserved regions from multiple sequence alignments, which can then be used to detect distant homologs in genomic or transcriptomic data with greater sensitivity than pairwise methods like BLAST [56].

The standard workflow for HMM-based NBS gene identification comprises several key stages:

- **Data Collection**: Curated reference NBS protein sequences
- **Multiple Sequence Alignment**: Using tools like MUSCLE or MAFFT
- **HMM Construction**: Building profile HMMs using HMMER
- **Database Searching**: Scanning target genomes/proteomes
- **Validation**: Confirming domain architecture via complementary tools

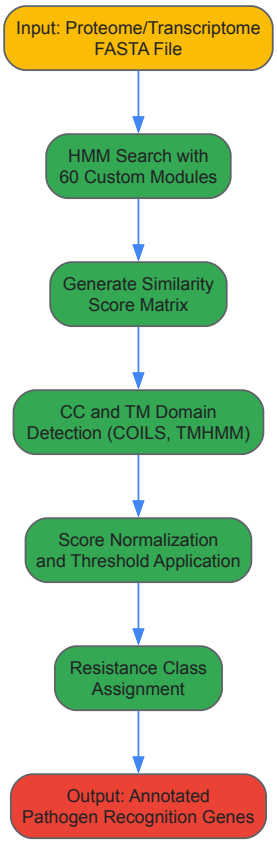
DRAGO2: An Advanced Implementation

The **DRAGO2 (Pathogen Recognition Genes Analysis and Gene Orthology)** pipeline represents a sophisticated implementation of HMM methodology specifically designed for plant resistance gene annotation [56]. This tool utilizes 60 custom HMM modules to detect domains including LRR, Kinase, NBS, and TIR, computing alignment scores based on a BLOSUM62 matrix [56]. DRAGO2 incorporates additional domain detection using COILS 2.2 for coiled-coil domains and TMHMM 2.0c for transmembrane domains, providing comprehensive architectural annotation [56].

Table 2: Key Computational Tools for NBS Gene Identification

Tool Name	Methodology	Key Features	Reference
DRAGO2	HMM-based	60 HMM modules, integrated domain prediction, orthology analysis	[56]
PRGminer	Deep Learning	Dipeptide composition features, two-phase classification	[58]
Standard HMMER	HMM-based	Pfam domain searches, customizable thresholds	[75]
Standard BLAST	Sequence alignment	Homology-based identification, rapid screening	[72]

The following diagram illustrates the complete DRAGO2 workflow for pathogen recognition gene annotation:



Experimental Protocols for HMM Optimization

Protocol 1: Construction and Optimization of HMM Profiles

Objective: To create updated, high-specificity HMM profiles for comprehensive NBS gene identification.

Materials and Reagents:

- **Reference sequences**: Curated set of experimentally validated R genes from PRGdb [56]
- **Multiple sequence alignment tool**: MUSCLE v3.6 or MAFFT v7.0 [56] [5]
- **HMM construction software**: HMMER v3 package [56] [75]
- **Validation datasets**: Proteomes from well-annotated species (e.g., Arabidopsis thaliana)

Methodology:

- **Sequence Curation and Classification**:
  - Collect protein sequences of cloned R genes from public databases
  - Classify sequences into four major categories: CNL, TNL, RLP, and RLK [56]
  - Perform multiple sequence alignment for each category using MUSCLE with default parameters [56]
- **HMM Construction**:
  - Build HMM modules using hmmbuild command from HMMER package
  - Apply filtering criteria: minimum BLOSUM62 score of +1 and minimum of 10 amino acids in length [56]
  - Validate initial HMMs by searching against the original FASTA files using hmmsearch
- **Domain-Specific HMM Refinement**:
  - Extract domain-specific sequences from multiple sequence alignments
  - Test each HMM module against domain-specific FASTA files for accurate labeling [56]
  - Use jackhmmer for unassigned modules to identify matches in UniProt database
- **Threshold Determination**:
  - Execute DRAGO2 pipeline with cloned R-gene FASTA files as input
  - Calculate normalization value as the absolute smallest similarity score across all domains [56]
  - Establish minimum score thresholds from the smallest similarity score per domain

Protocol 2: Validation and Benchmarking Framework

Objective: To validate prediction accuracy and benchmark against established methods.

Materials and Reagents:

- **Test proteomes**: Arabidopsis thaliana and Oryza sativa
- **Comparison tools**: InterProScan with PfamA-26.0 and Coils-2.2 databases [56]
- **Accuracy metrics**: Sensitivity, specificity, Matthews Correlation Coefficient (MCC)

Methodology:

- **Tool Execution**:
  - Run DRAGO2 on Arabidopsis thaliana proteome
  - Perform parallel analysis with InterProScan using default parameters [56]
- **Result Comparison**:
  - Compare domain predictions between DRAGO2 and InterProScan
  - Resolve discrepancies through manual curation and reference to literature
- **Performance Quantification**:
  - Calculate true positives, false positives, true negatives, and false negatives
  - Compute MCC, accuracy, sensitivity, and specificity metrics
  - DRAGO2 achieved MCC of 0.91 on independent testing [56]

The Scientist's Toolkit: Essential Research Reagents

Table 3: Key Research Reagents and Computational Tools for NBS Gene Analysis

Category	Specific Tool/Resource	Function/Application	Key Features
Database Resources	PRGdb 3.0	Repository of validated R genes	153 reference R genes, 177,072 candidate PRGs [56]
	Pfam Database	Protein family collection	NB-ARC domain (PF00931) for initial identification [75]
HMM Tools	HMMER Suite	HMM construction and searching	hmmbuild, hmmsearch for profile creation and application [56]
	Custom HMM Modules	Domain-specific detection	60 modules for LRR, Kinase, NBS, TIR domains [56]
Domain Prediction	COILS/PCOILS	Coiled-coil domain prediction	Probability score ≥0.9 for CC domains [75]
	TMHMM	Transmembrane domain detection	Identifies transmembrane helices [56]
Alignment Tools	MUSCLE	Multiple sequence alignment	Creates MSAs for HMM construction [56]
	MAFFT	Multiple sequence alignment	Alternative for large datasets [5]
Orthology Analysis	OrthoFinder	Orthogroup identification	Gene family evolutionary analysis [5]

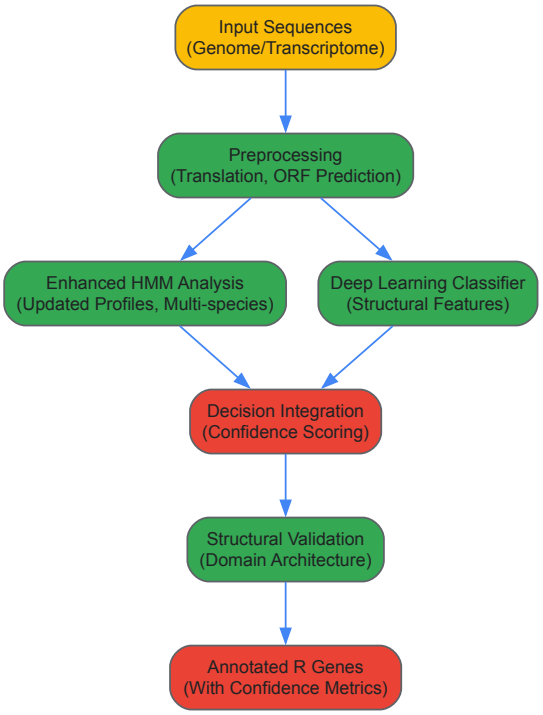
Pathway to DRAGO3: Future Development Priorities

Current Limitations and Enhancement Opportunities

While DRAGO2 represents a significant advancement, several limitations present opportunities for improvement in DRAGO3. The reliance on sequence homology, though sensitive, may miss highly divergent NBS genes [58]. Additionally, the current implementation focuses primarily on domain presence without fully incorporating spatial relationships and structural constraints. The integration of **deep learning approaches** similar to PRGminer, which achieved 95.72% accuracy using dipeptide composition features [58], could enhance prediction of non-canonical resistance genes.

Proposed DRAGO3 Architecture and Implementation

The envisioned DRAGO3 system would incorporate a hybrid architecture combining the strengths of HMM methodology with emerging machine learning techniques:



Key enhancements for DRAGO3 would include:

- **Expanded HMM Profiles**: Incorporation of newly identified NBS domain architectures from large-scale comparative studies [5] [76]
- **Deep Learning Integration**: Implementation of a classifier similar to PRGminer for detecting remote homologs [58]
- **Structural Validation**: Incorporation of protein structure prediction to validate domain arrangements
- **Cross-Species Orthology Mapping**: Enhanced orthology prediction to facilitate comparative genomics studies [5]

The accuracy of NBS gene prediction is fundamentally linked to the diversity of the underlying training data and the sophistication of the computational methods employed. The ongoing diversification of NBS genes across plant species, as revealed by comparative genomic studies, necessitates continuous refinement of HMM profiles and analytical tools [5] [76]. The DRAGO framework represents a robust platform for this purpose, with the proposed DRAGO3 enhancements offering the potential for substantially improved prediction accuracy. As genomic data continue to expand, such computational advances will be crucial for unlocking the full diversity of plant resistance genes and harnessing them for crop improvement strategies.

Strategies for Linking Genetic Variation to Phenotypic Resistance

The diversification of nucleotide-binding site (NBS) domain genes represents a fundamental evolutionary strategy for plant adaptation against rapidly evolving pathogens. These genes, particularly those encoding NBS-leucine-rich repeat (NLR) proteins, constitute the largest and most versatile class of intracellular immune receptors in plants, capable of recognizing diverse pathogen effectors to trigger robust immune responses [5] [77]. The extensive genetic variation within NBS-encoding gene families across plant species creates both a challenge and opportunity for researchers seeking to link specific genetic polymorphisms to phenotypic resistance outcomes. Understanding these relationships is critical for developing durable disease resistance in crops, particularly as pathogens continue to evolve new virulence mechanisms.

Plant immunity operates through a sophisticated two-tiered system where cell surface receptors detect pathogen-associated molecular patterns (PAMPs) to activate PAMP-triggered immunity (PTI), while intracellular NLR receptors mediate effector-triggered immunity (ETI) through recognition of specific pathogen effectors [77] [78]. The NBS domain serves as a critical molecular switch within NLR proteins, hydrolyzing ATP to initiate conformational changes that activate downstream defense signaling [79]. Recent structural studies have revealed that NLRs can assemble into resistosome complexes that trigger calcium influx and programmed cell death at infection sites, providing crucial mechanistic links between genetic variation and phenotypic resistance [77].

Identification and Classification of NBS Domain Genetic Variation

Genome-Wide Identification Approaches

The first critical step in linking genetic variation to phenotypic resistance involves comprehensive identification and annotation of NBS-encoding genes across target species. Hidden Markov Model (HMM) profiles derived from conserved domain databases (e.g., PF00931 for NB-ARC domains)

provide the most reliable method for systematic identification [5] [11] [79]. Typical workflows involve HMMER searches against target genomes with stringent E-value thresholds (e.g., <1e-20) followed by domain architecture validation using PfamScan, SMART, and CDD tools [5] [11]. This approach identified 12,820 NBS-domain-containing genes across 34 plant species in a recent pan-genomic study, revealing significant diversity from bryophytes to higher plants [5].

Advanced computational tools like PRGminer now leverage deep learning algorithms to improve R-gene prediction accuracy, achieving up to 98.75% accuracy in distinguishing resistance genes from non-resistance genes through dipeptide composition analysis [58]. This is particularly valuable for identifying atypical NBS-domain architectures that may be missed by traditional homology-based approaches.

Classification of NBS Domain Architectures

NBS-encoding genes display remarkable structural diversity, which can be systematically classified based on domain architecture:

Table 1: Classification of NBS Domain Gene Architectures

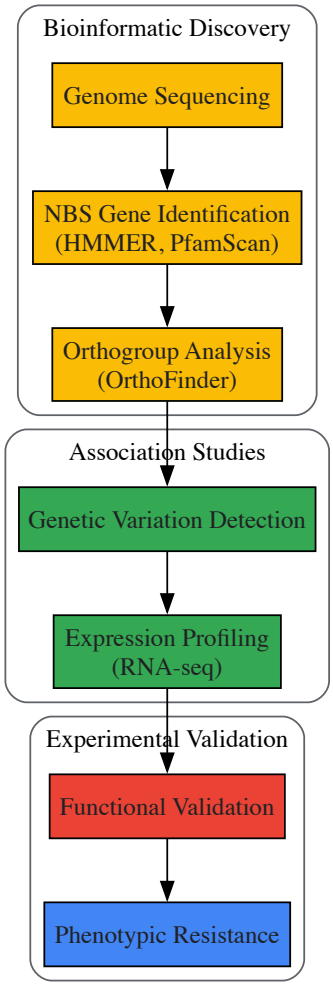
Architecture Type	Domain Composition	Functional Role	Species Examples
TNL	TIR-NBS-LRR	Pathogen recognition, resistosome formation	Arabidopsis, Tobacco [11] [77]
CNL	CC-NBS-LRR	Pathogen recognition, resistosome formation	Rice, Wheat [77] [60]
RNL	RPW8-NBS-LRR	Helper NLR, signaling transduction	Solanaceae species [5] [77]
TN	TIR-NBS	Regulatory/adaptor functions	Nicotiana benthamiana [11]
CN	CC-NBS	Regulatory/adaptor functions	Nicotiana benthamiana [11]
NL	NBS-LRR	Pathogen recognition	Multiple species [11]
N	NBS only	Regulatory functions	Multiple species [11]
Atypical NLRs	Integrated domains (WRKY, HMA, zf-BED)	Expanded recognition specificity	Rice (XA1, XA14) [77]

Comparative analyses across species reveal intriguing evolutionary patterns. For instance, TNL subfamilies show marked reduction or complete loss in monocot species like rice and wheat, while undergoing significant expansion in gymnosperms like Pinus taeda [79]. Similarly, medicinal plants like Salvia miltiorrhiza display substantial contraction of TNL and RNL subfamilies, with only 2 TIR-domain-containing proteins identified among 196 NBS-encoding genes [79].

Methodological Framework for Connecting Genotype to Phenotype

Orthogroup Analysis and Evolutionary Studies

Orthogroup (OG) analysis using tools like OrthoFinder provides a powerful framework for tracing the evolutionary relationships among NBS-encoding genes across species [5]. This approach groups genes into orthologous clusters based on phylogenetic relationships, enabling identification of core conserved orthogroups (e.g., OG0, OG1, OG2) versus species-specific expansions. In a comprehensive analysis of 34 plant species, researchers identified 603 orthogroups, with certain core OGs showing conserved expression patterns across taxonomic boundaries [5]. Tandem duplication events represent a major driver of NBS gene diversification, creating clusters of closely related genes that undergo neofunctionalization to recognize evolving pathogen effectors [5] [58].



Expression Profiling and Transcriptomic Analysis

Gene expression signatures provide crucial intermediate phenotypes connecting genetic variation to resistance outcomes. Recent studies reveal that functional NLRs frequently exhibit high steady-state expression levels in uninfected plants, contrary to the historical assumption that NLR expression must be tightly repressed [60]. In fact, known functional NLRs are significantly enriched among the top 15% of highly expressed NLR transcripts across multiple species, suggesting that expression level can serve as a predictive signature for functional NLR identification [60].

RNA-seq analysis of NBS genes across tissues, developmental stages, and stress conditions provides critical insights into functional specialization. For example, comprehensive expression profiling of orthogroups in cotton revealed that OG2, OG6, and OG15 show upregulated expression in different tissues under various biotic and abiotic stresses in both susceptible and tolerant cotton accessions [5]. Tissue-specific expression patterns are particularly informative, as demonstrated by the helper NLR NRC6, which shows root-specific expression in tomato cultivars despite its low expression in leaves [60].

Table 2: Genomic and Transcriptomic Approaches for NBS Gene Characterization

Method	Key Applications	Technical Considerations	Representative Findings
RNA-seq Expression Profiling	Tissue-specific expression, stress responsiveness, identification of highly expressed NLRs	Normalize by FPKM/TPM; include multiple biological replicates; relevant tissue selection	Functional NLRs enriched in top 15% of expressed transcripts [60]
Orthogroup Analysis	Evolutionary conservation, functional inference, cross-species comparisons	Use OrthoFinder with MCL clustering; include diverse species representatives	603 orthogroups identified across 34 species with core conserved OGs [5]
GWAS	Linking natural variation to resistance phenotypes, candidate gene identification	High-density SNP markers; diverse germplasm; appropriate statistical models	Soybean PRSR resistance associated with chromosome 3 region containing NBS-LRR genes [80]
Promoter cis-Element Analysis	Identification of regulatory motifs, understanding expression patterns	Analyze 1.5kb upstream regions; use PlantCARE database; validate	29 shared cis-elements identified in NBS-LRR promoters with stress-responsive

Method	Key Applications	Technical Considerations	Representative Findings
		experimentally	motifs [11]

Genetic Mapping and Association Studies

Genome-wide association studies (GWAS) and quantitative trait locus (QTL) mapping provide powerful approaches for linking natural genetic variation to resistance phenotypes. In soybean, GWAS of 205 accessions using a 180K SNP array identified 19 significant SNPs associated with resistance to Phytophthora sojae, with a key region on chromosome 3 containing multiple NBS-LRR genes and serine-threonine protein kinases [80]. Haplotype analysis further refined these associations, identifying Glyma.03g036500 as a strong candidate gene with expression patterns correlating with resistance phenotypes [80].

Genetic variation analysis between susceptible and tolerant accessions can reveal functionally significant polymorphisms. In Gossypium hirsutum, comparison of Coker 312 (susceptible) and Mac7 (tolerant) accessions identified 6,583 unique variants in NBS genes of the tolerant line versus 5,173 in the susceptible line, highlighting the potential contribution of these polymorphisms to resistance differences [5].

Experimental Validation of Functional Resistance

Protein-Ligand and Protein-Protein Interaction Studies

Validating the molecular mechanisms through which NBS domain genes confer resistance requires direct experimental evidence of protein interactions. Protein-ligand interaction assays demonstrate that functional NBS domains bind ATP/ADP, with the nucleotide-bound state regulating activation status [5] [11]. Protein-protein interaction studies further reveal that resistant NBS proteins can directly bind pathogen effectors or interact with other components of the immune signaling cascade. For cotton leaf curl disease, molecular docking approaches showed strong interactions between putative NBS proteins and core proteins of the cotton leaf curl disease virus, providing mechanistic insights into recognition specificity [5].

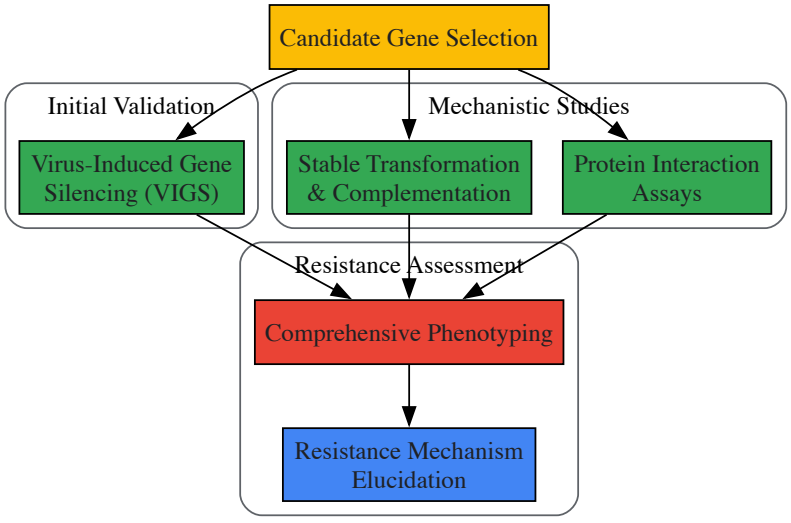
Functional Genetic Approaches

Several established functional genomic approaches provide direct evidence for gene function in disease resistance:

Virus-Induced Gene Silencing (VIGS): VIGS enables rapid functional characterization of candidate NBS genes in species with established transformation protocols. Silencing of GaNBS (OG2) in resistant cotton demonstrated its essential role in limiting virus titer, confirming its function in cotton leaf curl disease resistance [5].

Transgenic Complementation: Stable transformation with candidate NLR genes can validate function through complementation of susceptible genotypes. Interestingly, some NLRs require multiple copies for full resistance, as demonstrated with barley Mla7, where single-copy transgenics failed to confer resistance while multicopy lines showed strong resistance to Blumeria hordei [60]. This challenges conventional assumptions about NLR expression thresholds and has important implications for engineering resistance.

High-Throughput Transformation Arrays: Recent advances enable systematic functional screening of NLR libraries. A groundbreaking approach expressing 995 NLRs from diverse grass species in wheat identified 31 new resistance genes (19 against stem rust, 12 against leaf rust), demonstrating the power of high-throughput functional screening [60].



The Scientist's Toolkit: Essential Research Reagents and Solutions

Table 3: Essential Research Reagents for NBS Gene Functional Analysis

Reagent Category	Specific Examples	Applications and Functions
Bioinformatic Tools	HMMER, OrthoFinder, PRGminer, MEME, PlantCARE	Domain identification, phylogenetic analysis, motif discovery, promoter element prediction [5] [11] [58]
Expression Analysis Platforms	RNA-seq libraries, qPCR assays, Promoter-reporter constructs	Expression profiling, tissue-specific localization, stress responsiveness [5] [60]
Genetic Transformation Systems	VIGS vectors, Agrobacterium strains, High-throughput transformation protocols	Functional gene validation, complementation assays, large-scale screening [5] [60]
Protein Interaction Assays	Yeast two-hybrid systems, Co-immunoprecipitation kits, Molecular docking software	Protein-protein interactions, pathogen effector recognition, resistosome formation [5] [77]
Phenotyping Resources	Pathogen isolates, Disease scoring systems, Growth facilities	Resistance assessment, symptom development, quantitative trait measurement [5] [80]

Linking genetic variation to phenotypic resistance requires integrated approaches that combine comparative genomics, expression profiling, genetic mapping, and experimental validation. The extensive diversification of NBS domain genes across plant species represents a rich source of variation for engineering disease resistance in crops. By employing the systematic strategies outlined in this technical guide—from initial genome-wide identification through orthogroup analysis to functional validation—researchers can accelerate the discovery and deployment of effective resistance genes. The continuing development of high-throughput methods for NLR identification and validation, coupled with advanced genome editing technologies for precise modification of both R genes and susceptibility genes, promises to revolutionize crop improvement for durable disease resistance [60] [78]. As these approaches mature, they will increasingly enable researchers to not only understand the link between genetic variation and phenotypic resistance but also to engineer these relationships for sustainable crop protection.

Optimizing High-Throughput Functional Screens in Complex Plant Systems

High-throughput functional screening represents a transformative approach for interrogating gene function in complex plant systems. Within the context of nucleotide-binding site (NBS) domain gene research, these methodologies enable researchers to systematically analyze the extensive diversification of this critical gene family across plant species. NBS domain genes constitute one of the largest resistance (R) gene superfamilies, encoding proteins central to plant immune responses against pathogens [5]. Recent comparative analyses have identified 12,820 NBS-domain-containing genes across 34 plant species, classifying them into 168 distinct classes with both canonical and species-specific architectural patterns [5]. This remarkable diversity presents both a challenge and opportunity for functional characterization, necessitating sophisticated screening platforms that can efficiently link genetic diversity to biological function. The optimization of high-throughput functional screens is therefore paramount for elucidating the mechanistic roles of diversified NBS genes in plant immunity, stress adaptation, and evolutionary success.

Core Principles of High-Throughput Screening in Plant Systems

Phenotype-Driven Screening Strategies

High-throughput screening in plant systems employs two complementary approaches: phenotype-based screening and target-directed screening. Phenotype-based screens offer an unbiased alternative to classical genetic approaches, allowing researchers to identify small molecules that induce specific physiological responses without prior knowledge of their molecular targets [81]. This strategy is particularly valuable for studying NBS gene function, as it can circumvent the functional redundancy often present in large gene families and avoid the lethal effects of essential gene disruption through conditional, reversible, and dosage-dependent perturbation of biological systems [81]. The fundamental advantage of phenotype-based screening lies in its capacity to reveal novel gene functions and genetic interactions without predetermined hypotheses about molecular mechanisms.

Differential Genetic Screening Frameworks

Differential genetic screening represents a powerful enhancement to conventional phenotypic screening, enabling direct comparison of multiple genotypes within primary screens. This approach utilizes isogenic plant lines differing only at specific genetic loci of interest—such as DNA repair mutants or NBS gene variants—to identify chemical compounds or genetic interactions that produce genotype-specific phenotypes [81]. In practice, wild-type and mutant Arabidopsis seedlings are grown separately in microtiter plates containing small molecules, with internal positive and negative controls establishing thresholds for altered versus healthy phenotypes [81]. This differential framework significantly improves screening efficiency by simultaneously eliminating general growth effectors while highlighting genetic context-specific interactions, making it particularly suitable for dissecting the functional contributions of specific NBS gene variants to disease resistance pathways.

Experimental Design Considerations for Plant Systems

Robust high-throughput screening in plants requires careful optimization of growth conditions and experimental parameters. Key considerations include:

- **Growth medium selection**: Liquid media often induce more pronounced growth phenotypes compared to solid media, enhancing detection sensitivity for subtle phenotypic variations [81].
- **Vessel format**: Twenty-four-well plates typically provide superior plant growth and imaging compatibility compared to 96-well formats, with documented improvements in true leaf development (3.6 vs. 2.2 leaves per plant) and better visualization of roots and leaves at a single focal point [81].
- **Replication strategy**: Growing multiple seedlings per well (typically three) provides biological replicates within each experimental unit while accounting for potential germination variability [81].
- **Environmental control**: Consistent light availability and gas exchange are critical for photosynthetic microorganisms and must be rigorously maintained throughout screening procedures [82].

Table 1: Key Experimental Parameters for High-Throughput Plant Screening

Parameter	Optimal Condition	Impact on Screening Quality
Culture Format	Liquid medium	Enhanced phenotypic resolution for growth alterations
Plate Format	24-well plates	Improved plant development (3.6 vs. 2.2 true leaves) and imaging fidelity
Replication	3 seedlings per well	Biological redundancy with minimal germination failure impact
Light Intensity	50-500 μmol m^{-2} s^{-1}	Maintains consistent photosynthetic activity without light limitation
Light Uniformity	<5% variability across platform	Minimizes position effects on growth rate and gene expression

Technical Implementation of High-Throughput Screening Platforms

Automated Imaging and Phenotypic Analysis

Advanced image processing pipelines constitute the technological foundation of modern high-throughput plant screening, enabling quantitative assessment of plant growth and development at scale. Convolutional neural networks (CNNs) have revolutionized this domain through their capacity for automated feature extraction from raw image data, dramatically accelerating the analysis of large chemical libraries [81]. Residual neural network (ResNet) architectures have demonstrated particular efficacy for classifying seedling images into normal or altered growth categories with up to 100% accuracy in controlled conditions [81]. These systems can be further enhanced through complementary segmentation approaches that separately quantify root and aerial structures (leaves and hypocotyl), providing multidimensional phenotypic profiles from a single imaging session. The integration of these machine learning tools has transformed previously qualitative morphological assessments into rigorous, quantitative datasets suitable for statistical analysis and hypothesis testing.

Integrated Lighting Systems for Photosynthetic Organisms

Specialized illumination systems represent a critical engineering consideration for high-throughput screening of photosynthetic organisms like plants and algae. Custom-designed LED arrays that maintain consistent light intensity and spectrum across cultivation platforms are essential for reproducible experimental outcomes [82]. Optimal systems should provide even illumination adjustable between 50-500 μmol m^{-2} s^{-1} across all sample positions, with less than 5% variability in light intensity to minimize position effects on growth rates [82]. Protein economy models of cyanobacteria indicate that the most significant metabolic variability occurs under light-limiting conditions (<100 μmol m^{-2} s^{-1}), while higher intensities yield more consistent growth rates and metabolic activities [82]. These lighting systems must conform to standard automation form factors (e.g., Society for Laboratory Automation and Screening specifications) for integration into robotic incubators and handling systems, enabling parallel processing of hundreds to thousands of individual cultures under precisely controlled conditions.

Data Processing and Quantitative Analysis Frameworks

The statistical rigor of high-throughput screening outcomes depends on appropriate data processing and analysis methodologies. For within-individual comparisons—where the same quantitative variable is measured multiple times on each experimental unit—case-profile plots effectively visualize temporal patterns and response trajectories [83]. When comparing two observations per individual (e.g., pre- and post-treatment), calculating difference scores for each plant followed by construction of histogram distributions provides robust visualization of response variability [83]. Numerically, the mean difference and standard deviation of differences should be computed directly from the individual change scores rather than derived from summary statistics of the separate measurements [83]. This approach preserves the paired nature of the data and provides accurate estimates of treatment effects. For multi-group comparisons, analysis of variance (ANOVA) with appropriate follow-up tests (e.g., F-protected least significant difference or Tukey's honestly significant difference) maintains appropriate type I error rates while enabling specific hypothesis testing [84].

Application to NBS Domain Gene Functional Analysis

Expression Profiling and Orthogroup Validation

High-throughput screening methodologies have been successfully applied to characterize the functional roles of NBS domain genes in plant immunity and stress responses. Expression profiling across orthogroups—evolutionarily related gene sets descended from a common ancestor—has revealed distinct patterns of regulation in response to biotic and abiotic challenges [5]. Orthogroup-based classification of NBS genes has identified 603 distinct groups, with certain core orthogroups (OG0, OG1, OG2) demonstrating conserved functions across species, while unique orthogroups (OG80, OG82) exhibit species-specific specialization [5]. Functional validation through virus-induced gene silencing (VIGS) of specific NBS genes (e.g., GaNBS in OG2) has confirmed their essential roles in pathogen response, particularly in reducing viral titers in resistant cotton plants challenged with cotton leaf curl disease [5]. These systematic approaches demonstrate how high-throughput functional screening can bridge the gap between gene sequence diversity and biological function.

Genetic Variation Mapping in Tolerant and Susceptible Accessions

Comparative analysis of genetic variation in NBS genes between disease-tolerant and susceptible plant accessions provides powerful insights into resistance mechanisms. In Gossypium hirsutum, comprehensive variant identification has revealed 6,583 unique variants in tolerant (Mac7) accessions compared to 5,173 in susceptible (Coker 312) lines [5]. These natural variations, when coupled with protein-ligand and protein-protein interaction studies, demonstrate specific binding affinities between NBS proteins and cotton leaf curl disease virus components [5]. High-throughput screening platforms enable systematic evaluation of how these genetic variations influence molecular interactions and ultimately determine resistance phenotypes, providing a functional roadmap for prioritizing candidate genes for crop improvement programs.

Table 2: Essential Research Reagents for High-Throughput NBS Gene Screening

Reagent/Condition	Function in Screening	Application Example
Prestwick Chemical Library	Off-patent drug collection for phenotype induction	Identification of genotype-specific growth effectors [81]
Virus-Induced Gene Silencing (VIGS)	Transient gene knockdown validation	Functional testing of GaNBS (OG2) in cotton leaf curl disease resistance [5]
Orthogroup Classification	Evolutionary relationship mapping	Functional comparison of 603 NBS gene groups across species [5]
Differential Growth Media	Phenotypic enhancement	Liquid media for robust growth inhibition detection [81]
Custom LED Illumination	Controlled photosynthetic conditions	Uniform light intensity (50-500 μmol m^{-2} s^{-1}) for consistent growth [82]

Experimental Protocols for High-Throughput NBS Gene Characterization

Protocol 1: Differential Chemical Genetic Screening

This protocol outlines a high-throughput phenotype-directed chemical screening method for identifying small molecules that produce genotype-specific effects in plant systems:

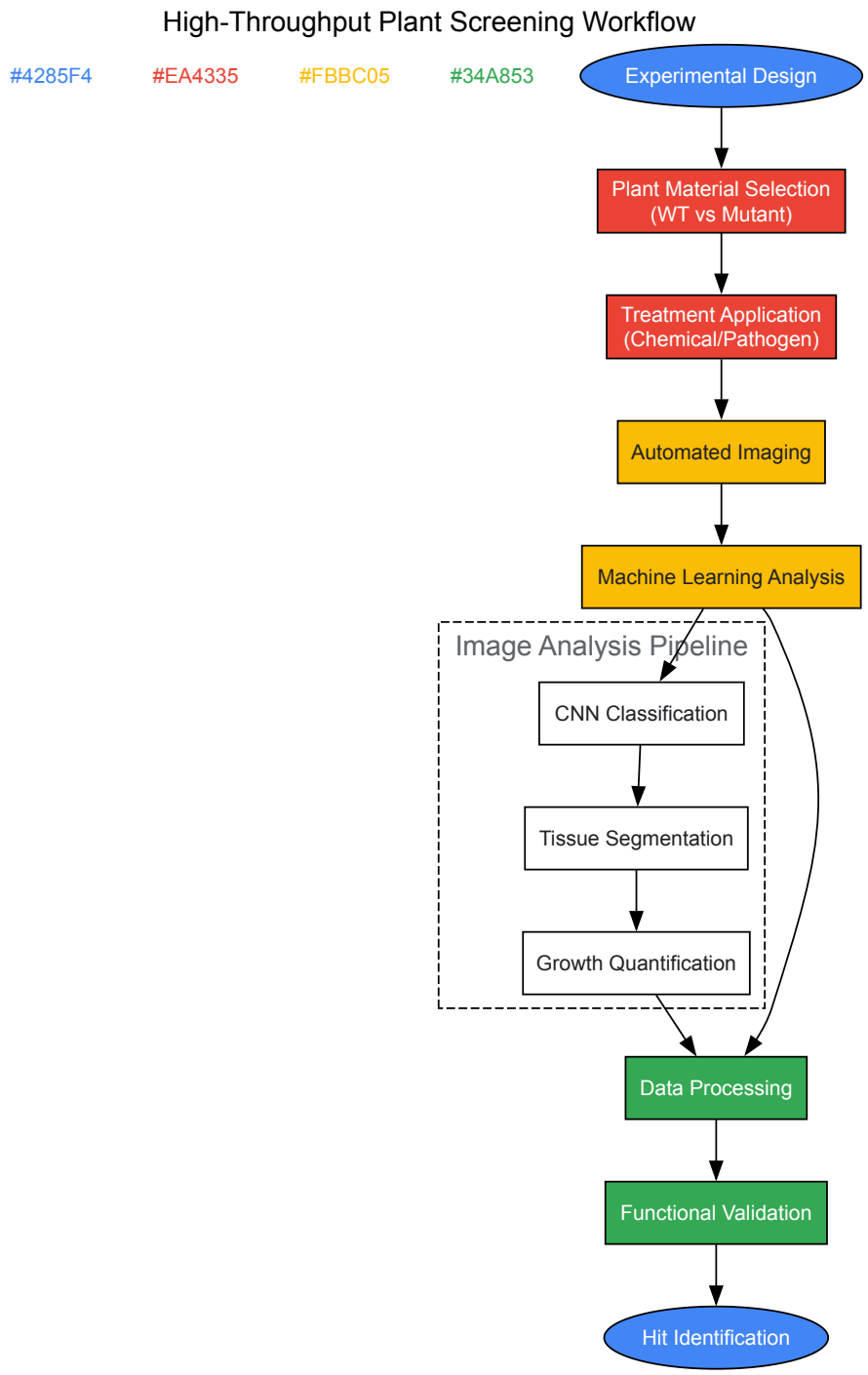
- **Plant material preparation**: Sow sterilized seeds of wild-type and mutant genotypes (e.g., mus81 DNA repair mutant) in separate 24-well microtiter plates containing liquid growth medium.
- **Chemical treatment**: Add small molecules from screening libraries (e.g., Prestwick library) to respective wells, including DMSO negative controls and mitomycin C positive controls for altered growth phenotypes.
- **Growth conditions**: Incubate plates under controlled environmental conditions (photoperiod, temperature, humidity) with consistent illumination (50-500 μmol m^{-2} s^{-1}) for 10-14 days.
- **Image acquisition**: Capture high-resolution images of seedlings in each well using a light macroscope, ensuring consistent focal points for roots and aerial structures.
- **Machine learning analysis**: Process images through pre-trained convolutional neural networks (ResNet architecture) to classify seedlings as normal or altered growth based on probability thresholds.
- **Hit identification**: Compare growth responses between genotypes to identify compounds producing differential effects, with confirmation through follow-up dose-response experiments [81].

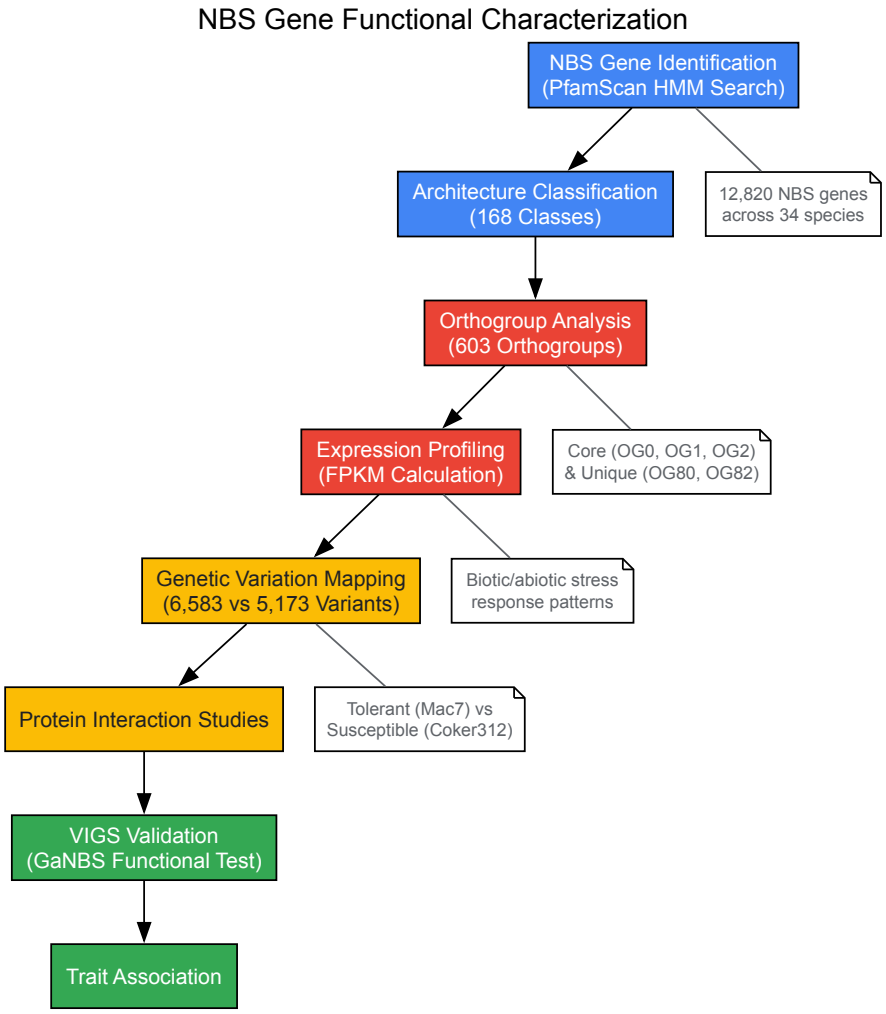
Protocol 2: NBS Gene Expression Profiling Under Stress Conditions

This protocol describes transcriptomic analysis of NBS gene expression patterns in response to biotic and abiotic stresses:

- **Treatment application**: Expose susceptible and tolerant plant accessions to pathogen inoculation (e.g., cotton leaf curl virus) or environmental stresses (drought, salinity, temperature extremes).
- **Tissue collection**: Harvest root, leaf, stem, and floral tissues at multiple time points post-treatment (0, 6, 12, 24, 48, 72 hours) with biological replication.
- **RNA extraction and sequencing**: Isolve total RNA using column-based purification methods, assess quality, and prepare RNA-seq libraries for Illumina sequencing.
- **Data processing**: Calculate Fragments Per Kilobase of transcript per Million mapped reads (FPKM) values from raw sequencing data to quantify gene expression levels.
- **Orthogroup analysis**: Classify expressed NBS genes into orthogroups using OrthoFinder v2.5.1 with DIAMOND for sequence similarity searches and MCL for clustering.
- **Expression visualization**: Generate heatmaps of NBS gene expression patterns across tissues, time points, and treatment conditions to identify stress-responsive orthogroups [5].

Workflow Visualization





The continuing optimization of high-throughput functional screens promises to dramatically accelerate the characterization of diversified NBS domain genes across plant species. Integration of advanced machine learning platforms with automated cultivation systems creates unprecedented capacity for linking genetic diversity to biological function at scale. Future methodological developments will likely focus on enhancing three-dimensional phenotyping capabilities, integrating multi-omics data streams, and establishing more sophisticated genotype-phenotype prediction models. These technological advances, applied within the framework of differential genetic screening, will ultimately illuminate the evolutionary mechanisms driving NBS gene diversification and enable targeted harnessing of these critical genetic elements for crop improvement and sustainable agriculture. The functional validation of NBS genes through optimized screening platforms represents a crucial step toward understanding plant adaptation mechanisms and developing durable disease resistance strategies in a changing global environment.

Bridging Discovery and Application: Functional Validation and Cross-Kingdom Insights

Virus-Induced Gene Silencing (VIGS) has emerged as a powerful reverse genetics tool for rapid functional genomic analysis in plants. This technology leverages the plant's innate RNA-based antiviral defense mechanism to silence target genes of interest. Within the broader context of researching the diversification of Nucleotide-Binding Site (NBS) domain genes across plant species, VIGS provides an indispensable methodology for validating the function of candidate genes identified through genomic studies. This technical guide details the application, protocols, and key case studies of VIGS in cotton, with additional insights from related species, providing a framework for researchers investigating plant resistance gene evolution and function.

VIGS Technology: Core Principles and Vectors

Mechanism of Action

VIGS operates through **Post-Transcriptional Gene Silencing (PTGS)**, an RNA-mediated defense mechanism [85] [86]. When a recombinant viral vector carrying a fragment of a host plant gene infects the plant, the plant's immune system recognizes and degrades the viral RNA. This process generates **small interfering RNAs (siRNAs)** that guide the sequence-specific degradation of complementary endogenous mRNA transcripts,

leading to knocked-down expression of the target gene [87] [85]. This knockdown allows researchers to observe resulting phenotypes and infer gene function.

Established VIGS Vectors for Cotton

Two primary viral vector systems have been successfully deployed for gene silencing in cotton:

1. Tobacco Rattle Virus (TRV)-Based Vectors:

- **Structure:** Comprises two components: **TRV1** (encoding replication and movement proteins) and **TRV2** (encoding coat protein and hosting the insert fragment) [85] [86].
- **Advantages:** Offers high silencing efficiency, long duration, mild viral symptoms, and effective silencing in meristems and various tissues [87] [88]. It is the most widely used VIGS system in cotton.

2. Cotton Leaf Crumple Virus (CLCrV)-Based Vectors:

- **Structure:** A bipartite begomovirus (single-stranded DNA) with **DNA-A** (replication-associated genes) and **DNA-B** (movement proteins) components [85] [86].
- **Application:** First reported for VIGS via particle bombardment and useful for specific applications where RNA viruses are less effective [85].

Table 1: Comparative Analysis of Primary VIGS Vectors Used in Cotton

Vector Type	Genome Type	Key Components	Advantages	Primary Delivery Method
TRV	RNA virus	TRV1, TRV2	High efficiency, mild symptoms, broad tissue range	Agrobacterium infiltration
CLCrV	DNA virus	DNA-A, DNA-B	-	Particle bombardment, Agrobacterium

Visible Marker Genes for Silencing Efficiency

Monitoring silencing efficiency is crucial for successful VIGS experiments. Several visible marker genes are used as positive controls, each with distinct advantages and limitations.

Table 2: Visible Marker Genes for Monitoring VIGS Efficiency in Cotton

Marker Gene	Biological Function	Silencing Phenotype	Limitations/Advantages
CLA1	Chloroplast development	Leaf albinism, wilting, plant death	Lethal; not suitable for long-term studies [87]
PDS	Carotenoid biosynthesis	Photobleaching of tissues	Lethal; not suitable for long-term studies [85] [86]
GoPGF/PGF	Pigment gland formation	Reduced pigment gland number	Non-lethal; ideal for tracing silencing throughout lifecycle [87] [85]
ANS	Anthocyanin biosynthesis	Brownish plant phenotype	Non-lethal; mild marker [85] [86]

The **GoPGF** gene is a particularly advanced marker. Its silencing results in a visible reduction of pigmented gossypol glands without affecting plant viability, enabling researchers to monitor silencing efficacy from seedling stages through to boll development and fiber maturation [87] [85]. This is a significant improvement over early markers like *CLA1* and *PDS*, whose silencing causes lethal photobleaching.

Experimental VIGS Protocols in Cotton

Standard Agrobacterium-Mediated Cotyledon Infiltration

This is the most common method for VIGS delivery in cotton [88].

Detailed Methodology:

- **Vector Construction:** Clone a 200-300 bp fragment of the target gene into the **TRV2** vector using homologous recombination or traditional restriction-ligation.
- **Agrobacterium Preparation:**
  - Transform recombinant **pTRV1** and **pTRV2** (with insert) vectors separately into *Agrobacterium tumefaciens* strain **GV3101**.

    - Grow individual cultures overnight in LB medium with appropriate antibiotics (e.g., kanamycin, rifampicin).
- **Agro-infiltration Culture Preparation:**
    - Pellet bacterial cultures by centrifugation and resuspend in infiltration buffer (10 mM $MgCl_2$, 10 mM MES, 200 μM acetosyringone).
    - Adjust the OD_{600} of each culture to **1.5**.
    - Mix the pTRV1 and pTRV2 cultures in a **1:1** ratio and incubate at room temperature for 3-4 hours.
- **Plant Infiltration:**
    - Select cotton seedlings at the cotyledon expansion to two true-leaf stage.
    - Using a needleless syringe, infiltrate the mixed Agrobacterium culture into the abaxial side of the cotyledons.
- **Post-Infiltration Care:**
    - Maintain infiltrated plants in a growth chamber at **23°C** with a **16/8-hour light/dark** photoperiod.
    - Observe for the appearance of silencing phenotypes (e.g., glandless for *GoPGF*) in new leaves 2-3 weeks post-infiltration [87] [88].

Seed Soak Agroinoculation (SSA-VIGS)

For functional studies in very young seedlings and root tissues, a novel seed soak method has been developed [89].

Detailed Methodology:

- **Seed Preparation:** Delint cotton seeds to create "naked seeds."
- **Agrobacterium Inoculum:** Prepare Agrobacterium cultures carrying TRV vectors as described above, resuspending to an OD_{600} of **1.5** in infiltration buffer.
- **Inoculation:** Soak the naked seeds in the Agrobacterium culture for **90 minutes** with gentle agitation.
- **Germination and Growth:** Sow the treated seeds directly in soil or growth medium. Silencing phenotypes can be observed in emerging leaves and roots as early as **12-14 days** post-inoculation [89].

This method is particularly valuable for investigating genes involved in early seedling development and root biology, such as those responding to abiotic stresses.

Diagram 1: VIGS Experimental Workflow in Cotton (2/4)

Case Studies in Functional Validation

Case Study 1: Validating NBS Gene Function in Viral Resistance

Background: NBS domain genes are a major class of plant disease resistance (R) genes. A genome-wide study identified numerous NBS genes, and their diversification is a key research focus [90].

VIGS Application:

- The candidate NBS gene **GaNBS (OG2)** was silenced via VIGS in a CLCuD-resistant cotton genotype to confirm its functional role.
- **Method:** The standard TRV-VIGS protocol was used to silence *GaNBS*.
- **Result:** Silenced plants showed a significant increase in viral titer compared to controls, demonstrating that *GaNBS* is essential for resistance to Cotton Leaf Curl Disease [90]. This functional validation directly links a specific NBS gene from diversification studies to a concrete resistance phenotype.

Case Study 2: Elucidating Abiotic Stress Response

Gene: *GhBI-1* (Bax Inhibitor-1), implicated in salt stress response [89]. **VIGS Application:**

- **Method:** The SSA-VIGS protocol was employed to silence *GhBI-1* in cotton seedlings.
- **Functional Analysis:** After silencing, plants were exposed to salt stress. *GhBI-1*-silenced plants exhibited heightened sensitivity and more severe stress-induced cell death symptoms.
- **Conclusion:** VIGS validated that *GhBI-1* plays a protective role by suppressing cell death under salt stress [89].

Gene: *GhANK169*, a gene upregulated during heat stress [91]. **VIGS Application:**

- **Method:** Silenced via standard VIGS.
- **Functional Analysis:** Under heat stress (42°C), silenced plants showed poor thermotolerance, characterized by accelerated water loss, elevated reactive oxygen species (ROS), and reduced antioxidant capacity.
- **Conclusion:** *GhANK169* was confirmed as a critical positive regulator of heat tolerance [91].

Case Study 3: Investigating Reproductive Development

Gene: *GhDnaJ316*, a DnaJ family gene with preferential expression in anthers and filaments [92]. **VIGS Application:**

- **Method:** Silenced via standard VIGS.
- **Phenotype:** Silenced plants exhibited accelerated floral transition, budding **7.7 days** earlier and flowering **9.7 days** earlier than control plants.
- **Conclusion:** VIGS uncovered the role of *GhDnaJ316* as a negative regulator of flowering time, providing insights for breeding early-maturing varieties [92].

The Scientist's Toolkit: Essential Research Reagents

Table 3: Key Research Reagent Solutions for VIGS in Cotton

Reagent / Material	Specification / Example	Critical Function in VIGS
VIGS Vectors	pTRV1, pTRV2 (TRV system); DNA-A, DNA-B (CLCrV system)	Engine viral backbone for delivering host gene fragments and triggering silencing [87] [85].
Agrobacterium Strain	*A. tumefaciens* **GV3101**	Delivery vehicle; facilitates transfer of T-DNA containing viral vectors into plant cells [87] [88].
Antibiotics	Kanamycin, Rifampicin, Gentamicin	Selection pressure to maintain VIGS plasmids in bacterial and plant cells.
Infiltration Buffer	10 mM $MgCl_2$, 10 mM MES, 200 µM Acetosyringone	Maintains Agrobacterium viability and promotes T-DNA transfer during infiltration [87] [88].
Positive Control Plasmids	pTRV2-GoPGF, pTRV2-CLA1, pTRV2-PDS	Essential controls to confirm the system is working by producing a clear visual phenotype [87] [85].
RNA Extraction Kit	Biospin Plant Total RNA Extraction Kit or equivalent	Isolate high-quality RNA for validating target gene knockdown via qRT-PCR [87].

Technical Considerations and Limitations

While VIGS is a powerful technique, researchers must be aware of its limitations:

- **Silencing Efficiency Variability:** Efficiency can vary between cotton varieties and is generally higher in diploids (e.g., *G. arboreum*) than in tetraploids (e.g., *G. hirsutum*) due to ploidy level and gene duplication [85] [86].
- **Transient Nature:** Silencing is not stable across generations, limiting its use to single-generation functional studies.
- **Off-Target Effects:** Sequence similarity within gene families can lead to unintended silencing of non-target genes. Careful fragment design is essential.
- **Phenotype Interpretation:** In polyploid cotton, functional redundancy between homoeologs can mask silencing phenotypes, requiring simultaneous silencing of all copies [85] [93].

VIGS has established itself as a cornerstone technique for functional genomics in cotton, directly contributing to the validation of genes involved in stress responses, development, and, critically, disease resistance mediated by NBS genes. Its ability to provide rapid, high-throughput gene characterization without the need for stable transformation makes it ideally suited for bridging the gap between genomic sequencing/data mining and confirmed gene function. As research into the diversification of gene families like the NBS-LRR genes progresses, VIGS will remain an essential tool for moving from *in silico* predictions to validated biological understanding, ultimately accelerating crop improvement.

The nucleotide-binding site (NBS) domain constitutes a fundamental component of plant intracellular immune receptors, forming the core of one of the largest and most diverse gene families involved in pathogen recognition [5]. These genes, which often contain C-terminal leucine-rich repeat (LRR) domains, are collectively known as NBS-LRR genes or NLRs, and function as critical surveillance mechanisms in plant effector-triggered immunity [5] [94]. The diversification of NBS-encoding genes across plant species represents a dynamic evolutionary arms race between plants and their pathogens, resulting in remarkable structural and functional heterogeneity within and across species [5] [95]. This technical guide explores the genetic association between specific NBS haplotypes and disease phenotypes, providing methodologies for correlating sequence variation with susceptibility or tolerance traits, framed within the broader context of NBS gene diversification across plant species.

NBS Gene Diversity and Evolutionary Patterns

Classification and Domain Architecture

NBS-encoding genes display considerable diversity in their domain architecture, leading to their classification into distinct subgroups. Based on N-terminal domains, they are primarily categorized into:

- **TNLs**: Contain Toll/Interleukin-1 receptor (TIR) domains [5] [94]
- **CNLs**: Feature coiled-coil (CC) domains [5] [94]
- **RNLs**: Possess Resistance to Powdery Mildew 8 (RPW8) domains [5] [94]

Comprehensive analyses across plant species have identified both classical and species-specific structural patterns. A recent pan-species investigation identified 12,820 NBS-domain-containing genes across 34 species ranging from mosses to monocots and dicots, classifying them into 168 distinct classes based on domain architecture [5]. Beyond the classical patterns (NBS, NBS-LRR, TIR-NBS, TIR-NBS-LRR), researchers discovered several unusual architectures, including TIR-NBS-TIR-Cupin*1-Cupin*1 and TIR-NBS-Prenyltransf, highlighting the extensive diversification of this gene family [5].

Evolutionary Dynamics Across Plant Lineages

The evolutionary patterns of NBS-LRR genes vary significantly across plant families, reflecting distinct pathogen pressures and evolutionary histories:

Table 1: Evolutionary Patterns of NBS-LRR Genes Across Plant Families

Plant Family	Representative Species	Evolutionary Pattern	Gene Count Range
Rosaceae	Apple, Strawberry, Peach	Dynamic patterns including "continuous expansion," "first expansion then contraction," and "early sharp expanding to abrupt shrinking" [94]	Varied distinctively across species [94]
Solanaceae	Potato, Tomato, Pepper	"Consistent expansion" (potato), "expansion followed by contraction" (tomato), "shrinking" (pepper) [94]	Varies 2-6 fold between species [96]
Poaceae	Rice, Maize, Sorghum	"Contracting" pattern [94]	~600 in rice, ~129 in maize [94]
Fabaceae	Soybean, Common Bean	"Consistently expanding" pattern [94]	Not specified
Orchidaceae	*Dendrobium catenatum*, *Gastrodia elata*	"Early contraction to recent expansion" vs. "contraction" [96]	115 vs. 5 [94]

The genomic distribution of NBS genes is typically non-random and uneven, with a significant percentage occurring in clusters. In Ipomoea species, between 76.71% and 90.37% of NBS-encoding genes reside in clusters [96], facilitating sequence exchange through unequal crossing-over and gene conversion events that generate diversity [95].

Experimental Framework for NBS Haplotype-Disease Association

Genome-Wide Identification and Haplotype Characterization

Protocol 1: Identification and Classification of NBS-Encoding Genes

- **Sequence Retrieval**: Obtain whole genome sequences and annotated gene models from databases such as NCBI, Phytozome, Plaza, or species-specific databases [5] [96] [94].
- **HMMER Search**: Perform hidden Markov model (HMM) searches using HMMER v3.1b2 with the NB-ARC domain model (PF00931) from the PFAM database [5] [6].
- **Domain Validation**: Confirm identified genes through PFAM and NCBI Conserved Domain Database (CDD) searches for associated domains (TIR: PF01582; CC: via CDD; LRR: various PFAM models) [94] [6].
- **Classification**: Categorize genes based on domain architecture into classes (CN, CNL, N, NL, RN, RNL, TN, TNL) [6].

Protocol 2: Haplotype Variation Analysis

- **Plant Materials**: Select contrasting genotypes (susceptible and tolerant) for comparison. Example: For cotton leaf curl disease, use tolerant (Mac7) and susceptible (Coker 312) *Gossypium hirsutum* accessions [5].
- **Variant Identification**: Sequence NBS genes from multiple accessions and identify sequence variants (SNPs, indels). In cotton, this approach identified 6,583 unique variants in the tolerant Mac7 and 5,173 in the susceptible Coker312 [5].
- **Haplotype Reconstruction**: Group sequence variants into haplotypes, recognizing that two main types exist:
  - **Type I**: Extensive chimeras with frequent sequence exchanges that homogenize intron sequences but obscure orthologous relationships [95].
  - **Type II**: Exhibit infrequent sequence exchange with clear orthologous relationships across genotypes/species [95].

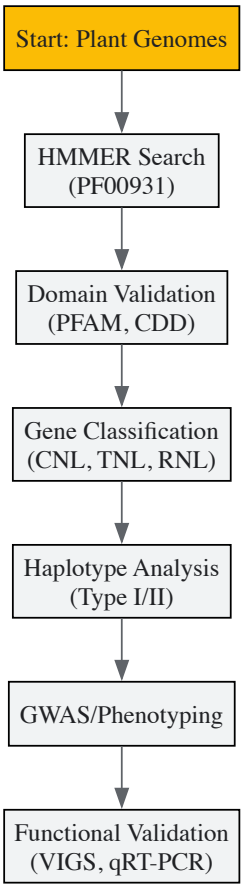


Figure 1: Experimental workflow for correlating NBS haplotypes with disease susceptibility and tolerance.

Genome-Wide Association Studies (GWAS) and Expression Analysis

Protocol 3: Genome-Wide Association Mapping

- **Phenotyping**: Conduct disease assays under controlled conditions. For late blight in potato, inoculate leaves with pathogen suspensions (e.g., *Phytophthora infestans* at 50,000 sporangia/mL) and score disease response using standardized scales [97].
- **Genotyping**: Generate high-quality SNPs and indels using various technologies (e.g., AFSM, GBS). A potato GWAS study identified 22,489 high-quality variants across 284 cultivars [97].
- **Population Structure Analysis**: Determine population subgroups using ADMIXTURE software and principal component analysis (PCA) to account for stratification [97].
- **Association Analysis**: Employ mixed linear models to identify significant associations between genetic variants and disease traits [97].

Protocol 4: Expression Profiling

- **RNA-Seq Data Collection**: Retrieve transcriptomic data from databases (e.g., IPF database, NCBI BioProjects) encompassing various tissues and stress conditions [5].
- **Differential Expression Analysis**: Process data through transcriptomic pipelines (HISAT2, Cufflinks/Cuffdiff) to identify differentially expressed genes (DEGs) under pathogen challenge [6].
- **qRT-PCR Validation**: Confirm expression patterns of candidate genes using quantitative reverse-transcription PCR with appropriate reference genes [96].

Case Studies and Data Interpretation

Cotton Leaf Curl Disease (CLCuD) Resistance

In a comprehensive study of NBS genes in cotton, researchers investigated the genetic variation between susceptible (Coker 312) and tolerant (Mac7) *Gossypium hirsutum* accessions:

Table 2: NBS Gene Variants Associated with CLCuD Tolerance in Cotton

Accession	Disease Response	Unique Variants in NBS Genes	Key Orthogroups	Functional Validation Results
Mac7	Tolerant	6,583 variants	OG2, OG6, OG15	Silencing of GaNBS (OG2) increased virus titer [5]
Coker 312	Susceptible	5,173 variants	Not specified	Not specified
G. arboreum	Resistant	Not specified	Not specified	Not specified

Expression profiling revealed putative upregulation of orthogroups OG2, OG6, and OG15 in different tissues under various biotic and abiotic stresses in both susceptible and tolerant plants [5]. Protein-ligand and protein-protein interaction analyses demonstrated strong interaction of some putative NBS proteins with ADP/ATP and different core proteins of the cotton leaf curl disease virus [5].

Sweet Potato and Ipomoea Species

A comparative analysis of NBS-encoding genes across four Ipomoea species revealed:

- Gene numbers: 889 in sweet potato (*Ipomoea batatas*), 554 in *I. trifida*, 571 in *I. triloba*, and 757 in *I. nil* [96]
- CN-type and N-type were more common than other NBS-encoding gene types [96]
- Distribution across chromosomes was non-random and uneven, with 83.13% of sweet potato NBS genes occurring in clusters [96]

Transcriptome analysis of sweet potato cultivars with contrasting responses to stem nematodes and *Ceratocystis fimbriata* identified 11 and 19 differentially expressed NBS genes, respectively [96]. qRT-PCR validation confirmed the expression patterns of six candidate DEGs [96].

Lactuca spp. and the RGC2 Locus

The RGC2 locus in lettuce represents one of the largest NBS-LRR clusters characterized in plants, with copy number varying from 12-32 per genome across seven genotypes [95]. Two evolutionarily distinct types of RGC2 genes were identified:

- **Type I genes**: Extensive chimeras formed by frequent sequence exchanges, homogenizing intron sequences but obscuring orthologous relationships [95].
- **Type II genes**: Exhibited infrequent sequence exchange with maintenance of obvious orthologous relationships across species [95].

Trans-specific polymorphism was observed for different groups of orthologs, suggesting balancing selection acting to maintain diversity [95].

The Scientist's Toolkit: Research Reagent Solutions

Table 3: Essential Research Reagents for NBS Haplotype-Disease Association Studies

Reagent/Resource	Function	Example Sources/Protocols
PF00931 HMM Profile	Identification of NB-ARC domains in protein sequences	PFAM Database [5] [6]
PRGminer	Deep learning-based prediction and classification of resistance genes	https://github.com/usubioinfo/PRGminer [58]
OrthoFinder	Orthogroup inference and comparative genomics	OrthoFinder v2.5.1 [5]
VIGS Vectors	Virus-induced gene silencing for functional validation	Tobacco rattle virus-based systems [5]
ADMIXTURE	Population structure analysis	ADMIXTURE software [97]
HISAT2-Cufflinks Pipeline	RNA-seq alignment and differential expression analysis	HISAT2 for alignment, Cufflinks for quantification [6]
KaKs_Calculator	Calculation of Ka/Ks ratios for selection pressure analysis	KaKs_Calculator 2.0 [6]

The correlation between NBS haplotypes and disease susceptibility represents a crucial interface between molecular genetics and plant breeding. The extensive diversification of NBS domain genes across plant species, driven by various evolutionary processes including tandem duplication, segmental duplication, and sequence exchanges, has generated a rich reservoir of genetic variation for disease resistance breeding. By employing integrated approaches combining genome-wide association studies, expression profiling, and functional validation, researchers can effectively mine this variation to identify superior haplotypes associated with disease tolerance. These efforts ultimately contribute to the development of durable disease resistance in crop plants, leveraging the natural diversity of NBS-encoding genes that has evolved through millennia of plant-pathogen interactions.

Nucleotide-binding site (NBS) domain genes constitute one of the largest and most critical gene families in plant innate immunity, encoding proteins that function as intracellular sensors for pathogen detection [5]. These genes exhibit remarkable diversification across plant species, resulting from

dynamic evolutionary processes including tandem duplications, gene conversions, and birth-and-death evolution [1] [7]. The NBS domain, more specifically known as the NB-ARC (Nucleotide-Binding Adaptor shared by APAF-1, R proteins, and CED-4) domain, serves as a molecular switch that alternates between ADP-bound (inactive) and ATP-bound (active) states to regulate defense signaling [1] [24]. Understanding the precise molecular mechanisms by which NBS proteins interact with pathogen effectors and nucleotides is fundamental to elucidating plant immunity and harnessing these genes for crop improvement. This technical guide provides an in-depth examination of experimental approaches and mechanistic insights into NBS protein interactions within the broader context of their diversification across plant species.

Structural and Evolutionary Context of NBS Proteins

Domain Architecture and Classification

NBS-containing proteins, particularly those belonging to the NBS-LRR (Nucleotide-Binding Site Leucine-Rich Repeat) class, are characterized by a conserved tripartite domain architecture:

- **N-terminal domain**: Typically consists of either a Toll/Interleukin-1 Receptor (TIR) domain or a Coiled-Coil (CC) domain, with some proteins containing Resistance to Powdery Mildew 8 (RPW8) domains [1] [7]. This domain is involved in initiating downstream signaling cascades.
- **Central NBS domain**: Contains conserved motifs (P-loop, RNBS-A, kinase-2, RNBS-B, RNBS-C, and GLPL) essential for nucleotide binding and hydrolysis [24].
- **C-terminal LRR domain**: Comprises variable leucine-rich repeats that facilitate protein-protein interactions and determine pathogen recognition specificity [98] [1].

Based on their N-terminal domains, NBS-LRR proteins are classified into three major subfamilies: TNL (TIR-NBS-LRR), CNL (CC-NBS-LRR), and RNL (RPW8-NBS-LRR) [7]. The evolutionary history of these subfamilies reveals distinct patterns of expansion and contraction across plant lineages, with TNLs completely absent from cereal genomes and CNLs representing the dominant subclass in many angiosperms [1] [7].

Genomic Distribution and Evolutionary Dynamics

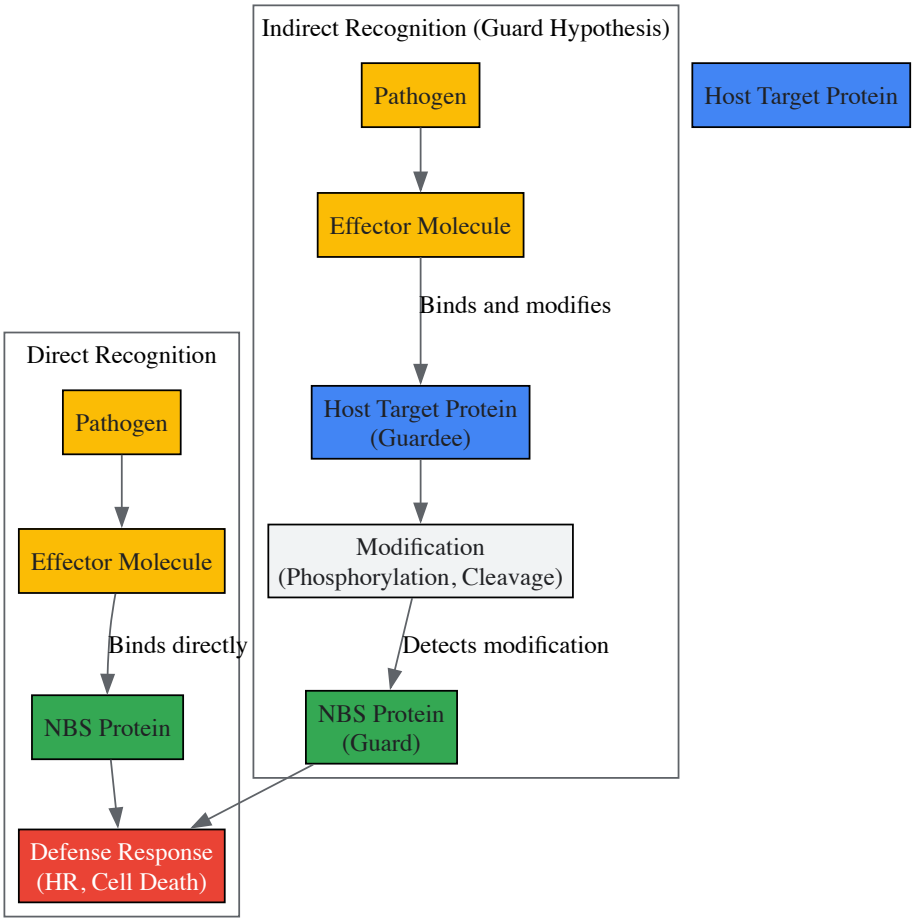
NBS-encoding genes are frequently organized as tandem arrays in plant genomes, with few existing as singletons [7]. Comparative genomic analyses across Solanaceae species (potato, tomato, and pepper) reveal diverse evolutionary patterns, from "consistent expansion" in potato to "shrinking" in pepper [7]. These dynamic evolutionary processes contribute to the species-specific repertoire of NBS genes, enabling adaptation to diverse pathogen pressures.

Table 1: NBS-LRR Gene Distribution Across Selected Plant Species

Plant Species	Total NBS Genes	TNL Genes	CNL Genes	RNL Genes	Reference
Arabidopsis thaliana	~150	62	88	Not specified	[1]
Oryza sativa (rice)	>400	0	>400	Not specified	[1]
Nicotiana benthamiana	156	5	25	4	[11]
Solanum tuberosum (potato)	447	Not specified	Not specified	Not specified	[7]
Capsicum annuum (pepper)	252	4	248	Not specified	[24]
Vernicia montana	149	12	98	Not specified	[99]

Mechanisms of Pathogen Recognition by NBS Proteins

NBS-LRR proteins employ two primary strategies for pathogen detection: direct and indirect recognition. The following diagram illustrates these fundamental mechanisms:



Direct Recognition Mechanisms

Direct recognition occurs when NBS proteins physically bind to pathogen effector molecules. This mechanism provides high specificity but requires continuous evolutionary adaptation to track rapidly evolving pathogen effectors. Key examples include:

- The rice NBS-LRR protein Pi-ta directly interacts with the effector AVR-Pita from the rice blast fungus *Magnaporthe grisea* [98].
- The flax L5, L6, and L7 proteins bind specifically to variants of the flax rust AvrL567 effector, as demonstrated through yeast two-hybrid experiments that recapitulate in vivo specificity [98].
- The wheat Ym1 CC-NBS-LRR protein confers resistance to wheat yellow mosaic virus (WYMV) by directly interacting with the viral coat protein [100].

Indirect Recognition Mechanisms

Indirect recognition, formalized in the "guard hypothesis," involves NBS proteins monitoring the status of host cellular proteins that are targeted by pathogen effectors. This strategy allows plants to detect multiple effectors that converge on the same host protein and reduces the evolutionary burden of developing new recognition specificities. Notable examples include:

- The Arabidopsis RPM1 protein guards the host protein RIN4, detecting its phosphorylation by the bacterial effectors AvrRpm1 and AvrB [98].
- Arabidopsis RPS2 recognizes the cleavage of RIN4 by the bacterial effector AvrRpt2 [98].
- The tomato Prf protein indirectly detects the effectors AvrPto and AvrPtoB through their interaction with the host kinase Pto [98].

Nucleotide Binding and Hydrolysis in NBS Proteins

The NBS domain functions as a molecular switch regulated by nucleotide-dependent conformational changes. In the inactive state, the domain binds ADP, maintaining the protein in an autoinhibited conformation. Upon pathogen recognition, ADP is exchanged for ATP, triggering a conformational change that activates downstream signaling [1].

The NBS domain contains several conserved motifs that facilitate nucleotide binding and hydrolysis:

Table 2: Conserved Motifs in the NBS Domain

Motif	Consensus Sequence	Function
P-loop	GxGGLGKT	Phosphate binding loop for ATP/GTP binding
RNBS-A	GxPLLF	Contributes to nucleotide binding pocket
Kinase-2	LVLDDVW	Mg^{2+} coordination and catalytic activity
RNBS-B	GSRIIITTRD	Differentiation between TNL and CNL subfamilies
RNBS-C	FLHIACF	Structural stabilization
GLPL	GLPLA	Nucleotide binding and hydrolysis
MHD	MHD	Regulatory function

Experimental evidence from tomato CNLs I2 and Mi demonstrates specific ATP binding and hydrolysis activity, with mutations in conserved motifs abolishing nucleotide binding and compromising resistance function [1]. The conformational changes associated with nucleotide exchange enable the NBS protein to transition from an autoinhibited to an activated state, facilitating interactions with downstream signaling components.

Experimental Approaches for Studying NBS Protein Interactions

Protein-Protein Interaction Assays

Yeast Two-Hybrid Systems

The yeast two-hybrid (Y2H) system has been instrumental in identifying direct interactions between NBS proteins and pathogen effectors.

Protocol for Yeast Two-Hybrid Analysis:

- Clone the coding sequence of the NBS protein into the pGBKT7 DNA-binding domain vector
- Clone the coding sequence of the candidate pathogen effector into the pGADT7 activation domain vector
- Co-transform both plasmids into yeast strain AH109
- Plate transformations on synthetic dropout media lacking leucine and tryptophan (-LW) to select for transformants
- Transfer positive colonies to media lacking leucine, tryptophan, histidine, and adenine (-LWAH) to test for interaction
- Include appropriate positive and negative controls
- Confirm interactions through β-galactosidase assays for additional stringency

This approach successfully identified the direct interaction between the rice Pi-ta protein and the AVR-Pita effector [98] and between flax L proteins and AvrL567 effectors [98].

Split-Ubiquitin System

The split-ubiquitin system is particularly useful for studying membrane-associated proteins or proteins with localization constraints.

Protocol:

- Fuse the NBS protein to the C-terminal ubiquitin fragment (Cub)
- Fuse the candidate effector to the N-terminal ubiquitin fragment (Nub)
- Co-express the fusion proteins in an appropriate yeast strain
- Monitor reconstitution of ubiquitin through cleavage of transcription factors and reporter gene activation

This method demonstrated interaction between the Arabidopsis RRS1 TNL protein and the bacterial effector PopP2 [98].

Nucleotide Binding Assays

In Vitro Binding Assays

Direct measurement of nucleotide binding to purified NBS domains provides quantitative data on binding affinity and specificity.

Protocol:

- Express and purify the recombinant NBS domain using E. coli expression systems
- Perform equilibrium binding experiments with radioactive or fluorescently-labeled nucleotides (ATP, ADP, GTP, GDP)
- Separate protein-bound nucleotides from free nucleotides through filter binding or gel filtration
- Determine dissociation constants (Kd) through saturation binding curves

- Conduct competition experiments to assess binding specificity

Studies using this approach with tomato I2 and Mi proteins confirmed specific ATP binding and hydrolysis activity [1].

Functional Validation Through Genetic Approaches

Virus-Induced Gene Silencing (VIGS)

VIGS provides a powerful method for functional characterization of NBS genes in plant systems.

Protocol:

- Clone a 200-500 bp fragment of the target NBS gene into a VIGS vector (e.g., TRV-based vectors)
- Transform the construct into Agrobacterium tumefaciens
- Infiltrate the Agrobacterium suspension into young leaves of the target plant species
- Monitor silencing efficiency through quantitative RT-PCR
- Challenge silenced plants with the target pathogen
- Assess disease symptoms and pathogen proliferation

This approach validated the function of GaNBS in cotton resistance to cotton leaf curl disease [5] and Vm019719 in Vernicia montana resistance to Fusarium wilt [99].

The Scientist's Toolkit: Research Reagent Solutions

Table 3: Essential Research Reagents for NBS Protein Interaction Studies

Reagent/Tool	Specific Examples	Application and Function
Yeast Two-Hybrid Systems	pGBKT7/pGADT7 vectors, AH109 yeast strain	Detection of direct protein-protein interactions between NBS proteins and effectors
Split-Ubiquitin System	Cub/Nub vectors	Study of membrane-associated protein interactions
VIGS Vectors	TRV-based vectors (pTRV1, pTRV2)	Functional validation through targeted gene silencing in plants
HMMER Software	HMMER3 suite	Identification of NBS domains in protein sequences using hidden Markov models
Domain Analysis Tools	PfamScan, InterProScan, SMART, MEME	Annotation of conserved domains and motifs in NBS proteins
Agrobacterium Strains	GV3101, LBA4404	Plant transformation for functional assays
Nucleotide Analogs	Fluorescent ATP/ADP, [γ-^{32}P]ATP	Measurement of nucleotide binding and hydrolysis kinetics
Phylogenetic Analysis Tools	OrthoFinder, MEGA7, FastTreeMP	Evolutionary analysis of NBS gene families

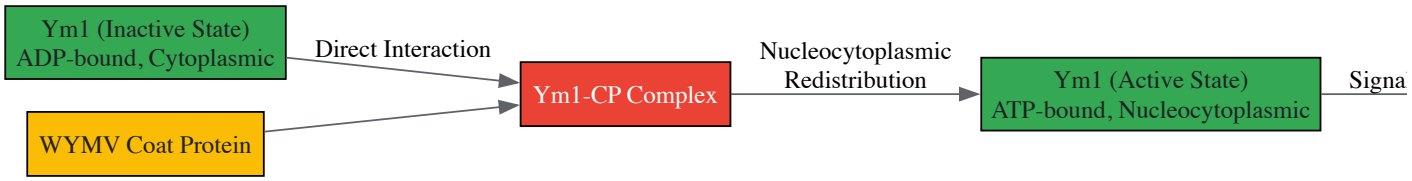
Case Study: Ym1-WYMV Coat Protein Interaction

A recent breakthrough in understanding NBS protein interactions comes from the cloning of the wheat Ym1 gene, which confers resistance to wheat yellow mosaic virus (WYMV) [100]. Ym1 encodes a typical CC-NBS-LRR protein that is specifically expressed in roots and induced upon WYMV infection.

Experimental Workflow:

- Map-based cloning identified Ym1 as a CC-NBS-LRR protein on wheat chromosome 2DL
- Bimolecular fluorescence complementation (BiFC) assays demonstrated direct interaction between Ym1 and WYMV coat protein (CP)
- Co-immunoprecipitation confirmed the physical association in planta
- Mutational analysis identified the CC domain as essential for triggering cell death
- VIGS-mediated silencing of Ym1 compromised WYMV resistance
- Overexpression of Ym1 enhanced WYMV resistance in wheat

This study revealed that the Ym1-CP interaction leads to nucleocytoplasmic redistribution of Ym1, representing a transition from an autoinhibited to an activated state [100]. The following diagram illustrates this activation mechanism:



The study of NBS protein interactions with pathogen effectors and nucleotides has revealed sophisticated mechanisms underlying plant immunity. The direct and indirect recognition strategies employed by NBS proteins, coupled with nucleotide-dependent conformational regulation, provide plants with a powerful surveillance system against diverse pathogens. The evolutionary diversification of NBS genes across plant species reflects an ongoing arms race with pathogens, with tandem duplications and birth-and-death evolution generating the genetic variation necessary for adapting to new pathogen threats.

Future research directions should focus on structural characterization of full-length NBS proteins in different nucleotide-bound states, high-throughput methods for mapping interaction networks between NBS proteins and pathogen effectors, and engineering NBS proteins with novel recognition specificities for crop protection. The integration of computational approaches like deep learning-based prediction tools (e.g., PRGminer) with experimental validation will accelerate the discovery and functional characterization of NBS genes across diverse plant species [58]. As our understanding of NBS protein interactions deepens, so does our potential to develop durable disease resistance in crops through informed breeding and biotechnology approaches.

Nucleotide-Binding Site (NBS) domains are ancient, evolutionarily conserved modules that function as molecular switches in diverse biological systems across kingdoms. In plants, they form the core of the NBS-LRR (Nucleotide-Binding Site Leucine-Rich Repeat) receptor family, which constitutes the primary mediators of effector-trigered immunity (ETI) against pathogens [79]. In humans, structurally similar domains are found in ATP-Binding Cassette (ABC) transporters, which mediate multidrug resistance in pathogens and cancer, and in NOD-Like Receptors (NLRs), which orchestrate innate immune responses. Despite their phylogenetic distance, these systems share remarkable mechanistic parallels in their dependence on nucleotide-dependent conformational changes for function. This whitepaper synthesizes recent advances in understanding plant NBS domain diversification to extract transferable principles for human immunology and transporter research, framing these insights within the broader context of nucleotide-binding domain gene evolution.

The modular architecture of NBS domains enables their functional diversification through domain shuffling and sequence evolution. Plant NBS-LRR genes have undergone dramatic expansion, with angiosperm genomes encoding hundreds to thousands of members [5]. A comprehensive analysis across 34 plant species identified 12,820 NBS-domain-containing genes classified into 168 distinct domain architecture patterns, revealing both classical (NBS, NBS-LRR, TIR-NBS-LRR) and species-specific structural variations [5]. Similarly, fungal NLR repertoires exhibit extraordinary domain assortment, with 4,613 NLRs identified across 82 Sordariales taxa demonstrating combinatorial associations of various N-terminal, NB, and C-terminal domains [101]. This evolutionary flexibility in domain architecture offers valuable insights for engineering synthetic immune receptors and transporters with novel specificities.

Table 1: Diversity of NBS Domain Architectures Across Kingdoms

Kingdom	Representative System	Domain Architecture Variants	Genomic Organization
Plants	NBS-LRR receptors	CNL, TNL, RNL, NL, CN, TN, N [5] [79]	Clustered, tandem arrays [102]
Fungi	NLRs	NACHT/NAIP-like, TLP1-like, various N-terminal domains [101]	Clustered organization [101]
Animals	NLRs, ABC transporters	STAND superfamily, NACHT domains, NB-ARC domains [101]	Variable, some clustering [101]

Evolutionary Diversification and Genomic Organization of NBS Domains

Patterns of Gene Family Expansion in Plants

The NBS-LRR gene family has expanded through both whole-genome duplication (WGD) and small-scale duplication events, with significant variation in duplication preferences across lineages. In sugarcane, WGD appears to be the primary driver of NBS-LRR gene number, while in other species, tandem duplications contribute significantly to the creation of expanded, rapidly evolving clusters [102]. This expansion is not correlated with genome size or total gene number, but rather with specific evolutionary pressures related to pathogen exposure [102]. Comparative analysis of four grass species (Saccharum spontaneum, Saccharum officinarum, Sorghum bicolor, and Miscanthus sinensis) revealed that conserved NBS-LRR genes maintain core structural features while acquiring species-specific variations through duplications and subsequent neofunctionalization.

The genomic organization of NBS-domain genes exhibits conserved features across kingdoms. In plants, NBS-LRR genes are frequently organized in clusters, a pattern also observed in fungal genomes [101] [102]. This clustered organization likely facilitates the generation of diversity through unequal crossing over and gene conversion. A strong correlation between the number of NLRs and the number of NLR clusters in Sordariales fungi suggests that organization in clusters contributes significantly to repertoire diversification [101]. Similarly, in plants, closely related NBS-LRR genes often reside in tandem arrays, allowing for the emergence of new specificities through recombination and diversifying selection.

Birth-and-Death Evolution and Adaptive Selection

NBS-domain genes undergo "birth-and-death" evolution, where new genes are created by duplication, and some duplicates are maintained while others degenerate or are lost [102]. This evolutionary dynamic generates substantial variation in gene content between even closely related species. Analysis of NBS-LRR genes in Salvia miltiorrhiza identified 196 NBS-domain-containing genes, but only 62 possessed complete N-terminal and LRR domains, indicating significant gene degeneration and partial gene retention [79]. Similarly, comparative analysis across Salvia species revealed a marked reduction and even complete loss of certain NLR subfamilies (TNL and RNL) in specific lineages [79].

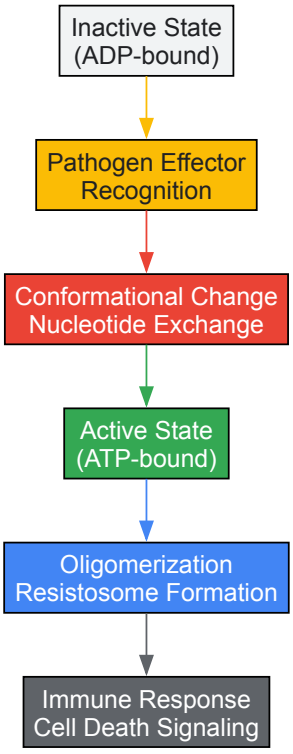
This birth-and-death process is driven by adaptive evolution, with analyses revealing a progressive trend of positive selection on NBS-LRR genes [102]. Positive selection primarily acts on the LRR domains responsible for pathogen recognition, while the NBS and other signaling domains remain under stronger purifying selection. This evolutionary pattern mirrors findings in human NLRs and ABC transporters, where substrate-binding regions show heightened variability while nucleotide-binding cores remain conserved. The identification of orthogroups with tandem duplications [5] provides evidence for lineage-specific expansions tailored to particular pathogen pressures, offering a model for understanding similar expansions in human immune gene families.

Molecular Mechanisms and Cooperative Signaling Systems

Nucleotide-Dependent Activation and Allosteric Regulation

NBS domains function as molecular switches that cycle between ADP-bound (inactive) and ATP-bound (active) states, a mechanism conserved across plant and animal systems [79]. In plant NBS-LRR proteins, the NBS domain binds and hydrolyzes ATP/GTP, with nucleotide exchange triggering conformational changes that activate downstream signaling [79]. Similarly, human NLRs and ABC transporters utilize nucleotide binding and hydrolysis for function—NLRs for oligomerization and signal initiation, and ABC transporters for substrate translocation. The recent improvement in annotation of the Helical Third section of fungal NLR nucleotide-binding domains [101] has revealed greater conservation between fungal and animal NACHT domains than previously recognized, suggesting deep evolutionary conservation of allosteric regulation mechanisms.

The modular architecture of NBS-domain proteins enables functional specialization through domain integration. Plant NBS-LRR proteins typically consist of three components: an N-terminal domain (TIR, CC, or RPW8), a central NB-ARC domain, and a C-terminal LRR domain [103] [79]. The NB-ARC domain (Nucleotide-Binding adaptor shared by APAF-1, plant R proteins, and CED4) serves as a signal transduction ATPase with numerous domains (STAND) [101], closely related to the NACHT domain (NAIP, CIIA, HET-E, TP1) found in animal NLRs [5]. This structural conservation enables comparative analyses to identify core functional principles.



NBS Domain Activation Pathway: Conserved mechanism of nucleotide-dependent activation in plant NBS-LRR proteins and human NLRs.

Paired NLR Systems and Cooperative Function

Recent research has revealed that plant NLRs often function in paired complexes rather than as solitary receptors, providing a powerful model for understanding protein cooperativity. The PmWR183 resistance locus from wild emmer wheat encodes two adjacent NLR proteins (PmWR183-NLR1 and PmWR183-NLR2) that function cooperatively—neither gene alone confers resistance, but their co-expression restores immunity, while disruption of either gene abolishes resistance [104]. Protein interaction assays demonstrate constitutive association between these paired NLRs, supporting their cooperative role in immune signaling [104]. This paired architecture often exhibits head-to-head genomic orientation, as seen in Arabidopsis RPS4/RRS1, rice RGA4/RGA5, and wheat RXL/Pm5e systems [104].

These cooperative NLR systems frequently employ "sensor-helper" divisions of labor, where one partner specializes in pathogen recognition while the other mediates signaling execution. This functional specialization enables more sophisticated immune recognition while constraining inappropriate activation. Similar cooperative arrangements exist in human NLR signaling complexes, suggesting convergent evolutionary solutions to the challenge of maintaining specificity while enabling amplified signal transduction. The study of these plant paired systems provides experimental templates for interrogating human NLR interactions and for engineering synthetic immune receptors with controlled activation thresholds.

Table 2: Experimentally Validated Paired NLR Systems in Plants

Paired System	Species	Genomic Arrangement	Functional Relationship
PmWR183-NLR1/NLR2	Triticum dicoccoides	Adjacent genes	Cooperative function, neither functions alone [104]
RPS4/RRS1	Arabidopsis thaliana	Head-to-head	Sensor-helper pair [104]
RGA4/RGA5	Oryza sativa	Head-to-head	Integrated decoy and executor [104]
Pm5e/RXL	Triticum aestivum	Head-to-head	Paired sensor and signaling NLR [104]
TdCNL1/TdCNL5	Triticum dicoccoides	Clustered	Coordinated function [104]

Experimental Approaches and Methodological Synergies

Genomic Identification and Diversity Analysis

The identification and characterization of NBS-domain genes across species employs standardized computational pipelines that could be adapted for comparative analyses of human ABC transporters and NLRs. A typical workflow begins with domain identification using Hidden Markov Models (HMM) from databases like Pfam, searching for NB-ARC (PF00931) and related domains with stringent e-value cutoffs (1.1e-50) [5] [103]. Subsequent orthogroup analysis using tools like OrthoFinder (v2.5.1) with DIAMOND for sequence similarity searches and MCL for clustering identifies conserved and lineage-specific NBS genes [5]. Phylogenetic analysis via MAFFT alignment and FastTreeMP or IQ-TREE construction reveals evolutionary relationships and diversification patterns.

Functional diversification is assessed through multiple complementary approaches. Domain architecture classification systems categorize NBS genes based on their domain combinations (N, L, CN, TN, NL, CNL, TNL, etc.) [76] [79]. Positive selection is detected by calculating non-synonymous to synonymous substitution rates (Ka/Ks) across orthogroups [102]. Expression profiling under various biotic and abiotic stresses, combined with genetic variation analysis between susceptible and resistant genotypes, identifies functionally important NBS genes [5]. These methodologies form a comprehensive toolkit for characterizing nucleotide-binding domain evolution that transcends kingdom boundaries.

Functional Validation Techniques

The functional characterization of plant NBS domains employs rigorous experimental approaches that provide models for validating human NLR and ABC transporter functions:

Stable Transformation and Complementation Assays: Functional validation of candidate NBS genes typically involves stable transformation into susceptible genotypes. For the PmWR183 locus, transformation experiments demonstrated that neither NLR1 nor NLR2 alone could confer resistance, but their co-expression restored immunity, establishing their cooperative function [104].

CRISPR/Cas9-Mediated Gene Knockout: Precise gene editing provides compelling evidence of gene function. Knockout of either PmWR183-NLR1 or PmWR183-NLR2 completely abolished resistance, confirming that both partners are essential [104]. This approach effectively establishes gene necessity rather than just sufficiency.

Virus-Induced Gene Silencing (VIGS): Transient silencing enables rapid functional assessment. Silencing of GaNBS (OG2) in resistant cotton demonstrated its role in virus tolerance [5]. VIGS is particularly valuable for screening multiple candidate genes and studying essential genes that might be lethal in stable knockouts.

Protein Interaction assays: Yeast two-hybrid, co-immunoprecipitation, and bimolecular fluorescence complementation assays validate physical interactions between NBS proteins. For PmWR183, protein interaction assays revealed constitutive association between NLR1 and NLR2, supporting their cooperative role [104].

Transcriptional Profiling: RNA-seq analysis of NBS gene expression patterns under pathogen challenge and in different tissues identifies condition-specific regulation. Studies in sugarcane revealed differential expression of NBS-LRR genes in response to multiple diseases, with more differentially expressed genes derived from the wild relative S. spontaneum than from the cultivated S. officinarum [102].

Experimental Workflow for NBS Domain Characterization: Integrated computational and experimental approaches for functional analysis.

The Scientist's Toolkit: Essential Research Reagents and Solutions

Table 3: Key Research Reagents for NBS Domain Studies

Reagent/Solution	Function/Application	Representative Use
HMMER Suite with Pfam HMM profiles	Domain identification and annotation	Identifying NB-ARC domains (PF00931) in genomic datasets [5] [103]
OrthoFinder with DIAMOND	Orthogroup inference and comparative genomics	Identifying conserved NBS genes across species [5]
CRISPR/Cas9 vectors	Targeted gene knockout	Validating essentiality of paired NLR components [104]
VIGS (Virus-Induced Gene Silencing) vectors	Transient gene silencing	Rapid functional assessment of NBS genes [5] [105]
Yeast Two-Hybrid System	Protein-protein interaction mapping	Testing constitutive association of paired NLRs [104]
RNA-seq libraries	Transcriptome profiling	Identifying NBS genes responsive to pathogens [5] [102]

Translational Applications and Future Directions

Insights for Human ABC Transporter Research

The extensive diversification of plant NBS domains offers valuable perspectives for understanding and overcoming multidrug resistance mediated by human ABC transporters. Plant ABC transporters like the CDR1-like proteins in Magnaporthe oryzae and Trichophyton mentagrophytes share conserved structures and functions with human ABCG transporters, including roles in multidrug resistance [106]. The identification of MoCDR1 and TmCDR1 as ABCG subfamily transporters involved in both drug resistance and pathogenicity [106] demonstrates how comparative analyses can reveal conserved functional modules. The systematic identification of 50 putative ABC transporter genes in M. oryzae and their classification into subfamilies [106] provides a methodological framework for comprehensive ABC transporter characterization in human pathogens.

Functional studies of plant and fungal ABC transporters reveal conserved mechanisms relevant to clinical drug resistance. Disruption of MoCDR1 in M. oryzae caused hypersensitivity to multiple drugs and impaired pathogenicity, while its homolog TmCDR1 mediated drug resistance and skin infection in T. mentagrophytes [106]. Complementation experiments demonstrated functional conservation, with MoCDR1 rescuing defects in ΔTmcdr1 strains and vice versa [106]. These findings highlight the potential for cross-kingdom analyses to identify structurally conserved regions that could be targeted with broad-spectrum inhibitors. The transcriptome analyses showing that disruption of both MoCDR1 and TmCDR1 caused analogous changes in gene expression related to MAPK signaling, transporter activity, and metabolic processes [106] suggest conserved regulatory networks that could be exploited therapeutically.

Implications for Human NLR Research and Therapeutic Development

Plant NBS-LRR research provides conceptual frameworks for understanding human NLR biology, particularly regarding receptor cooperativity, regulation, and signal amplification. The prevalence of paired NLR systems in plants [104] suggests that human NLRs may function in more complex cooperative networks than currently appreciated. The "sensor-helper" division of labor in plant NLR pairs offers a model for deconstructing functional specializations within human inflammasome complexes. Similarly, the identification of plant NLRs that integrate multiple pathogen sensors into unified signaling outputs [104] provides architectural principles for engineering synthetic immune receptors.

The study of plant NBS domain evolution also informs therapeutic strategies for human inflammatory and autoimmune diseases. The discovery that microRNAs target conserved nucleotide-binding sequences in plant NLRs [5] suggests similar regulatory mechanisms might operate in human NLRs. The developmental stage-dependent resistance mediated by PmWR183, with susceptibility at the seedling stage and strong resistance at the adult stage [104], demonstrates how NBS-mediated immunity can be temporally regulated—a concept relevant to age-associated inflammatory conditions in humans. Furthermore, the geographical and haplotype analyses showing that resistance loci often originate from wild relatives and exhibit

multiple haplotypes in cultivated species [104] highlight the importance of mining natural variation for therapeutic insights, paralleling human population genomics approaches.

The comparative analysis of plant NBS domains yields profound insights for human ABC transporter and NLR research, revealing conserved principles of nucleotide-dependent allosteric control, cooperative receptor function, and evolutionary diversification. The extensive functional characterization of plant NBS-LRR genes, including their genomic organization, paired architectures, and activation mechanisms, provides valuable models for interrogating human immune receptors and transporters. Methodological advances in domain annotation, evolutionary analysis, and functional validation in plants offer transferable approaches for human gene family studies. As structural and functional data accumulate across kingdoms, opportunities will expand for leveraging plant NBS domain knowledge to address human health challenges, particularly in overcoming multidrug resistance and modulating immune responses. The continued integration of comparative genomics with mechanistic studies will further illuminate the universal design principles of nucleotide-binding domain proteins while revealing lineage-specific adaptations that enable specialized functions.

The **nucleotide-binding site (NBS) domain** represents a critical evolutionary conserved module that functions as a molecular switch in numerous biological processes, ranging from plant immunity to human disease pathways. Within the broader context of research on the diversification of NBS domain genes across plant species, understanding their translational potential is paramount for informing modern drug discovery. These domains, particularly within **NBS-leucine rich repeat (NLR) proteins**, exhibit remarkable structural and functional diversity across evolutionary lineages, yet share conserved mechanisms in nucleotide-dependent activation and signaling [5] [101]. This technical guide explores how characterizing this natural diversification provides a framework for targeting analogous domains in human disease contexts, leveraging evolutionary insights to accelerate therapeutic development for cancer, infectious diseases, and immune disorders. The deep conservation of NBS domains across kingdoms, from plant NLRs to human STAND proteins, creates unique opportunities for cross-kingdom target identification and validation [101]. Furthermore, the modular architecture of these domains, often combined with various effector domains, presents multiple targeting strategies for small molecule interventions [107] [26].

Domain Architecture and Classification of NBS Proteins

Evolutionary Diversification of Domain Architectures

NBS-containing proteins exhibit remarkable architectural diversity resulting from evolutionary processes including gene duplication, domain shuffling, and functional diversification. The core NBS domain typically consists of approximately 300 amino acids involved in nucleotide-dependent activation and signal transduction [26]. This domain is frequently found in combination with various N-terminal and C-terminal domains that determine specific functional roles:

- **Coiled-coil (CC) NBS-LRR (CNL)**: Characterized by N-terminal coiled-coil domains, prevalent in plant immunity [26]
- **Toll/Interleukin-1 receptor (TIR) NBS-LRR (TNL)**: Feature TIR domains that mediate signaling interactions [26]
- **RPW8-NBS-LRR (RNL)**: Contain N-terminal RPW8 domains functioning in signal transduction [5]
- **Species-specific architectures**: Include novel combinations such as TIR-NBS-TIR-Cupin*1-Cupin*1 and TIR-NBS-Prenyltransf observed in diversified plant lineages [5]

Recent analyses across multiple kingdoms reveal that this combinatorial diversity extends beyond plants to fungal and animal NLRs, which display exceptional variety in their domain assortments, incorporating enzymatic domains including kinases, proteases, and amyloid motifs [101]. This natural diversification provides a rich repository of structural configurations that can inform targeted therapeutic development.

Quantitative Analysis of NBS Domain Distribution

Table 1: Genomic Distribution of NBS Domain Genes Across Species

Species Category	Number of Species Surveyed	Total NBS Genes Identified	Common Architectural Classes	Notable Features
Land plants (mosses to angiosperms)	34	12,820 [5]	CNL, TNL, NL, N, TN	168 distinct domain architecture classes identified [5]
Fabaceae crops	9	Substantial variation independent of genome size [76]	N, L, CN, TN, NL, CNL, TNL	Preferential co-occurrence of NB-ARC with specific LRR (IPR001611) [76]
Sordariales fungi	82	4,613 NLRs [101]	NACHT and NB-ARC domain types	Organization in clusters correlated with repertoire diversification [101]

Species Category	Number of Species Surveyed	Total NBS Genes Identified	Common Architectural Classes	Notable Features
Angiosperms (ANNA database)	304	>90,000 NLR genes [5]	18,707 TNL, 70,737 CNL, 1,847 RNL	Expansion primarily in flowering plants [5]

Table 2: NBS Domain Diversity in Select Plant Genomes

Plant Species	Total NBS Genes	TNL Subclass	CNL Subclass	RNL Subclass	Unique Features
Arabidopsis thaliana	~15% of 450 cloned R genes [26]	Present	Present	Present	Model for immune function
Wheat (*Triticum aestivum*)	~460 R genes documented [26]	Present	Present	Present	Resistance against rusts and powdery mildew
Rice (*Oryza sativa*)	46 R genes against *X. oryzae* alone [26]	Present	Present	Present	Includes Xa21, early-cloned R gene
Gossypium hirsutum (cotton)	6583 unique variants in tolerant accession [5]	Present	Present	Present	Association with CLCuD tolerance

Experimental Methodologies for NBS Domain Characterization

Genomic Identification and Classification

Comprehensive identification of NBS-encoding genes employs integrated computational pipelines combining sequence similarity searches, hidden Markov models (HMMs), and domain architecture analysis:

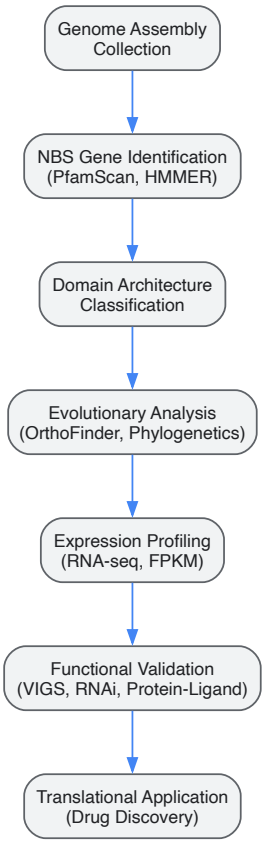
- **HMMER-based screening**: PfamScan with Pfam-A HMM models (default e-value 1.1e-50) using NB-ARC domain profiles [5]
- **Domain architecture classification**: Systems-based classification following established methods grouping genes with similar domain architectures [5]
- **Orthogroup analysis**: OrthoFinder v2.5.1 with DIAMOND for sequence similarity and MCL clustering algorithm with DendroBLAST for ortholog identification [5]
- **Phylogenetic reconstruction**: MAFFT 7.0 for multiple sequence alignment and FastTreeMP with 1000 bootstrap replicates for maximum likelihood trees [5]

Advanced annotation approaches have substantially improved characterization through examination of specific regions like the Helical Third section of nucleotide-binding domains, revealing finer evolutionary relationships [101].

Functional Validation Protocols

- **Expression profiling**: RNA-seq data processing through standardized transcriptomic pipelines with FPKM normalization; categorization into tissue-specific, abiotic stress-specific, and biotic stress-specific expression profiles [5]
- **Genetic variation analysis**: Comparison between susceptible and tolerant accessions to identify unique variants associated with resistance phenotypes [5]
- **Protein-ligand interaction studies**: Molecular docking and dynamics simulations to characterize interactions with nucleotides (ATP/ADP) and potential inhibitors [5] [108]
- **Functional perturbation**: Virus-induced gene silencing (VIGS) to validate gene function in resistant plants, demonstrating role in pathogen response [5]
- **High-throughput RNAi screening**: Double-stranded RNA treatment over extended periods (30 days) with monitoring of phenotypic consequences including attachment, movement, tissue degeneration, and mortality [109]

Visualization of NBS Domain Research Workflows



NBS Domain Research Pipeline

Modular Architecture of NLR Proteins

Translational Applications in Drug Discovery

Targeting NBS Domains in Human Disease

The conserved nature of NBS domains across kingdoms enables cross-application of insights from plant studies to human therapeutic development. Several strategic approaches have emerged:

- **Co-chaperone targeting**: Disruption of HOP-HSP90 interactions through TPR domain targeting to alter oncogenic client protein folding without triggering compensatory heat shock response [108] [107]
- **Nucleotide-binding site inhibition**: Small molecule interference with ATP/GTP binding and hydrolysis cycles in NBS-containing proteins [108]
- **Protein-protein interaction disruption**: Inhibition of complex formation between NBS proteins and their signaling partners [107]
- **Allosteric modulation**: Targeting of conformational states associated with nucleotide binding and hydrolysis [109]

Recent success in identifying the p97 ortholog in schistosomes demonstrates the utility of evolutionarily-informed target selection, where characterization of the D2 domain P-loop conformational change revealed novel allosteric binding sites for species-selective inhibitor development [109].

Computational and Structural Approaches

- **Proteome-wide Mendelian randomization**: Systematic screening to identify causal biomarkers and drug-targeting proteins through genetic evidence [110]
- **AI-driven structure prediction**: Leveraging AlphaFold models to extend domain-ligand annotations to proteins lacking experimental structures [111]
- **Molecular docking and dynamics**: Screening FDA-approved compounds for repurposing potential against NBS domains [108]
- **Cryo-EM structure elucidation**: Revealing conformational changes in nucleotide-binding motifs that create novel allosteric pockets [109]

These approaches are facilitated by resources like DrugDomain v2.0, which catalogs interactions with over 37,000 PDB ligands and 7,560 DrugBank molecules, integrating evolutionary domain classifications (ECOD) with ligand binding data [111].

The Scientist's Toolkit: Research Reagent Solutions

Table 3: Essential Research Resources for NBS Domain Studies

Resource Category	Specific Tools/Databases	Function and Application
Genomic Databases	Plaza, Phytozome, NCBI, CottonFGD, Cottongen [5]	Access to genome assemblies and annotations for diverse species
Domain Prediction	PfamScan, InterProScan, HMMER, NLR-Annotator [5] [26]	Identification of NBS and associated domains in protein sequences
Orthology Analysis	OrthoFinder, DIAMOND, MCL [5]	Evolutionary classification and orthogroup assignment
Expression Databases	IPF database, Cotton RNA-seq database, NCBI BioProjects [5]	Tissue- and stress-specific expression profiling data
Structural Resources	DrugDomain v2.0, PDB, AlphaFold DB [111]	Domain-ligand interactions and structural models
Functional Validation	VIGS constructs, dsRNA libraries, RNAi reagents [5] [109]	Gene silencing and functional characterization
Computational Analysis	RGAugury, DRAGO2/3, NLRtracker [26]	Genome-wide identification and classification of R genes
Specialized Databases	ANNA (Angiosperm NLR Atlas) [5]	Curated collection of >90,000 NLR genes from 304 angiosperms

The diversification of NBS domain genes across plant species provides not only fundamental insights into evolutionary immunology but also a robust foundation for translational drug discovery. The structural conservation of nucleotide-binding mechanisms across kingdoms enables cross-application of knowledge, where plant studies inform therapeutic targeting of human proteins containing analogous domains. The modular architecture of NBS proteins, with their combinatorial domain associations, presents multiple targeting opportunities through small molecule interference with nucleotide binding, allosteric regulation, or protein-protein interactions. As structural biology advances reveal increasingly detailed mechanisms of NBS domain function and regulation, and computational methods improve our ability to predict ligand interactions, the translational potential of this research area will continue to expand. Future directions will likely include more sophisticated bioinformatic pipelines integrating evolutionary and structural data, advanced screening platforms for NBS-targeted compounds, and innovative approaches to achieve species-selectivity in targeting pathogenic organisms while sparing host functions.

Conclusion

The diversification of NBS domain genes represents a cornerstone of plant adaptive immunity, driven by complex evolutionary processes that generate a vast repertoire for pathogen recognition. This review synthesizes how foundational genomics, advanced bioinformatics, and robust functional validation are converging to decode this complexity. The methodological frameworks and troubleshooting strategies discussed are critical for accurately annotating and harnessing these genes in crop improvement. Furthermore, the deep functional understanding of plant NBS domains offers profound comparative value, providing mechanistic insights into the operation of nucleotide-binding sites that are universal across kingdoms. Future research should focus on integrating multi-omics data and structural biology to predict resistance specificity and engineer novel immune receptors. For biomedical science, the principles gleaned from plant NBS gene evolution and function can illuminate the mechanisms of human nucleotide-binding proteins, including ABC transporters, and inform new strategies for targeting these proteins in disease, thereby opening exciting cross-disciplinary avenues for therapeutic development.

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